

# MightyConnect™ Security Platform

- Mesh P2P VPN
- Powered by WireGuard & FD.io VPP

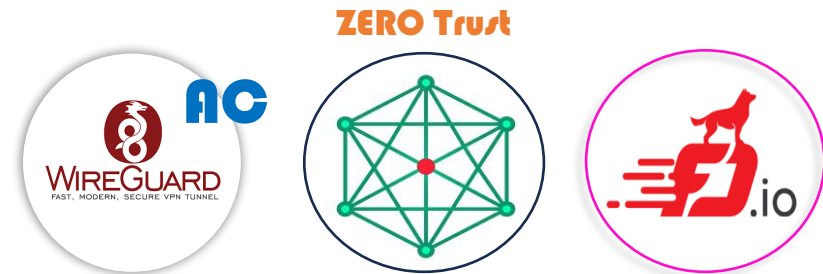
**MightyLink**

([www.mightyconn.com](http://www.mightyconn.com))

Doc. Revision: 1.00

# Table of Contents

1. Zero Trust P2P Security Platform
2. WireGuard
3. FD.io VPP
4. MightyConnect™ Security Box
5. MightyConnect™ Devices
6. References



(1) "WireGuard" and the "WireGuard" logo are registered trademarks of Jason A. Donenfeld  
(2) FD.io VPP and FD.io logo are registered trademarks of The Fast Data Project, a series of LF Projects, LLC

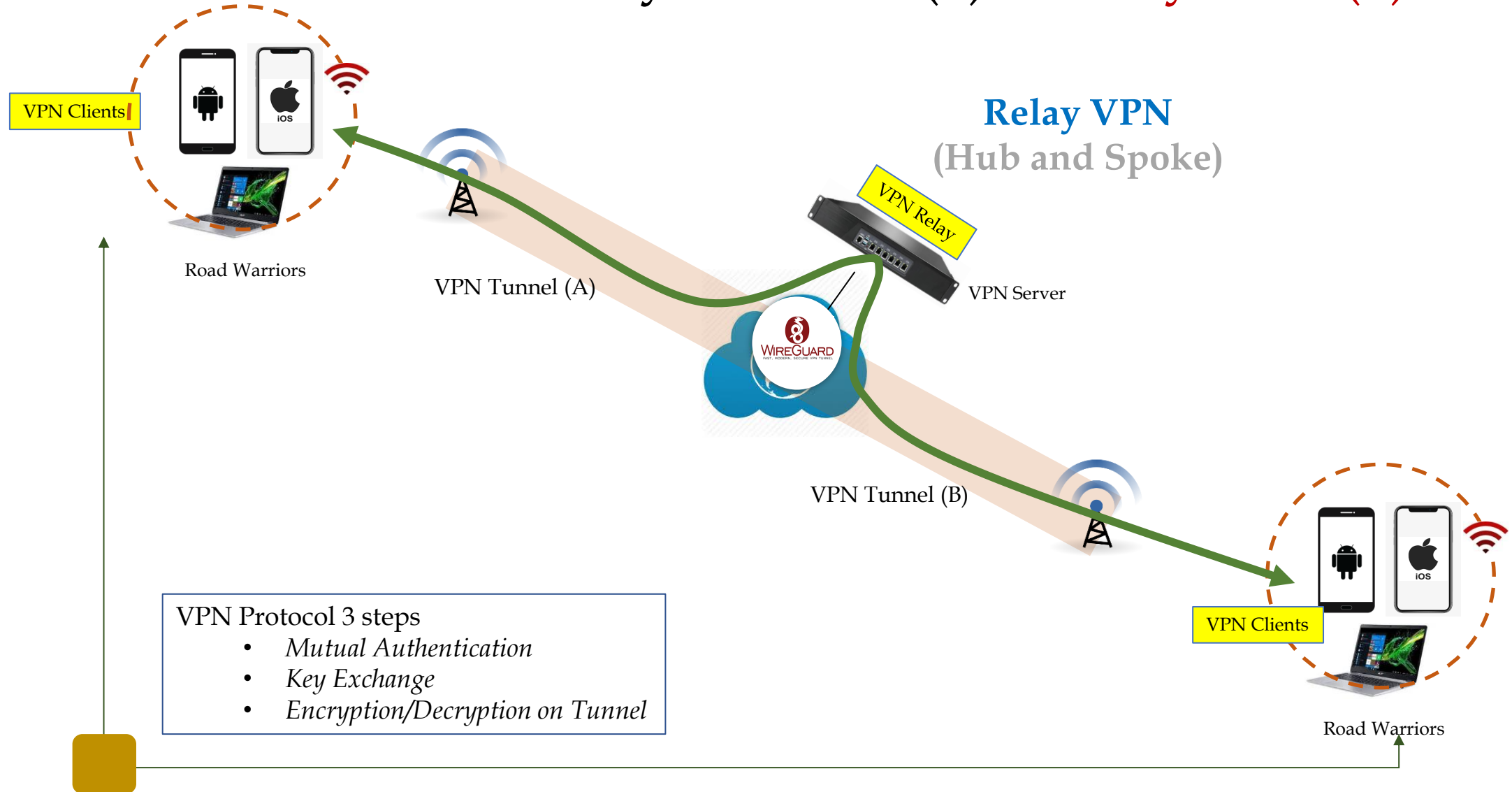
# 1. Zero Trust P2P Security Platform

## Zero Trust P2P Security Platform

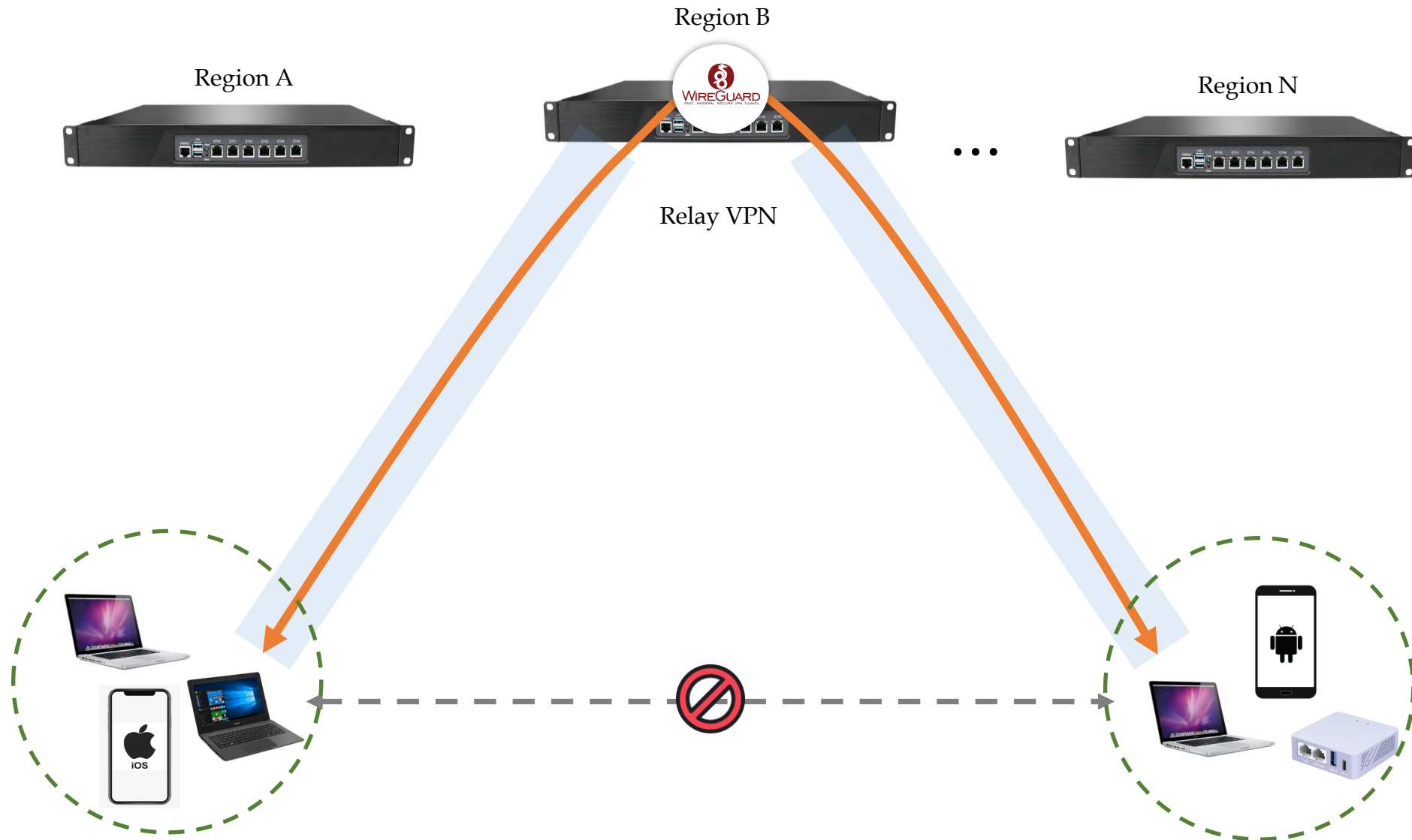
- ZTNA/Secure Network
- No modifications for any applications
- Relay Connect and Direct Connect



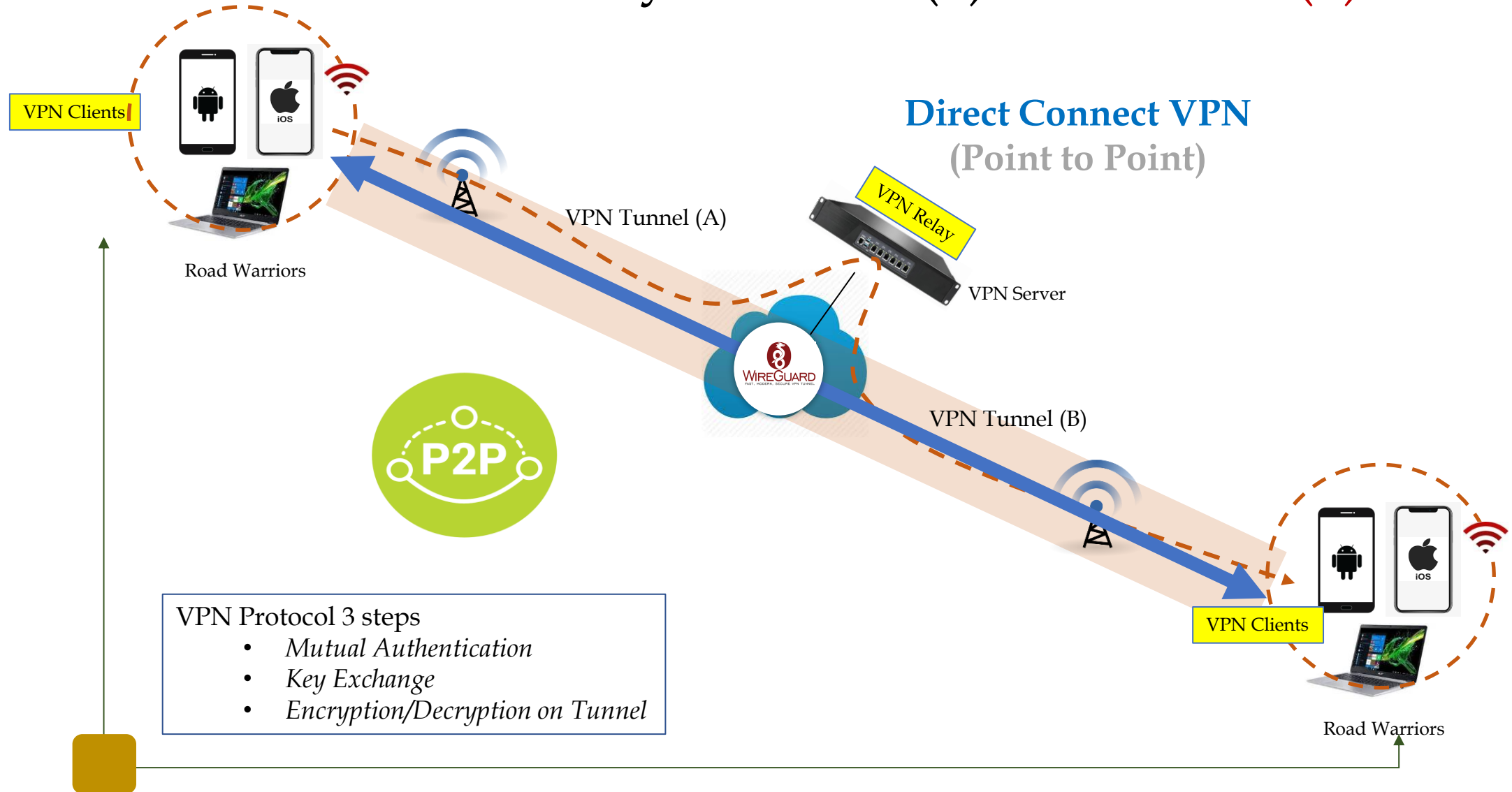
# 1. Zero Trust P2P Security Platform(1) – Relay VPN(1)



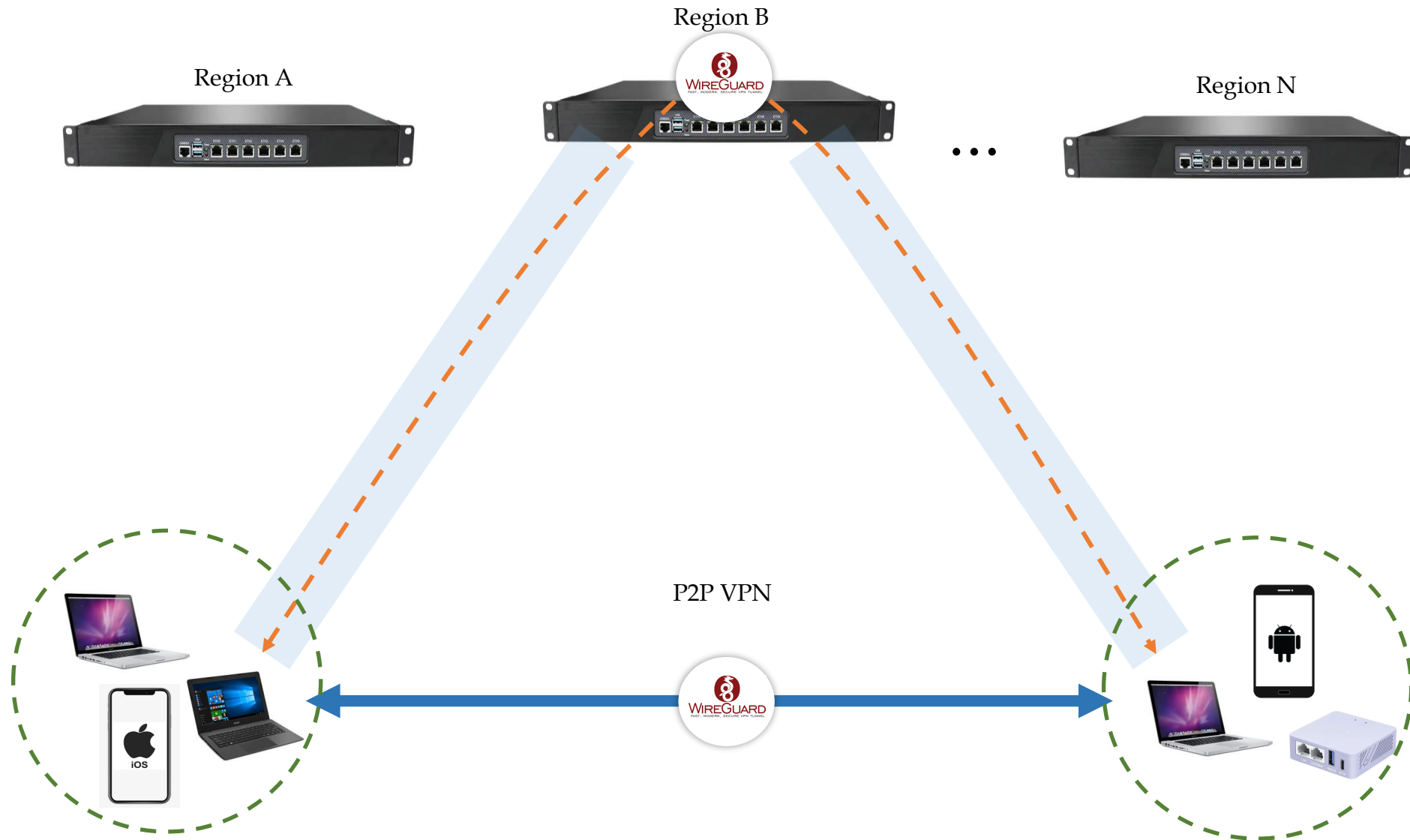
# 1. Zero Trust P2P Security Platform(1) – Relay VPN(2)



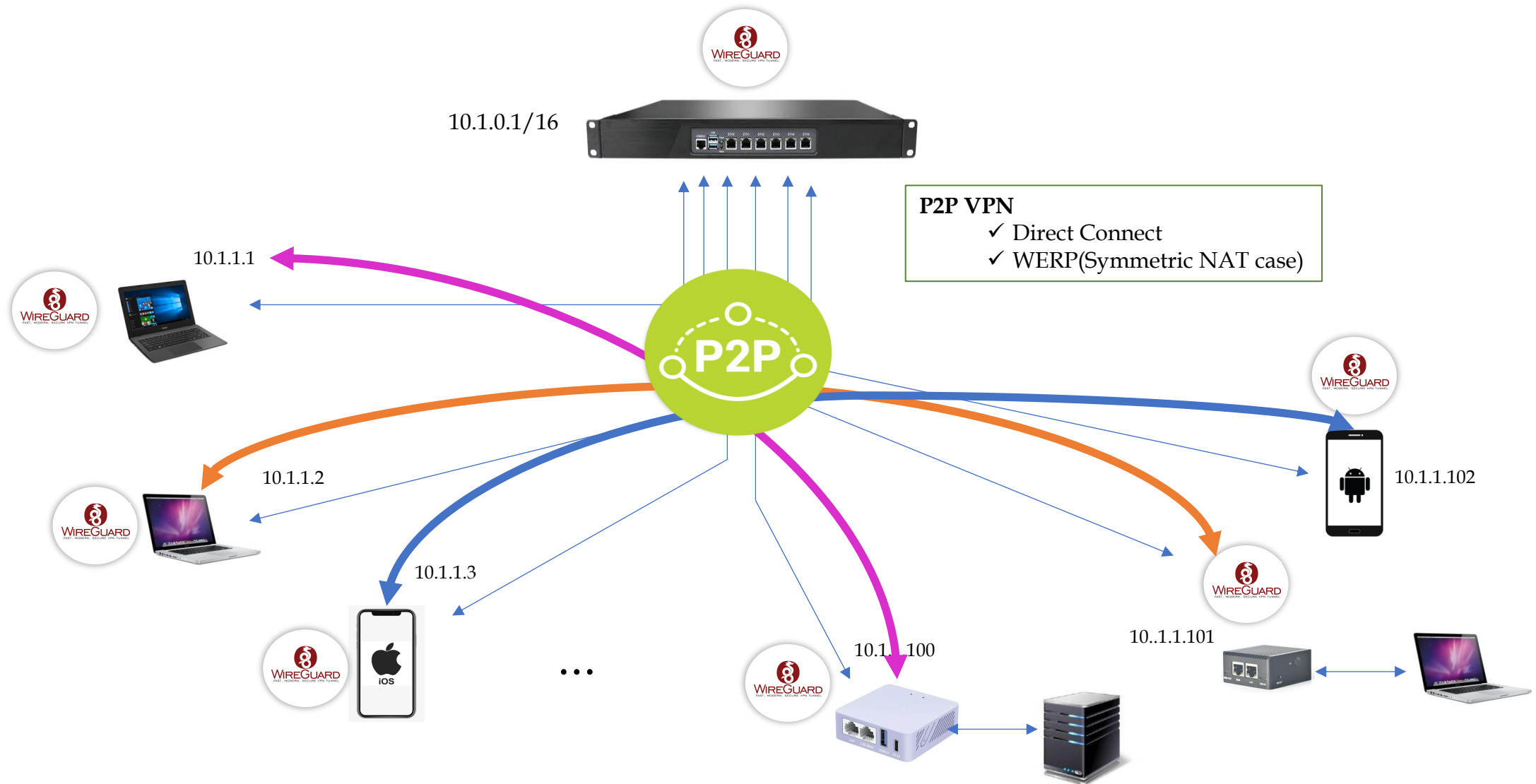
# 1. Zero Trust P2P Security Platform(2) – P2P VPN(1)



# 1. Zero Trust P2P Security Platform(2) – P2P VPN(2)



# 1. Zero Trust P2P Security Platform(2) – P2P VPN(3)





# 1. Zero Trust P2P Security Platform(3) – ZTNA(1-1)

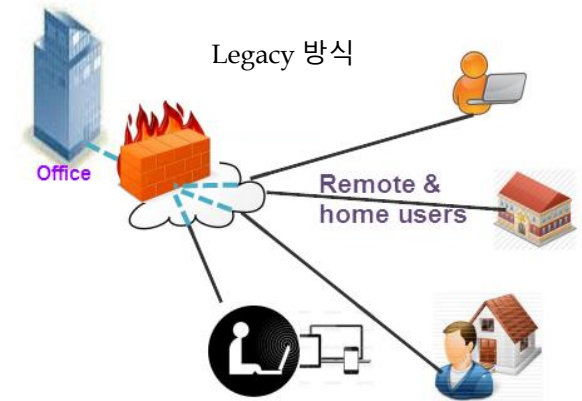
- Legacy 방식(Firewall/VPN/IPS/UTM)

- Internal network(Trusted Area)과 External network(Untrusted Area) 구분
- Internal network <=> Perimeter Security 제품 ⇔ External network : 안전한 내부망과 불안정한 외부망 경계 (Perimeter)에 보안 제품 설치
- Static Configuration, IP/Port 기반의 통제

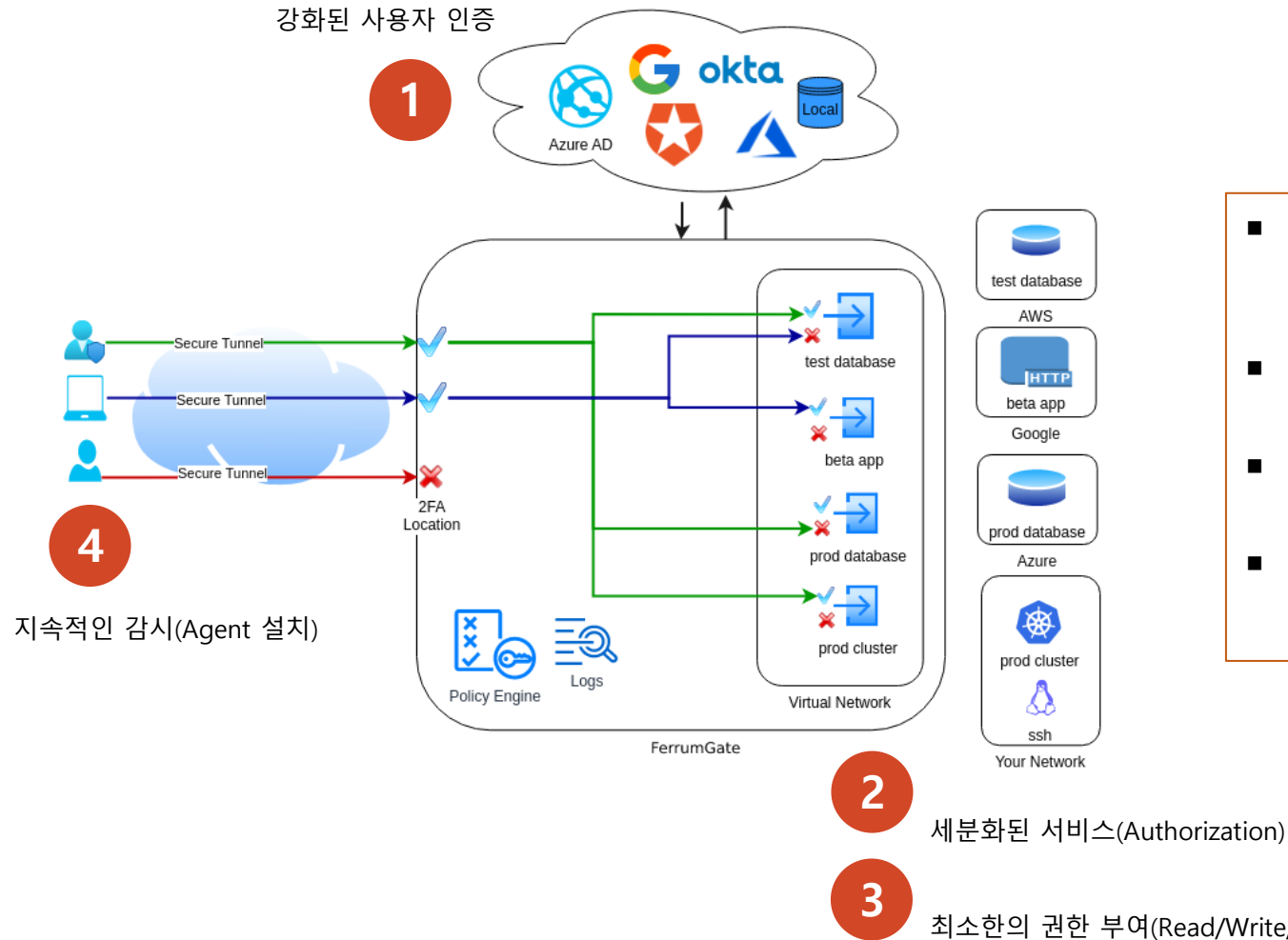
VPN is Dead

- ZTNA or SDP(Software Defined Perimeter) 방식

- 환경의 변화: Cloud, Mobile, BYOD
  - *이제는 중요한 것들이 Perimeter 바깥에 있는 세상이 되었다.*
- Zero Trust: "Internal, External network 모두 안전하지 않다"에서 출발
- S/W 방식의 중앙 Controller에 의한 접근 통제
  - *실제 network에 접근하기 전에, 인증(Authentication) & 접근 통제(Authorization)을 수행하는 개념*
  - *내부 Resource는 Appliance(Gateway)에 의해 모두 Hidden 상태, 인증을 통과한 후에 외부에 Open되는 구조. Dark Network(외부에 보이지 않는 network) 개념 출현*
- Dynamic Control, IP 기반이 아니라, 사용자 기반의 통제



# 1. Zero Trust P2P Security Platform(3) – ZTNA(1-2)

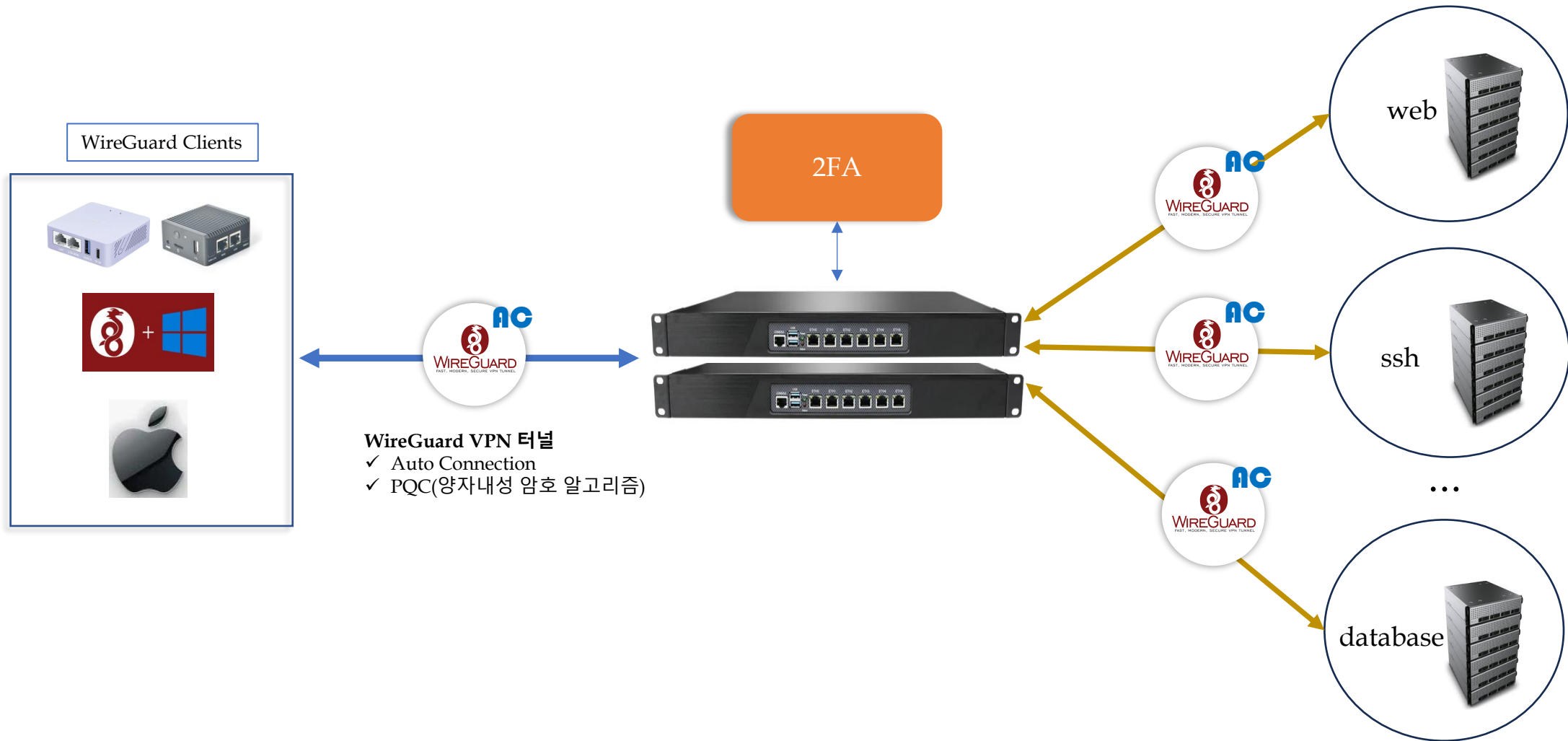


## ZTNA의 요건

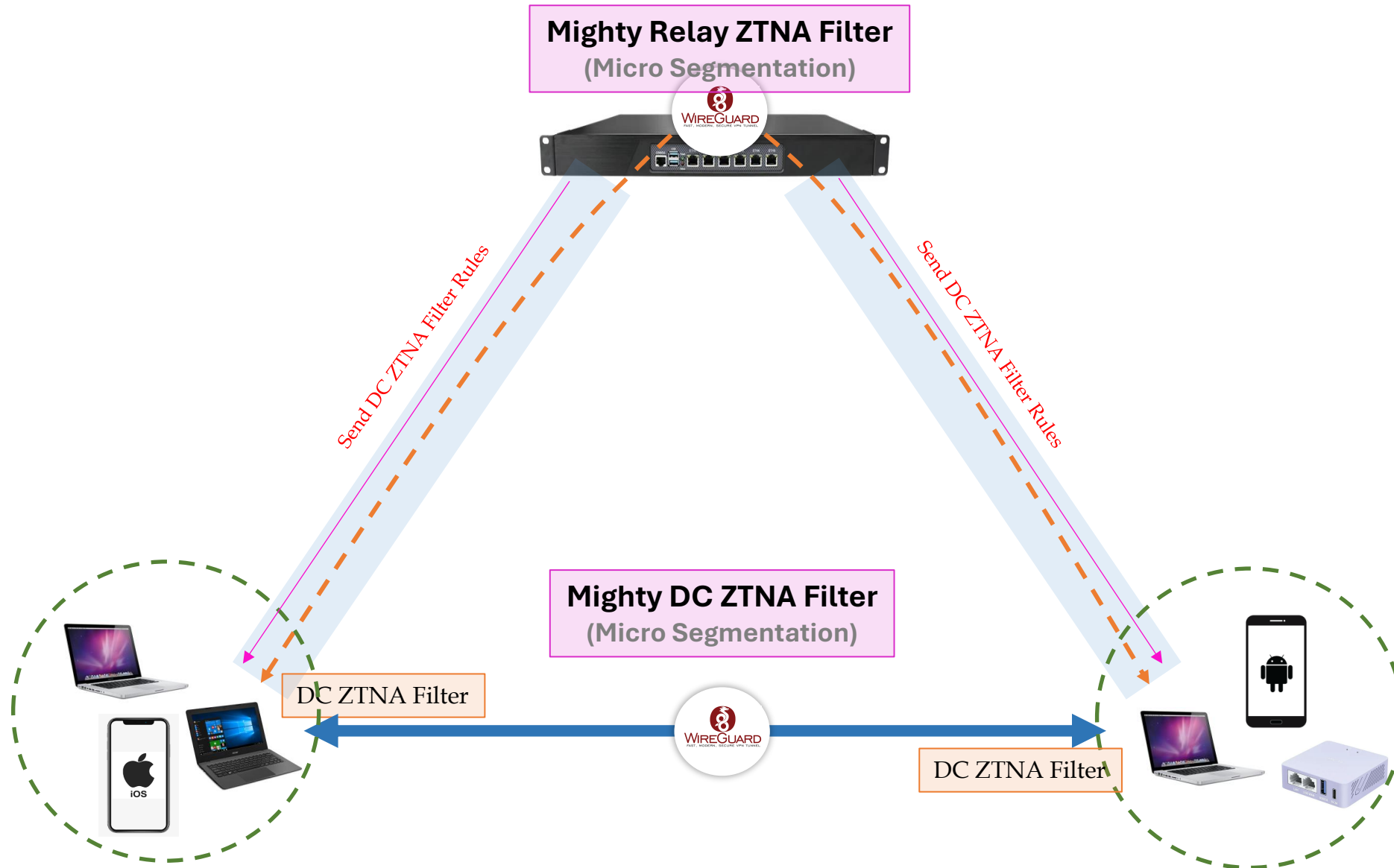
- More Authentication Options
  - MFA: SAML, OAuth2, LDAP, FIDO2
  - Location Check, Time Check, Device Posture Check
- Micro-segmentation
  - Virtual Application Networks
- Least privilege access
  - RBAC(Role Based Access Control)
- Continuous monitoring
  - Every packet, every flow, every connection

[출처] <https://ferrumgate.com/>

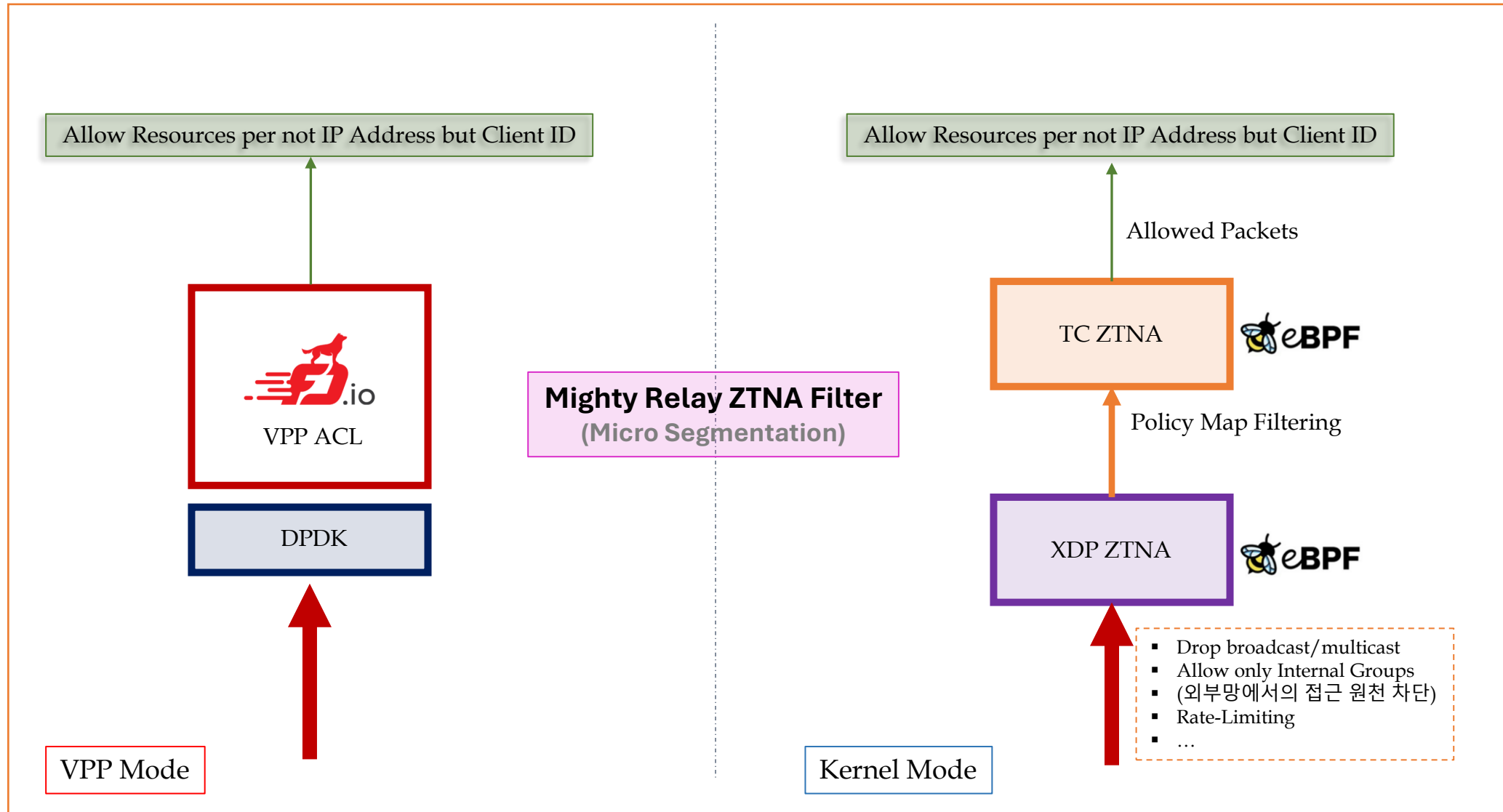
# 1. Zero Trust P2P Security Platform(3) – ZTNA(2)



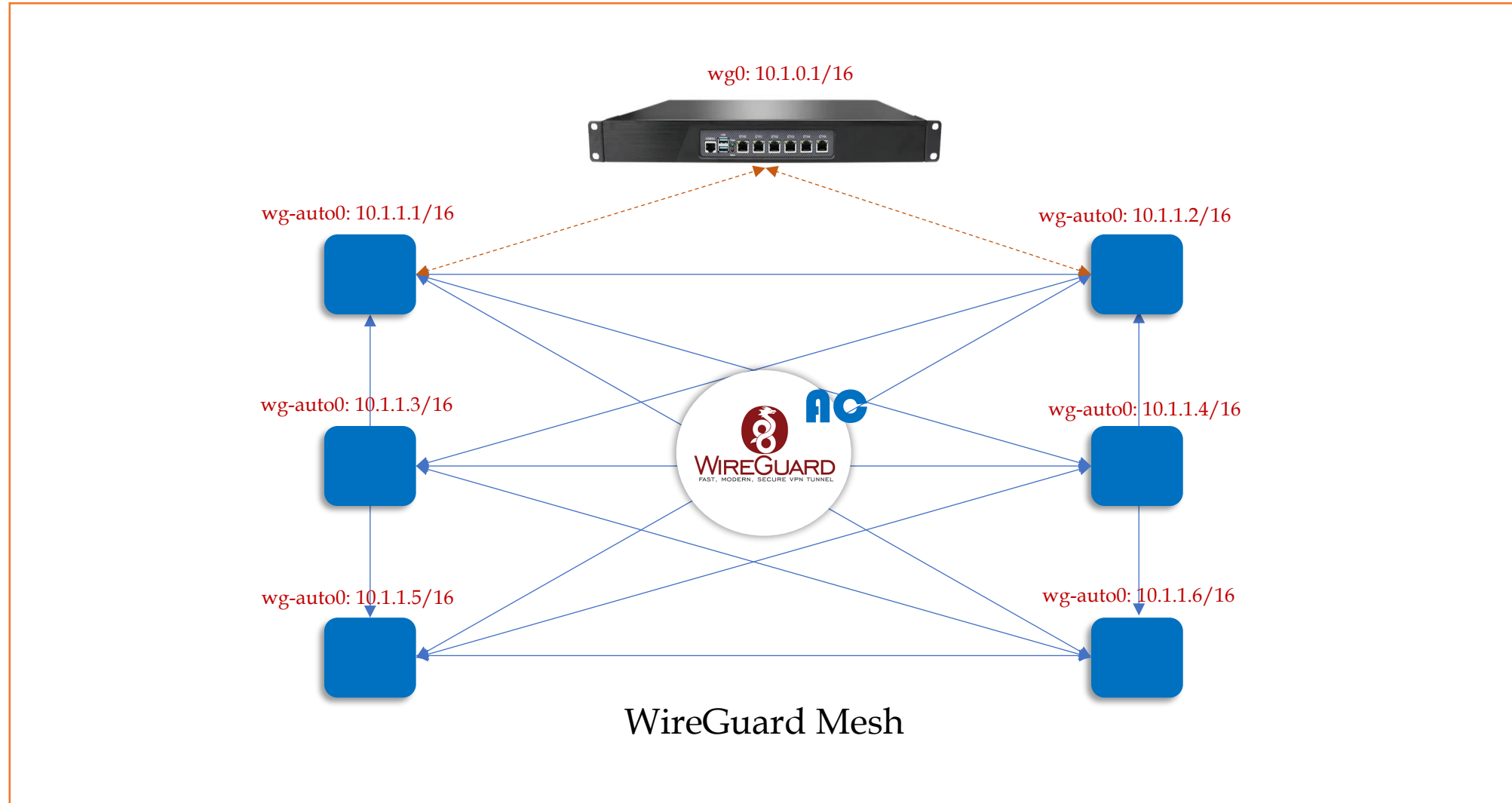
# 1. Zero Trust P2P Security Platform(3) – ZTNA(3-1)



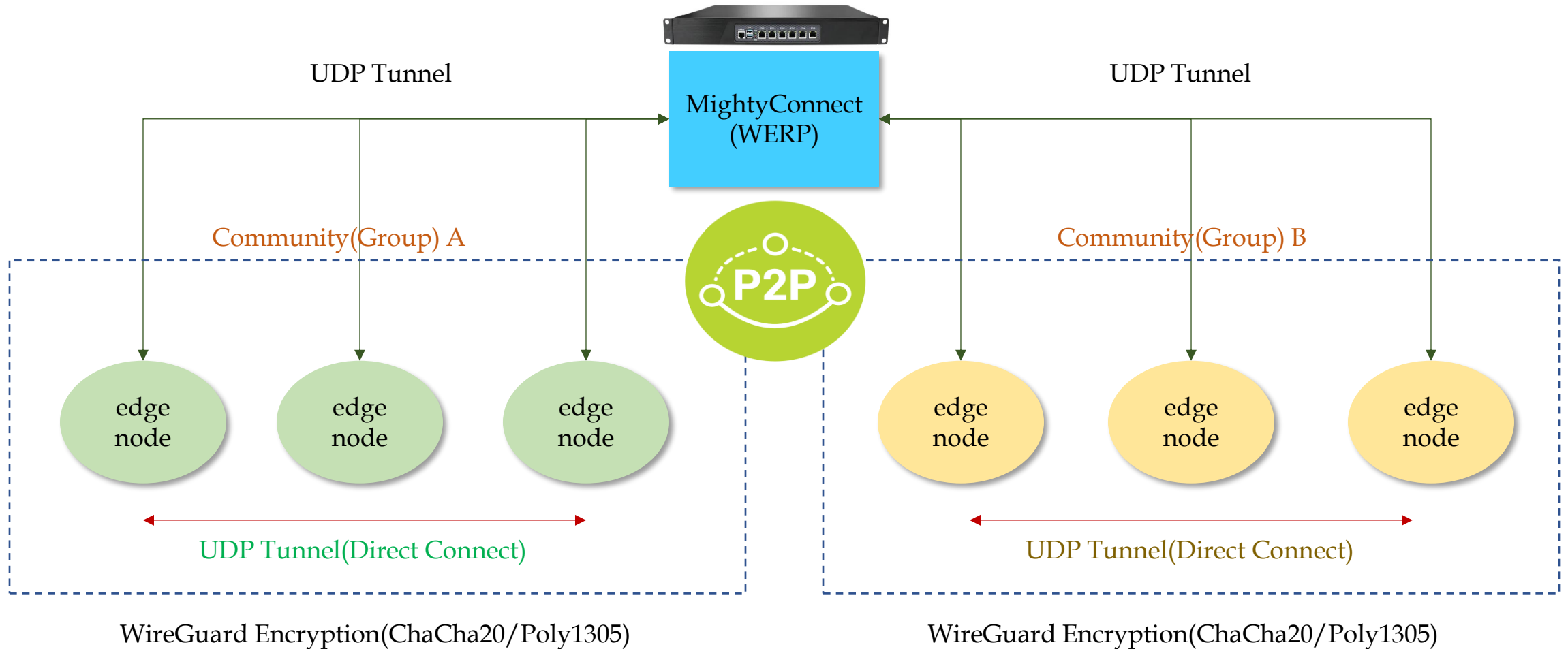
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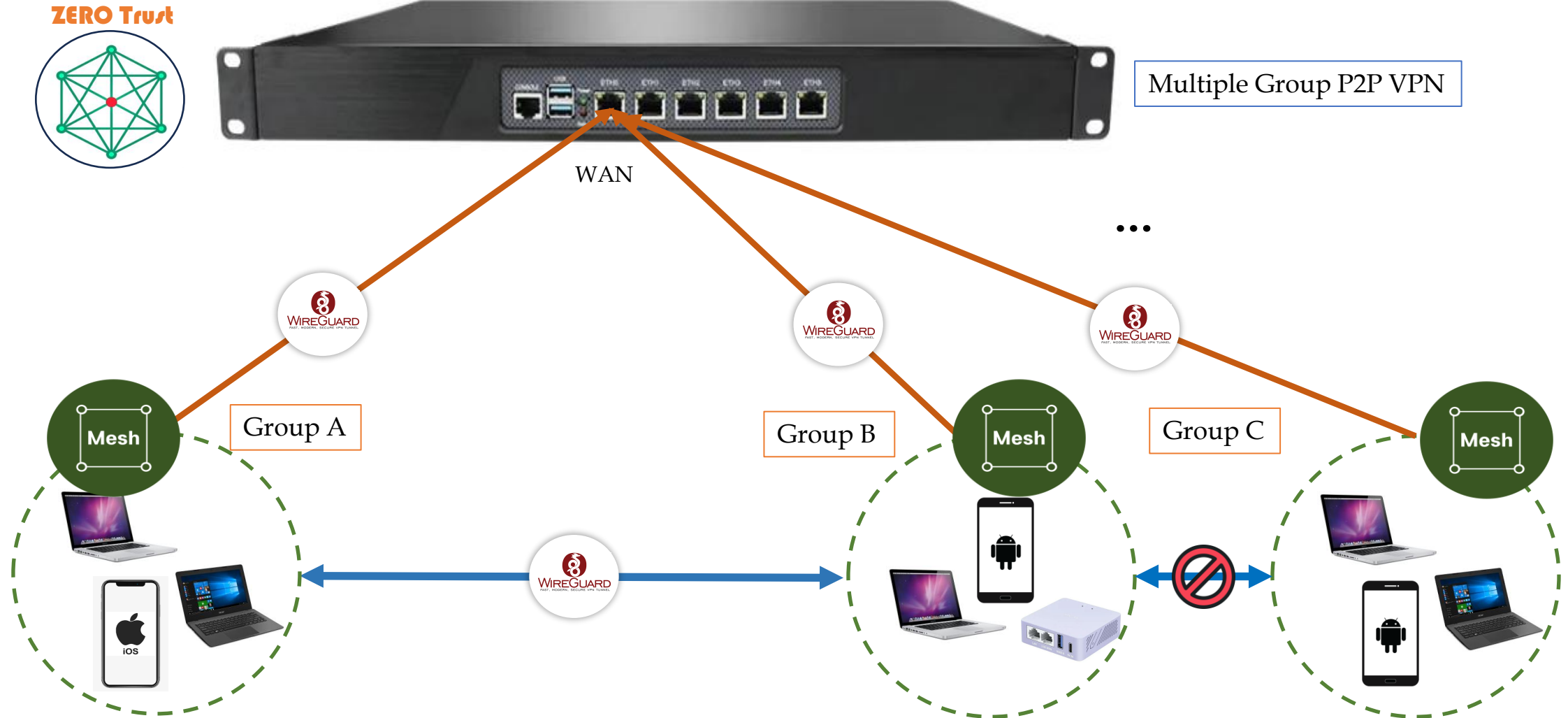
# 1. Zero Trust P2P Security Platform(4) - Mesh(1)



# 1. Zero Trust P2P Security Platform(4) – Mesh(2)

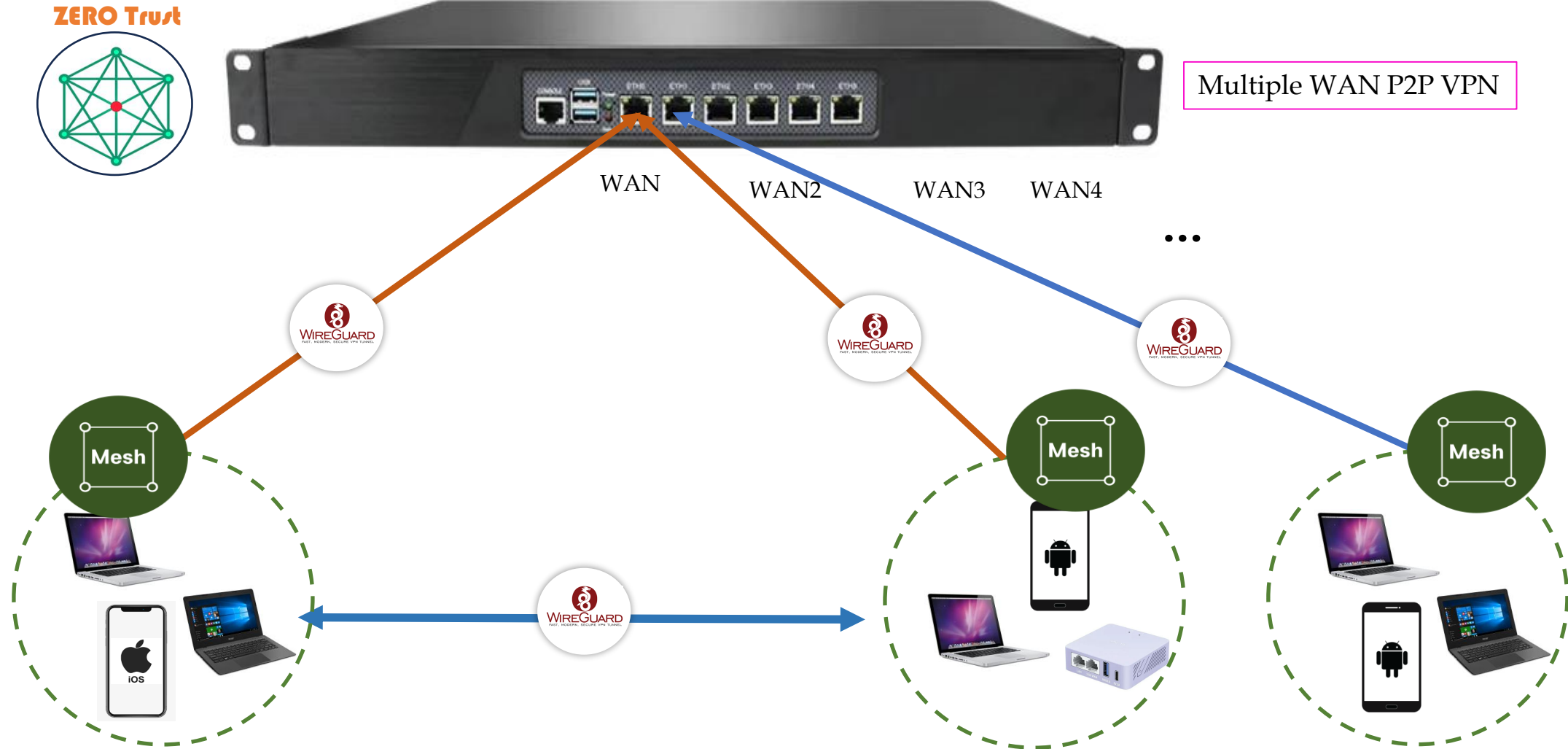


# 1. Zero Trust P2P Security Platform(4) – Mesh(3)

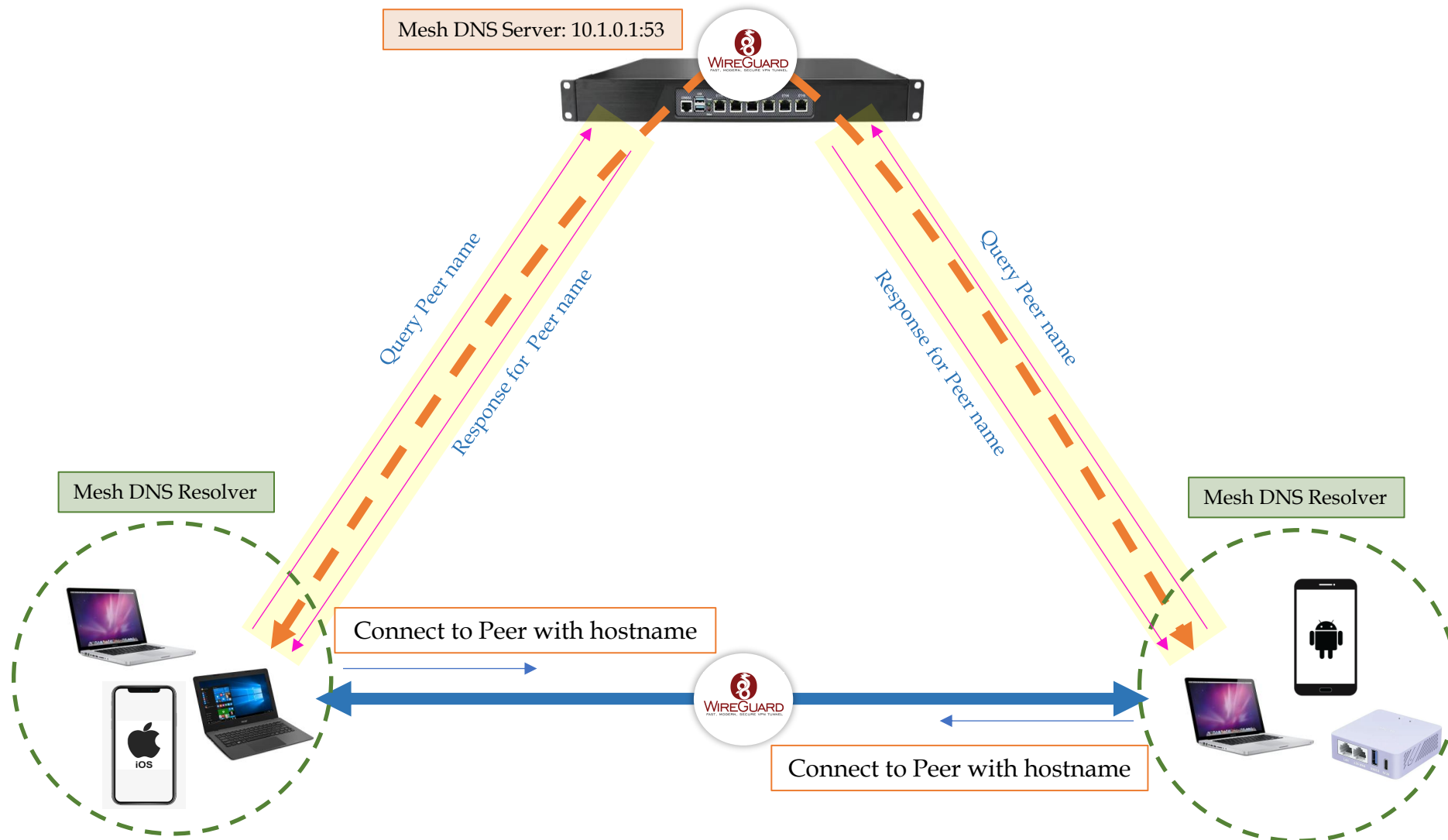




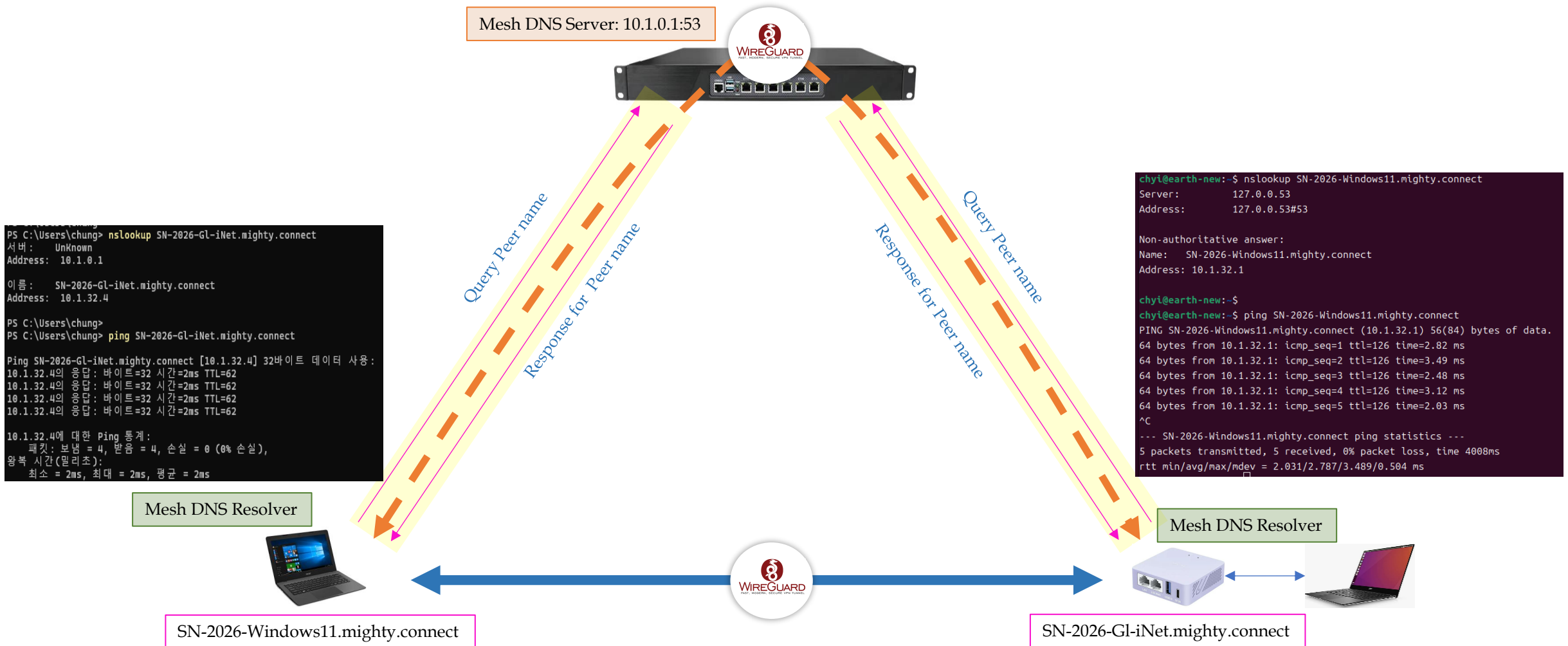
# 1. Zero Trust P2P Security Platform(4) - Mesh(4)



# 1. Zero Trust P2P Security Platform(5) – Mesh DNS(1)



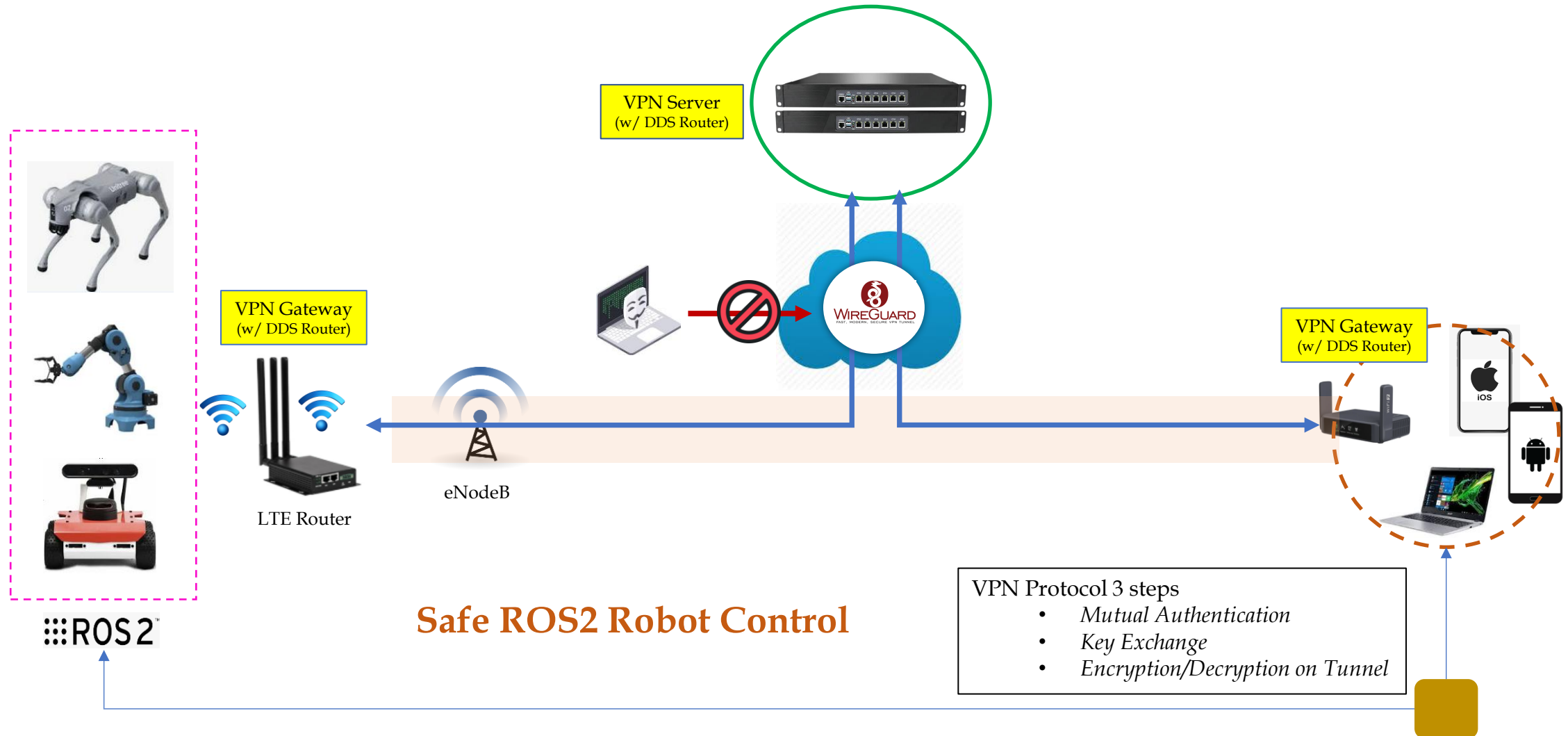
# 1. Zero Trust P2P Security Platform(5) – Mesh DNS(2)



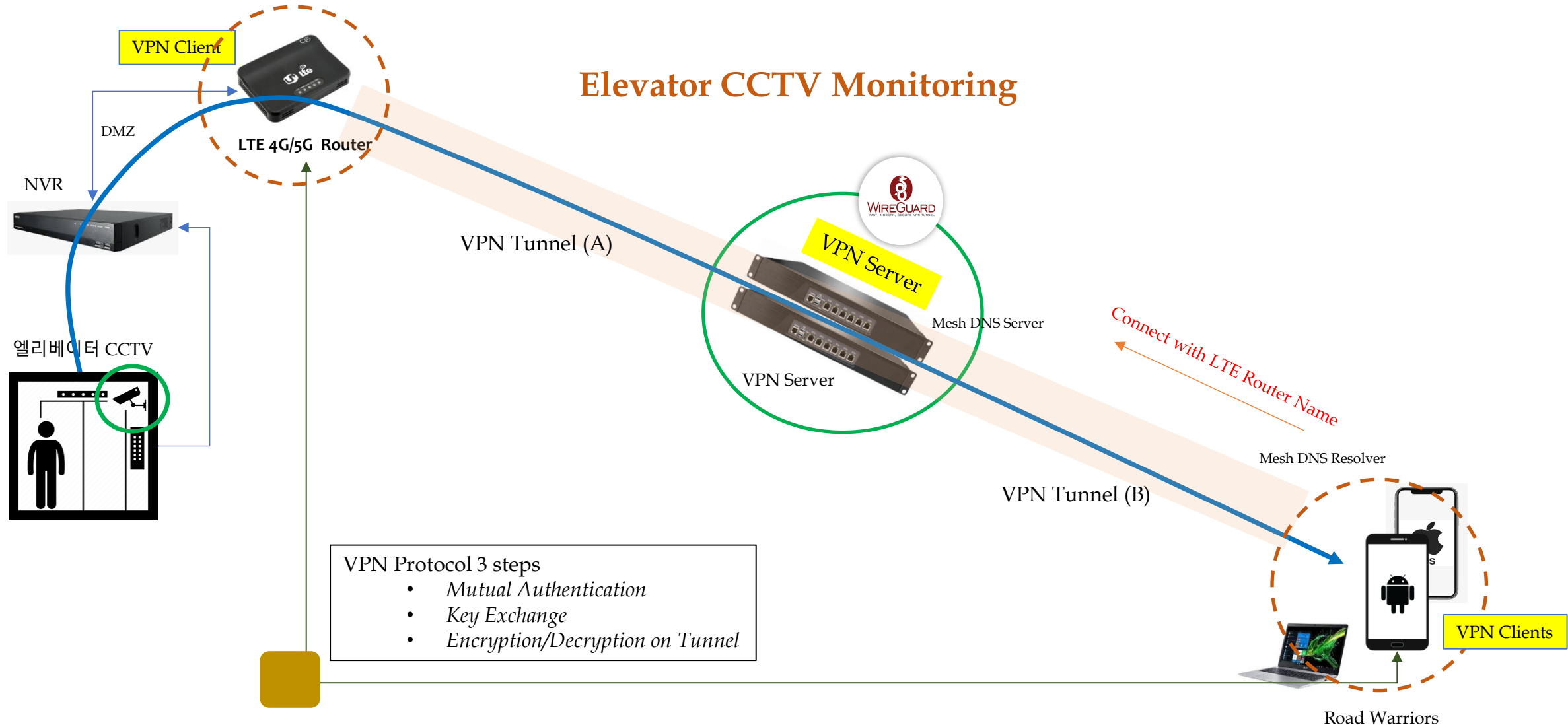
# 1. Zero Trust P2P Security Platform(6) – Use Cases(1)



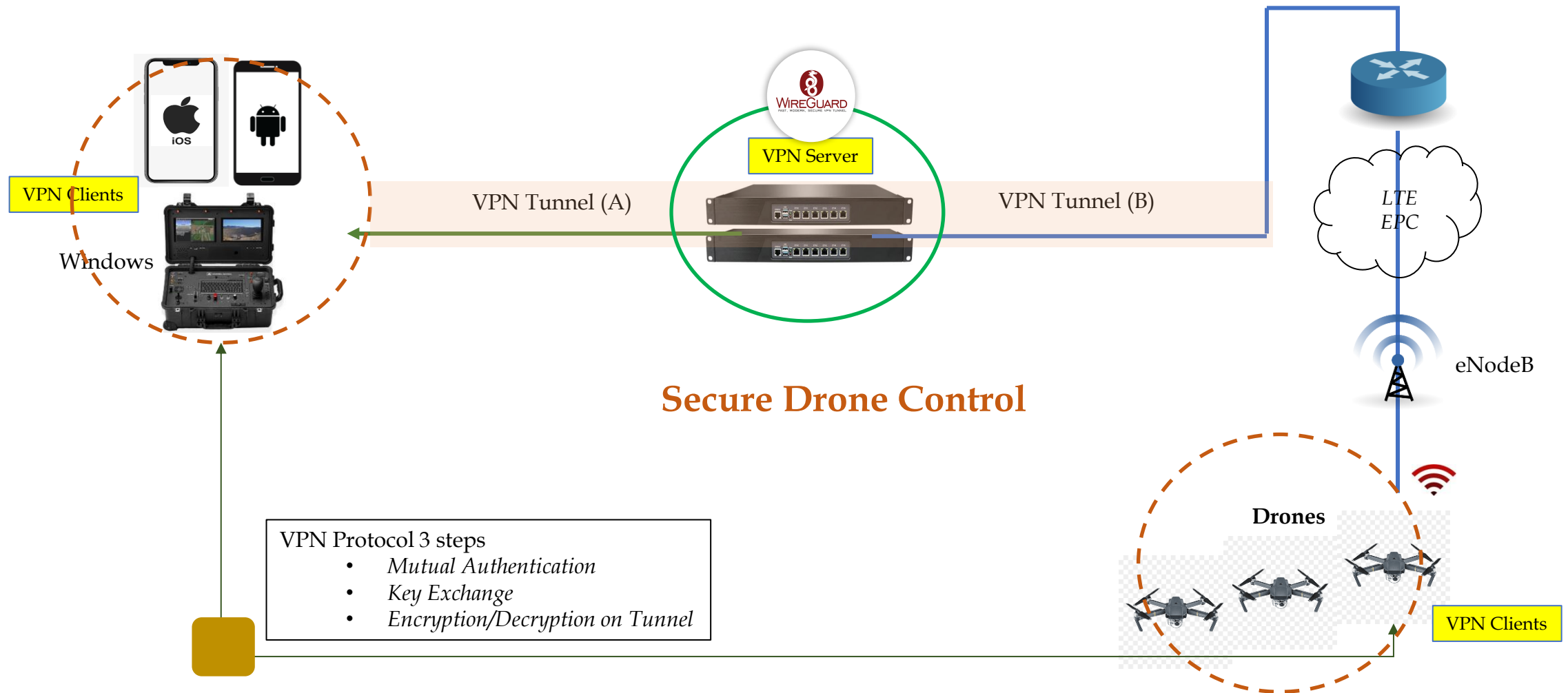
# 1. Zero Trust P2P Security Platform(6) - Use Cases(2)



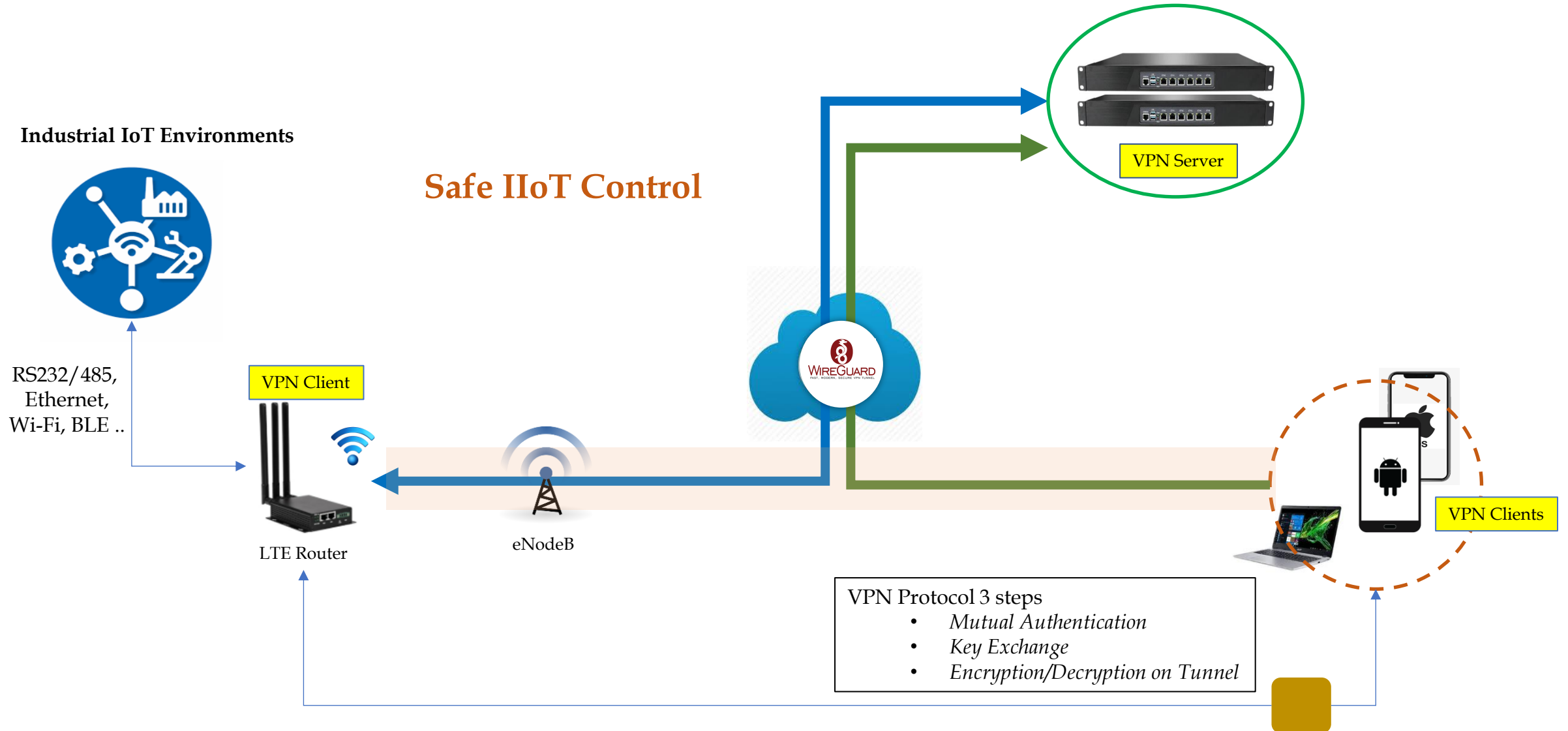
# 1. Zero Trust P2P Security Platform(6) - Use Cases(3)



# 1. Zero Trust P2P Security Platform(6) - Use Cases(4)

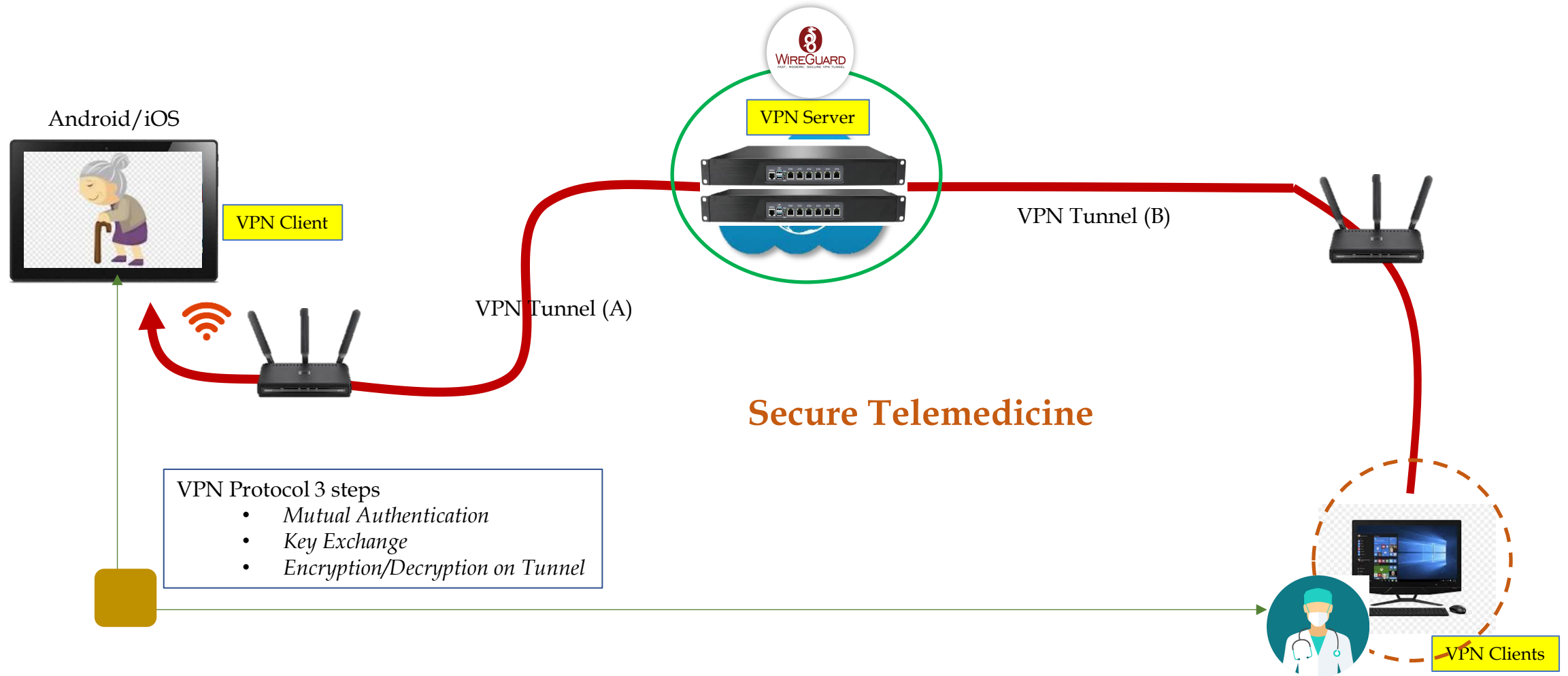


# 1. Zero Trust P2P Security Platform(6) - Use Cases(5)

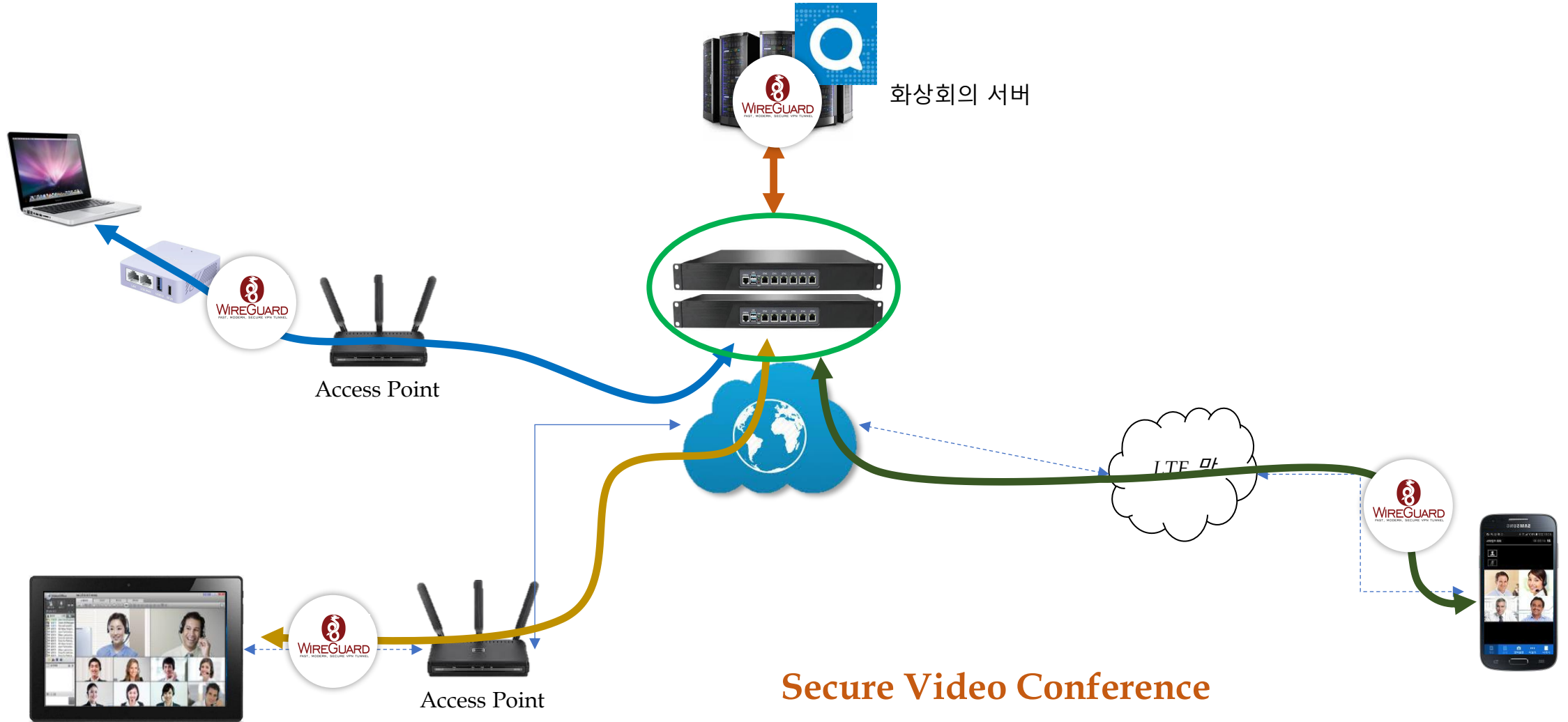




# 1. Zero Trust P2P Security Platform(6) - Use Cases(6)



# 1. Zero Trust P2P Security Platform(6) - Use Cases(7)



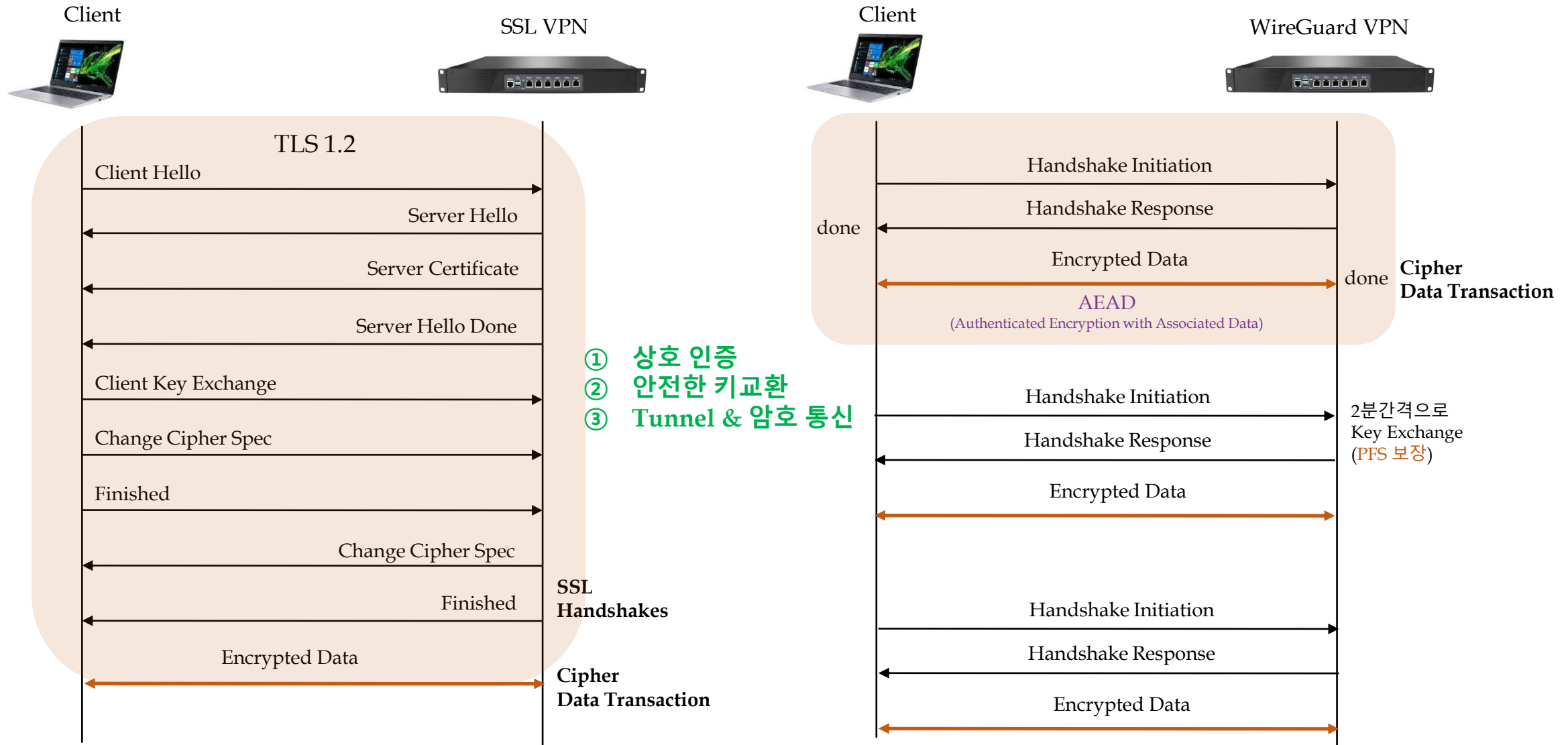
## 2. WireGuard

### New Generation VPN Protocol



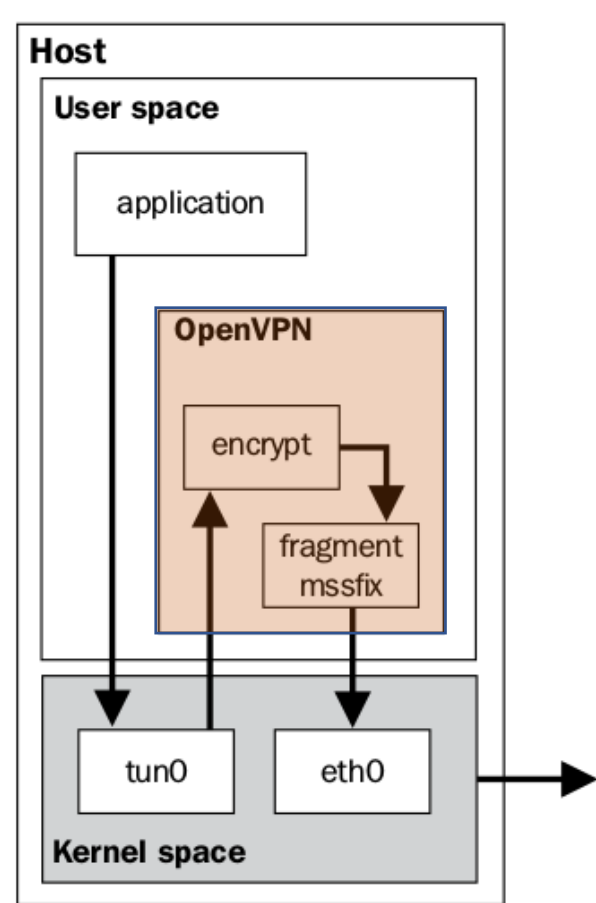
"WireGuard" and the "WireGuard" logo are registered trademarks of Jason A. Donenfeld

## 2. WireGuard(1) - Architecture(1)



## 2. WireGuard(1) - Architecture(1)

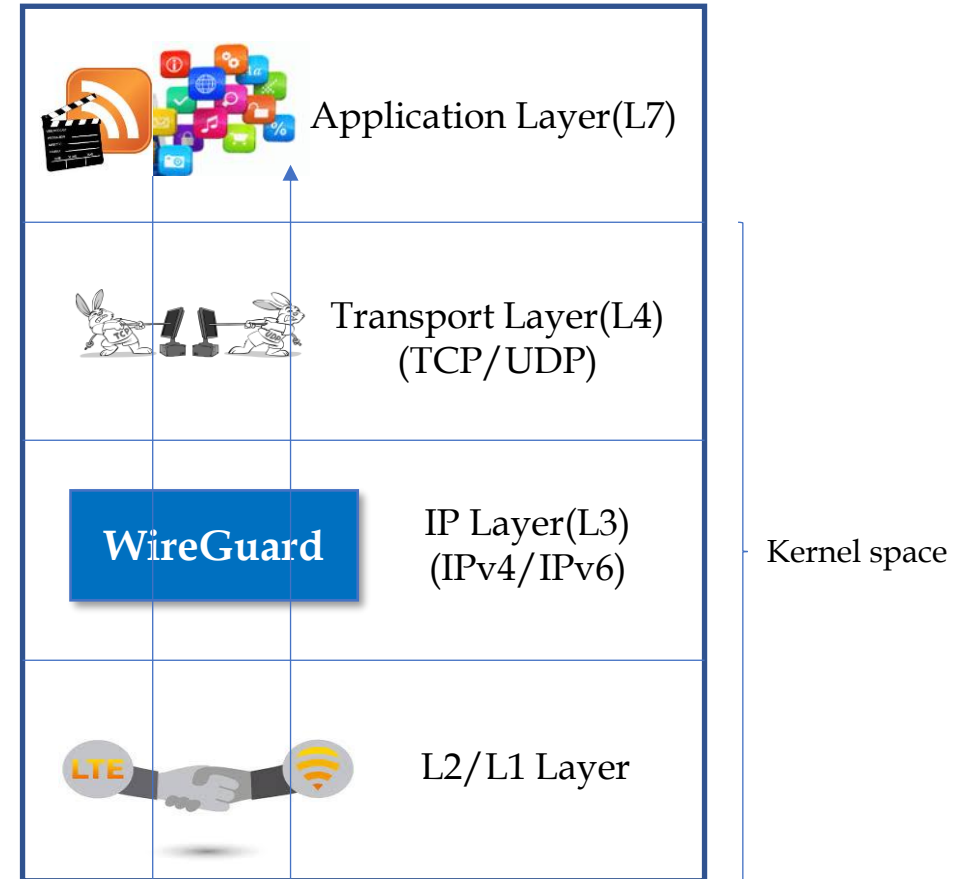
### Limitations of OpenVPN



SSL VPN Tunnel(Before DCO)

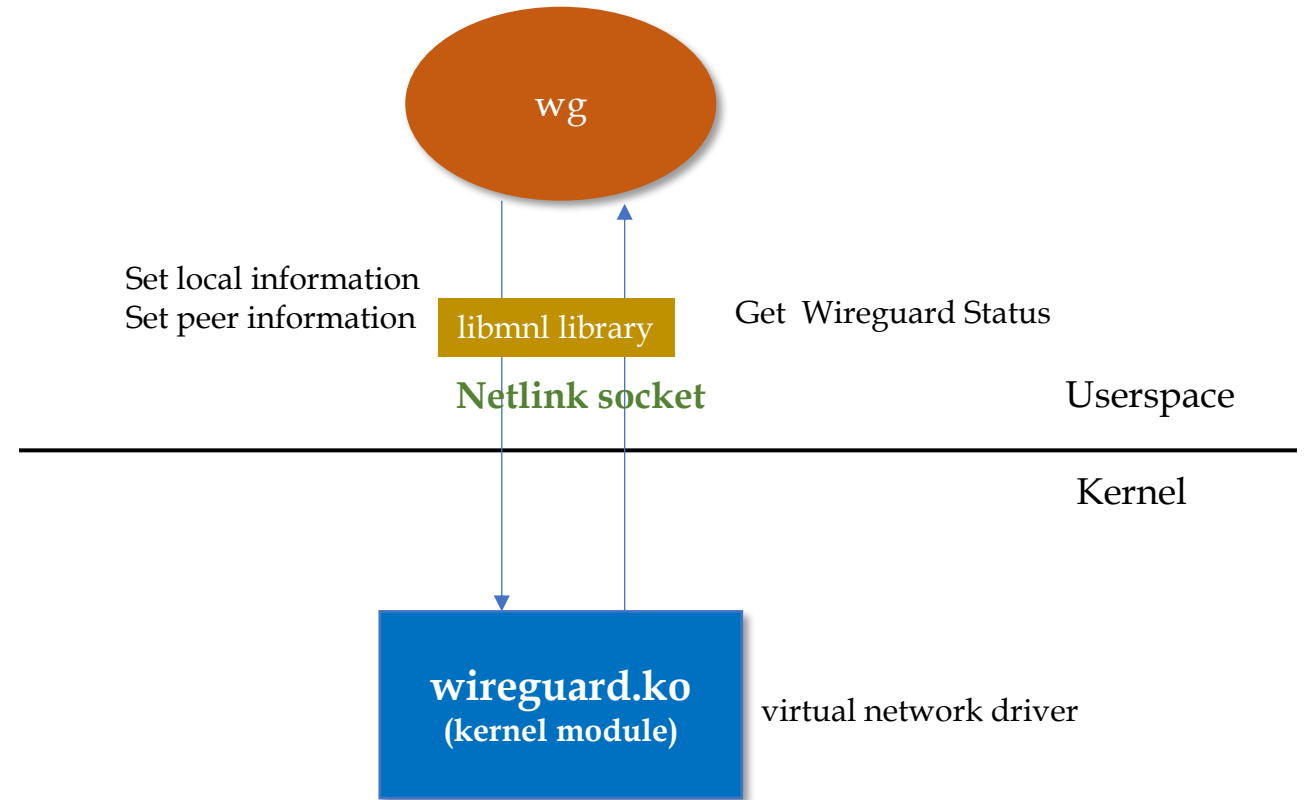
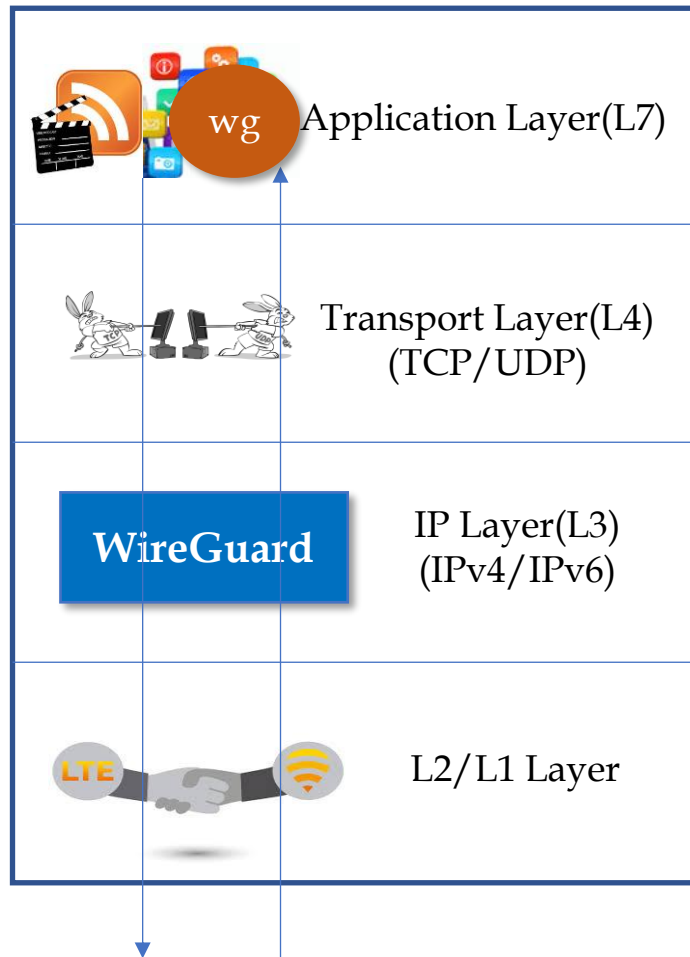
VS

### No Limitation. Cryptokey Routing

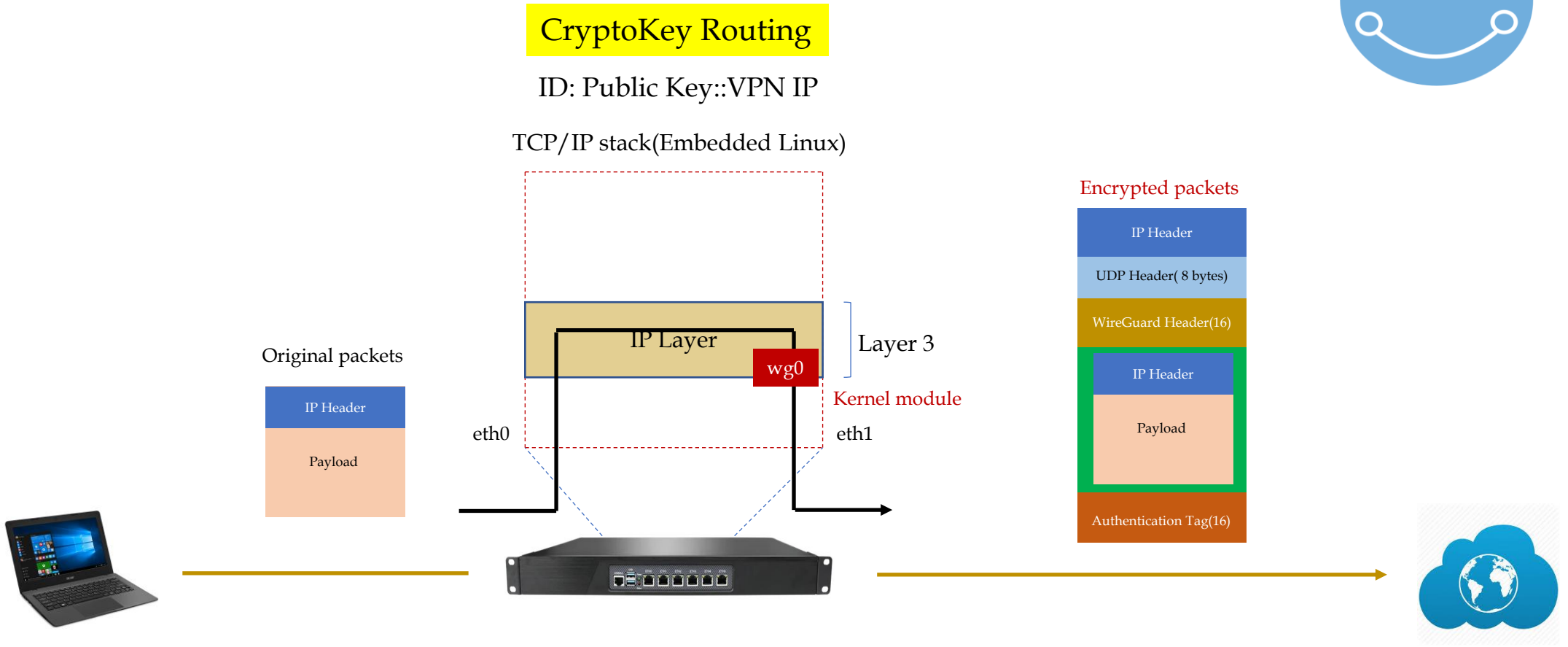


Why is WireGuard faster than OpenVPN?

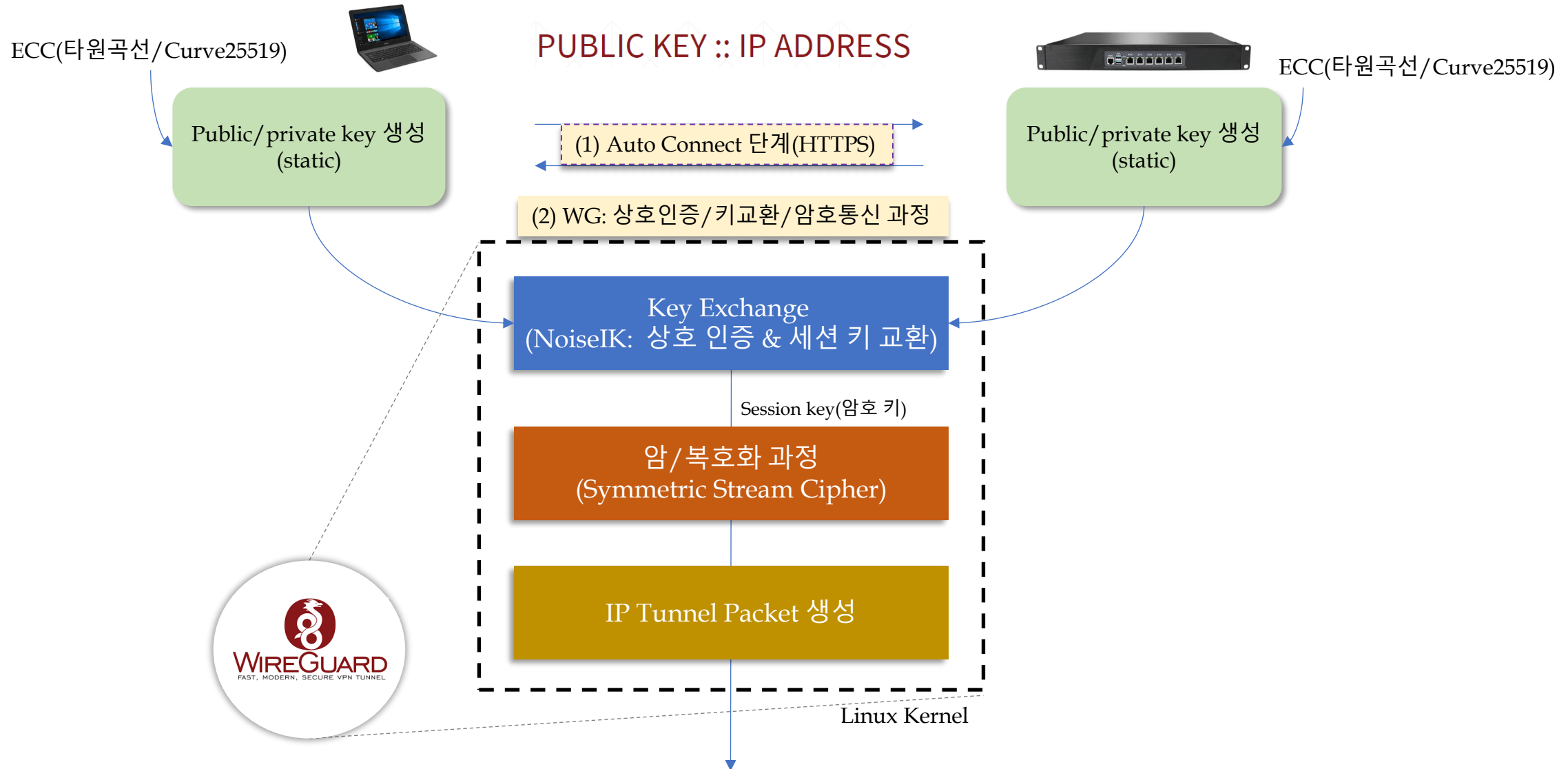
## 2. WireGuard(1) - Architecture(2)



## 2. WireGuard(1) - Architecture(3)

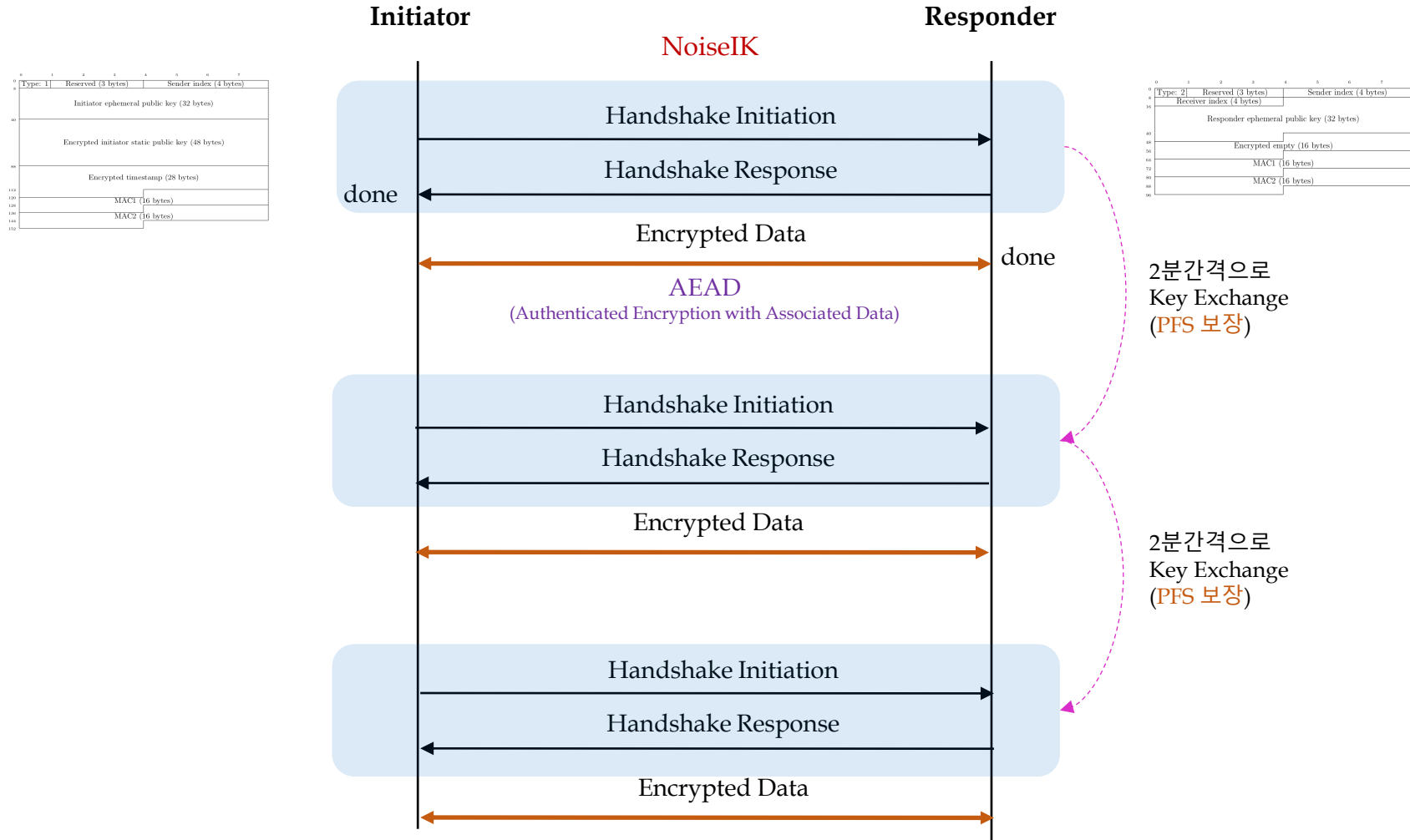


## 2. WireGuard(2) - Protocol(1)



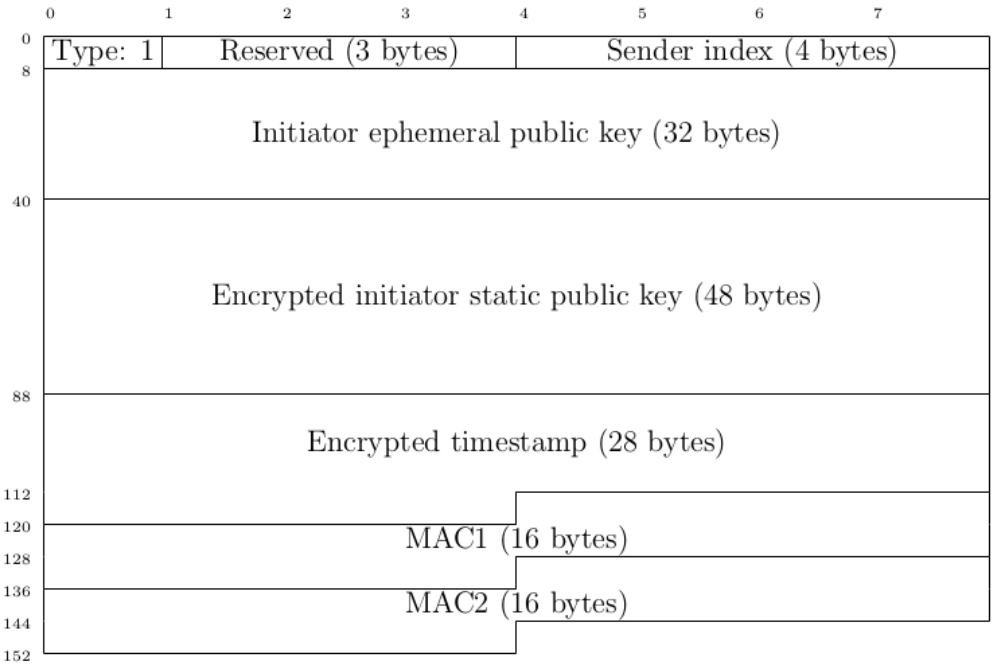


## 2. WireGuard(2) - Protocol(2)

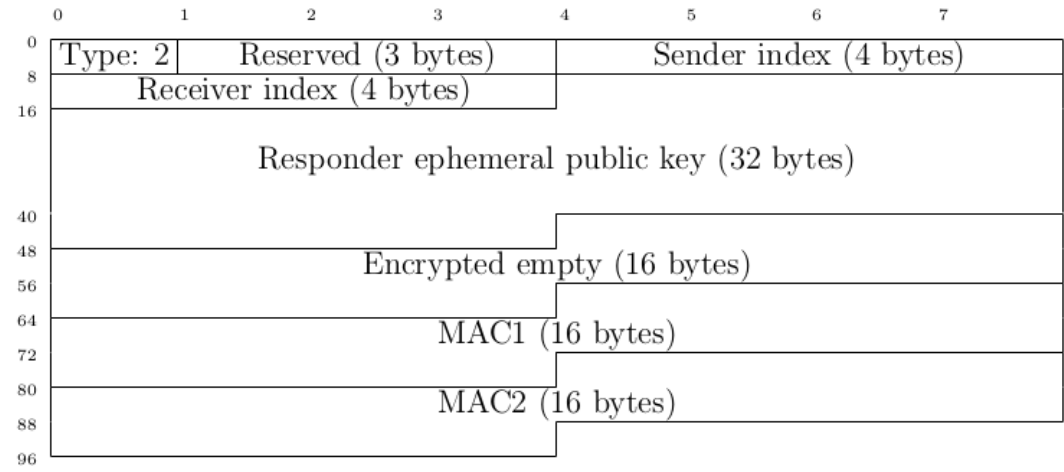


# 2. WireGuard(2) - Protocol(3)

Initiation Packet Format



Response Packet Format



| Type | Message Name         | Length     |
|------|----------------------|------------|
| 1    | Handshake Initiation | 148 bytes  |
| 2    | Handshake Response   | 92 bytes   |
| 3    | Cookie Reply         | 64 bytes   |
| 4    | Transport Data       | ≥ 32 bytes |

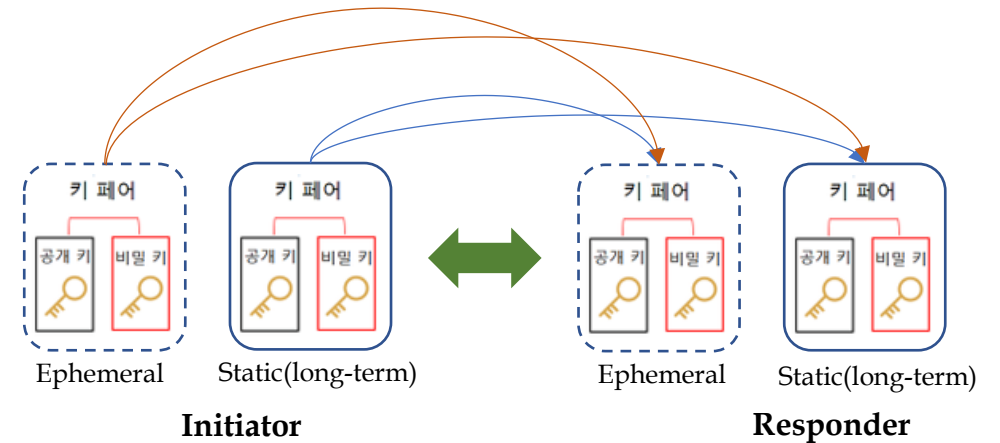
## 2. WireGuard(2) - Protocol(4)

Session keys = Noise(  
ECDH(ephemeral, static),  
ECDH(static, ephemeral),  
ECDH(ephemeral, ephemeral),  
ECDH(static, static)  
)

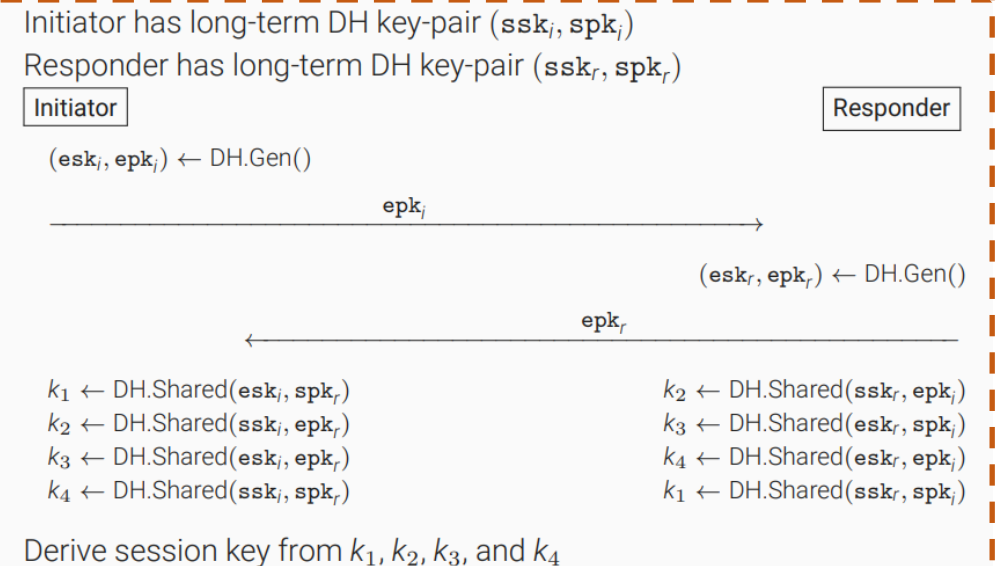
256bits(32bytes)



AEAD



상호 인증  
및  
키 교환



## 2. WireGuard(3) – Algorithms(1)

### Layer 3(IPv4, IPv6 지원) VPN Protocol

handshake pattern IKpsk2 (Noise Protocol Framework 기반)

DH function X25519

cipher function ChaCha20-Poly1305

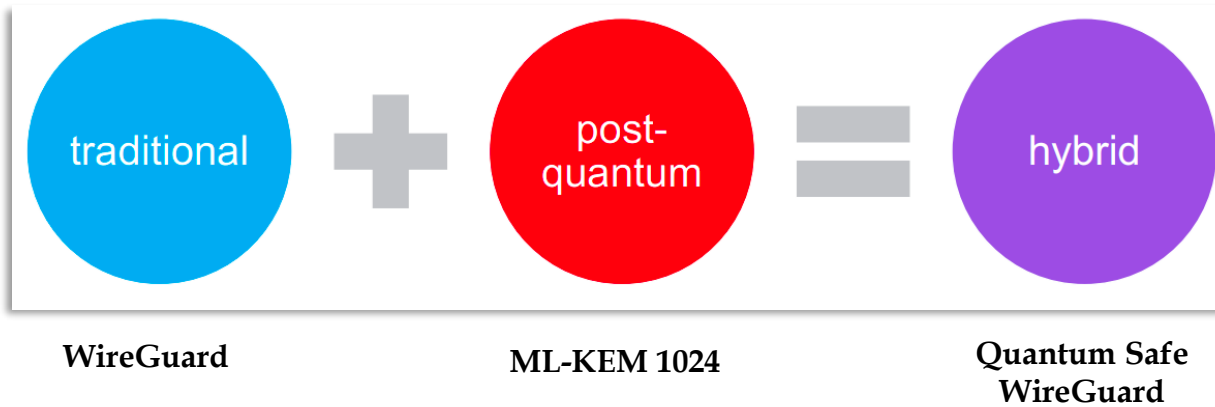
hash function BLAKE2s

prologue WireGuard v1 zx2c4 Jason@zx2c4.com (34 bytes)

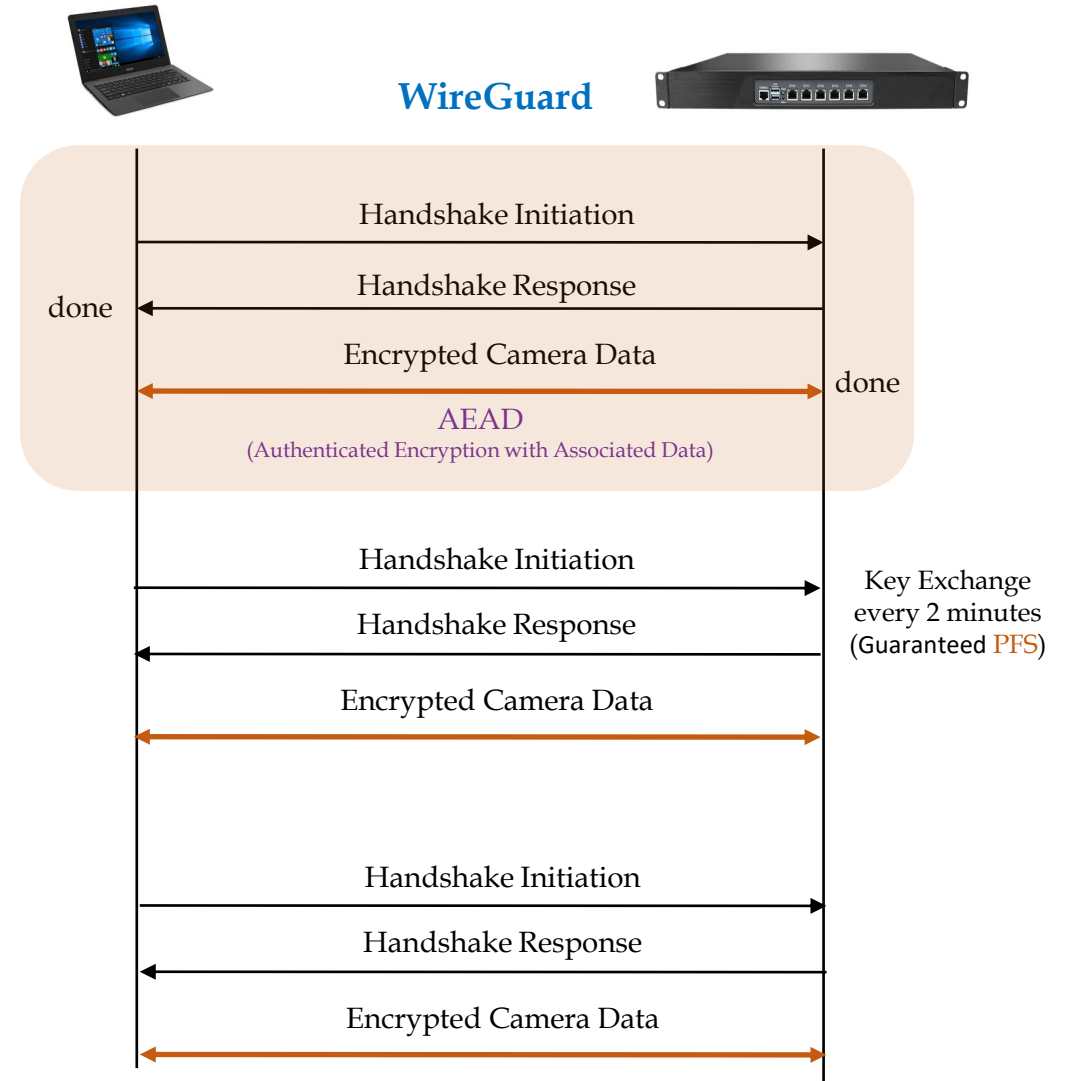
## 2. WireGuard(3) - Algorithms(2)

| 보안 알고리즘                                                   | 상세 내용                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>NoiseIK</b><br>Key 교환 방식<br>및<br>상호 인증                 | <b>NoiseIK handshake 방식(Noise IKpsk2)</b> <ul style="list-style-type: none"> <li>▪ ECDH(Diffie-Hellman) 기반</li> <li>▪ <b>Curve25519</b> Public key(32 byte)를 교환 후, 이를 통해 안전하게 shared secret 생성                             <ul style="list-style-type: none"> <li>▪ Static/Ephemeral private/public key(2 쌍) 이용</li> </ul> </li> <li>▪ Key 교환 시 아래 기능 보장                             <ul style="list-style-type: none"> <li>▪ <i>Perfect forward secrecy</i> 보장, <i>replay attack</i> 방지 기능</li> <li>▪ <i>Identity</i> 감춤 기능 제공, DoS 공격 완화 기능(Cookie),</li> <li>▪ PSK 사용 시, <i>post-quantum security</i> 보장</li> </ul> </li> </ul> <b>Hash 알고리즘</b> <ul style="list-style-type: none"> <li>▪ <b>BLAKE2</b> - fast secure hashing 알고리즘                             <ul style="list-style-type: none"> <li>▪ SHA series 보다 빠름. 즉 MD5 수준임. SHA-3 보안 강도.</li> </ul> </li> </ul> <b>PRF &amp; MAC 및 Key Derivation 함수</b> <ul style="list-style-type: none"> <li>▪ <b>SipHash4</b> (Hash table lookup 용으로 사용)</li> <li>▪ <b>HKDF</b> (HMAC-based Key Derivation Function) 사용</li> </ul> |
| 암호 알고리즘<br>Authenticator 알고리즘<br>(ChaCha20-Poly1305 AEAD) | <b>ChaCha20</b> - 256 bits stream cipher(20 round cipher Salsa20 기반) <ul style="list-style-type: none"> <li>▪ Stream cipher는 일반적인 block cipher(예: AES-256-CBC)에 비해 속도가 빠름</li> <li>▪ key(32 bytes)는 대칭키를 사용(즉, 암호화 용 키와 복호화 용 키 동일)</li> <li>▪ Video/Audio 등 stream 암호화에 적합</li> </ul> <b>Poly1305</b> - message authentication code 알고리즘(16 bytes output 생성)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

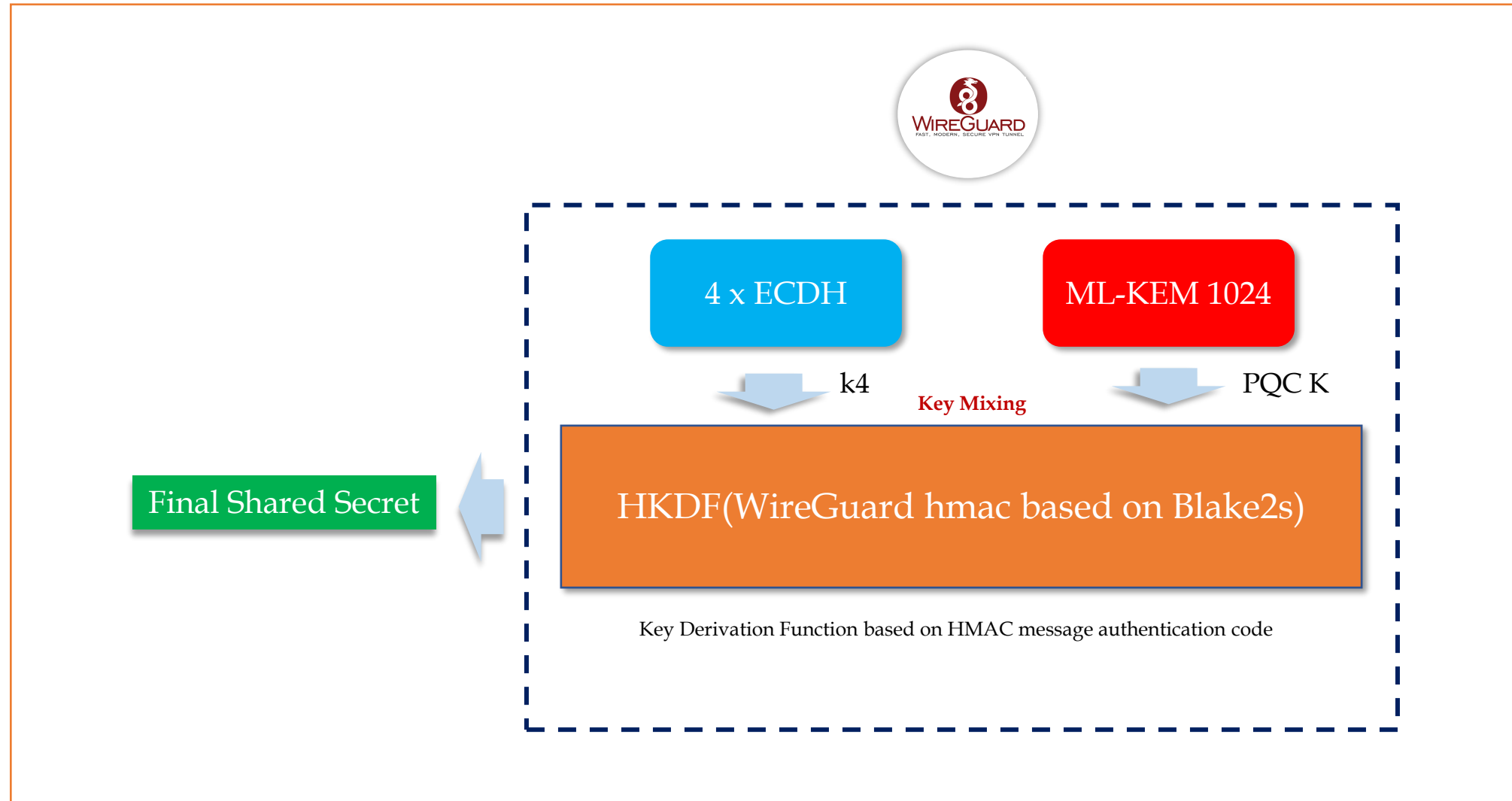
## 2. WireGuard(4) – PQC(1)



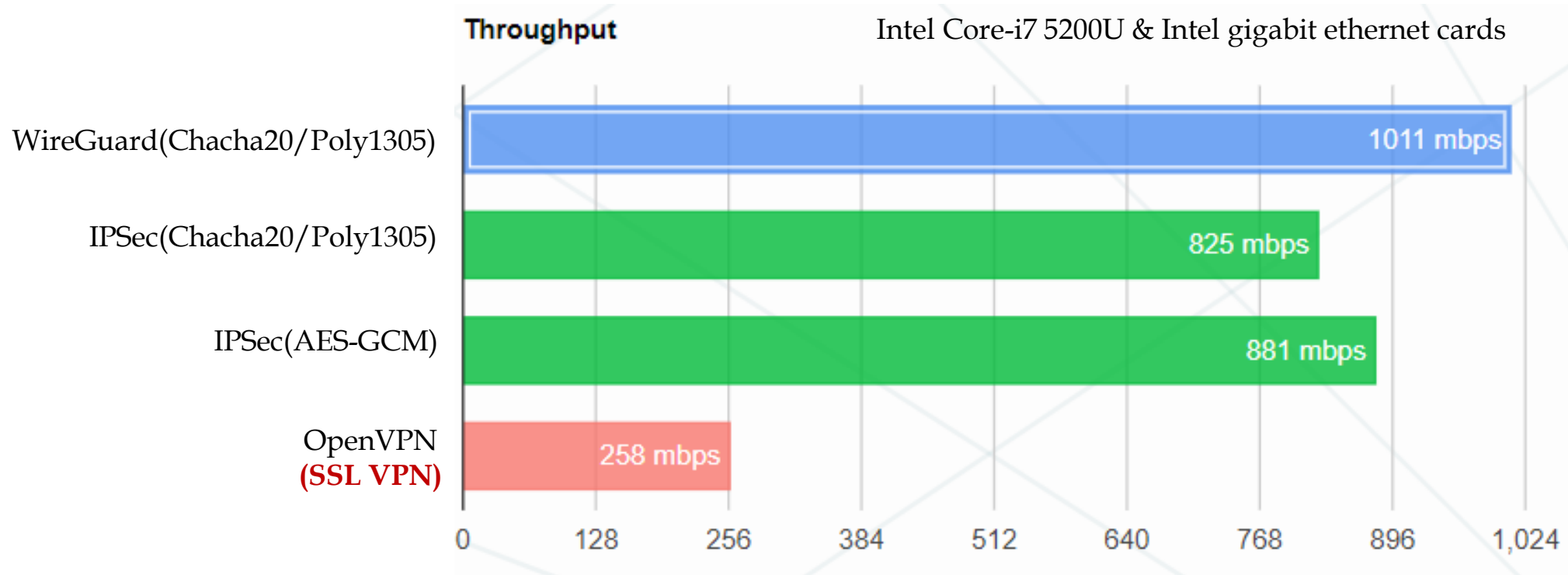
PQC 알고리즘이 통합된 WireGuard Protocol



## 2. WireGuard(4) – PQC(2)



## 2. WireGuard(5) – Performance(1)

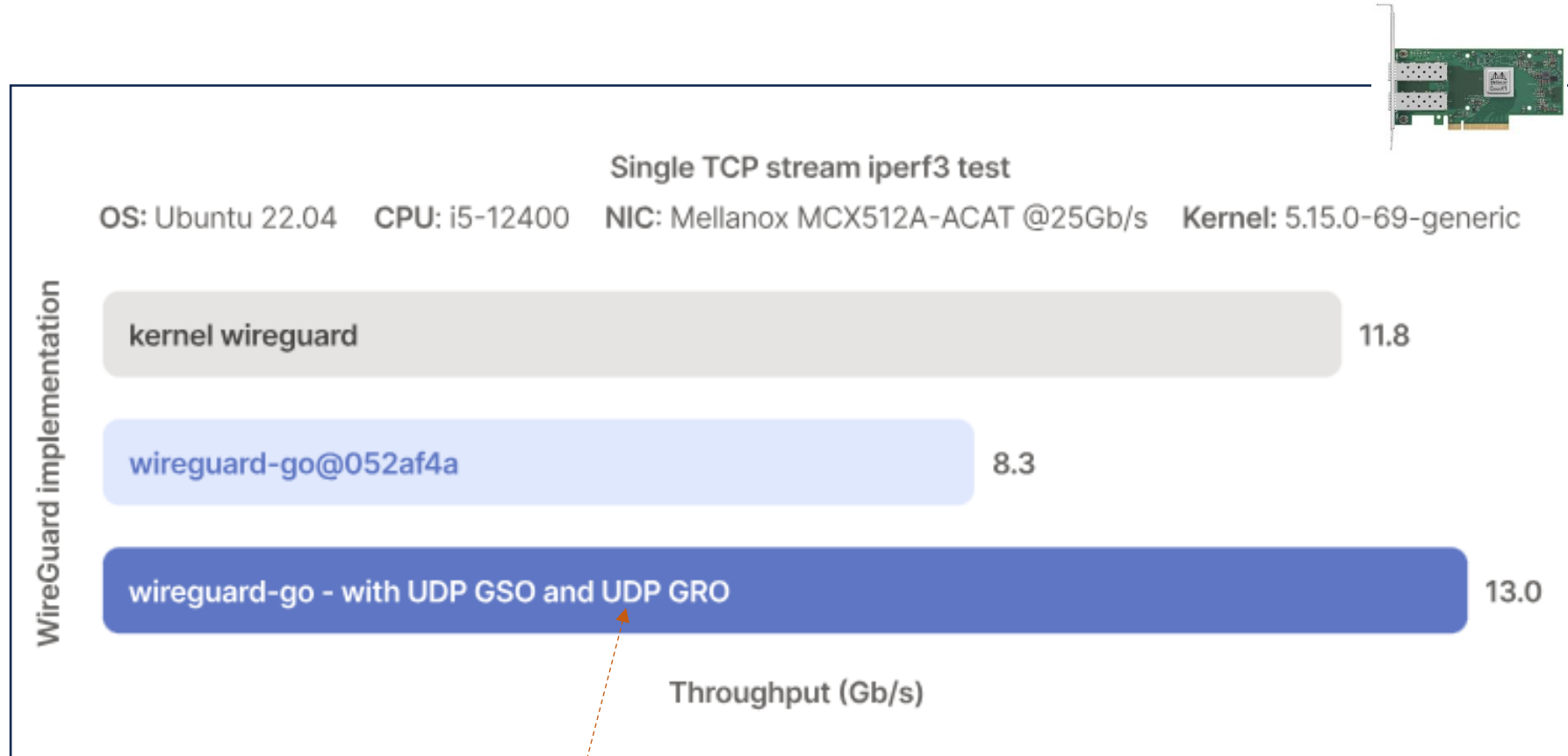


[출처] <https://www.wireguard.com/performance/>

[주의] OpenVPN DCO(Data Channel Offload)는 위의 그림보다는 훨씬 빠르다.



## 2. WireGuard(5) - Performance(2)



Packet Vector Processing  
UDP GRO/GSO  
Checksum Offload

그림 출처: <https://tailscale.com/blog/more-throughput>

## 2. WireGuard(5) - Performance(3)

### ■ WireGuard VPN의 우수한 성능 (Multi-Core 환경)

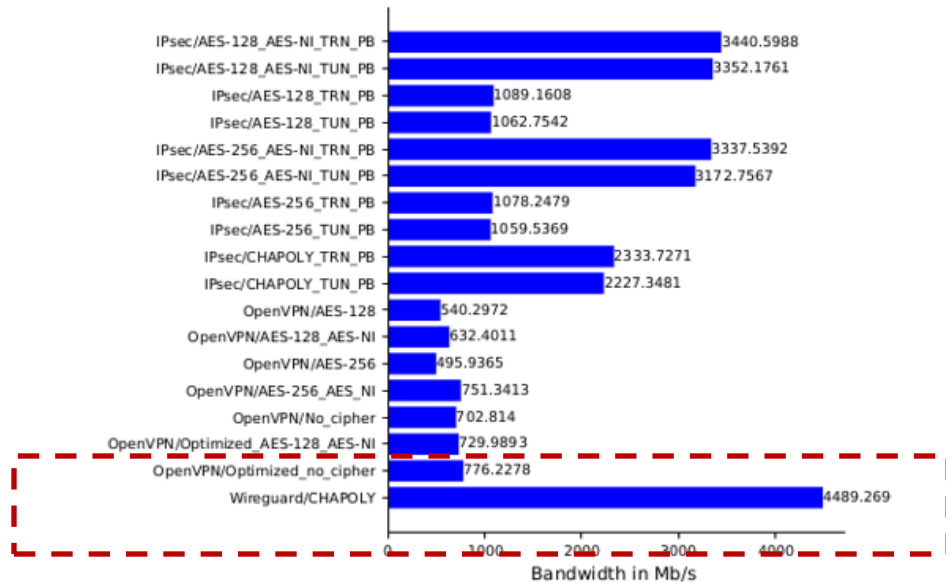
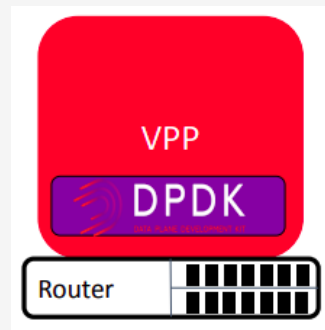


그림 출처: Performance Comparison of VPN Solutions, Lukas Osswald

- Signature 없이 인증 & 키 교환하는 AKE 기법 도입
- 최신 암호 알고리즘 및 암호 가속 기능으로 무장
  - Curve25519, Blake2s, ChaCha20, Poly1305, SipHash4
  - X86\_64 AVX2, AVX512
- Perfect Forward Secrecy 보장(2분 간격 Handshaking)
  - 1-RTT만에 Handshaking 완료
- Replay Attack 방지 기능(Counter 필드 유지/관리)
- Identity Hiding 기능
- DoS 공격 완화 기능(Cookie 활용)

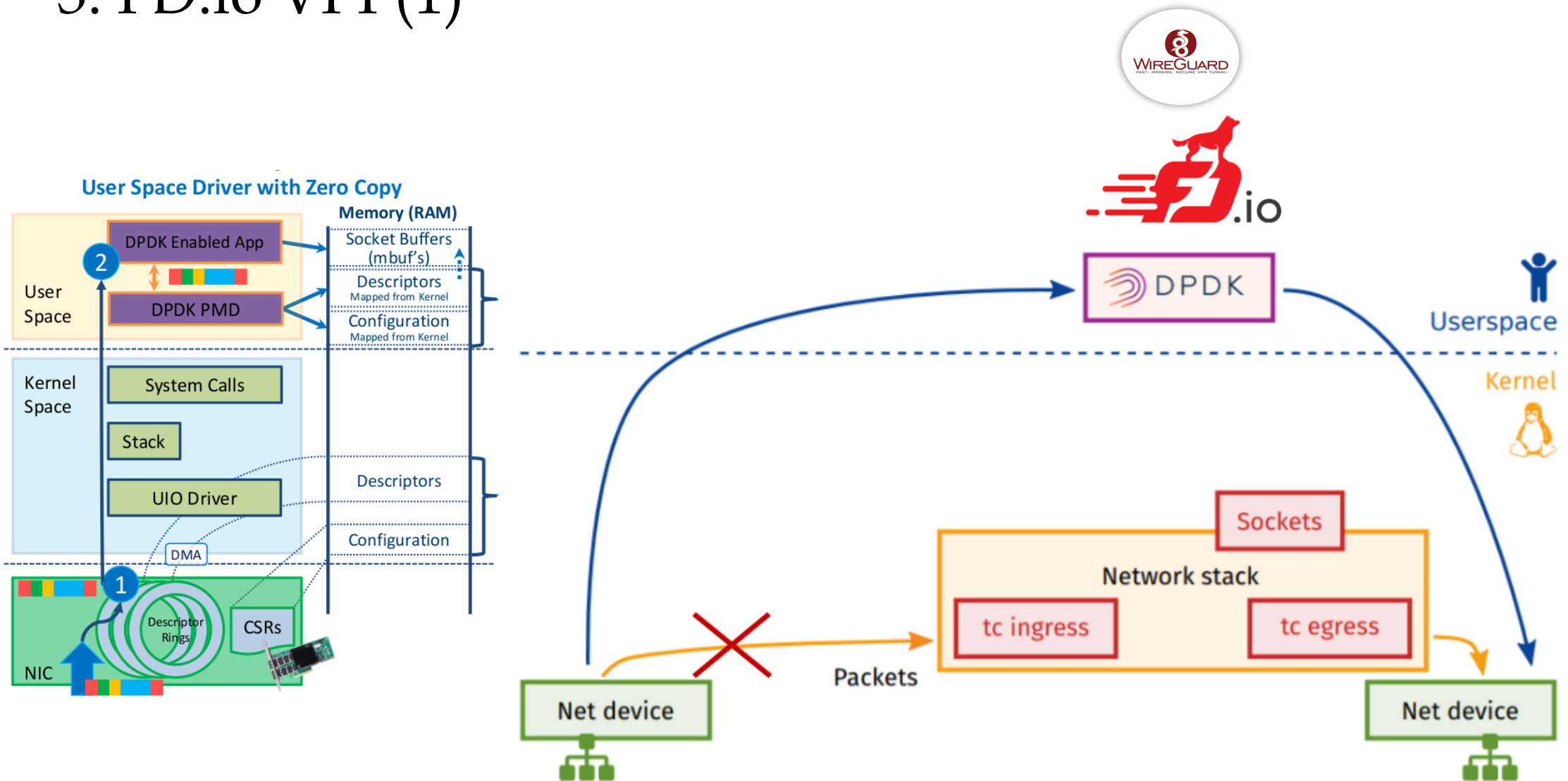
### 3. FD.io VPP



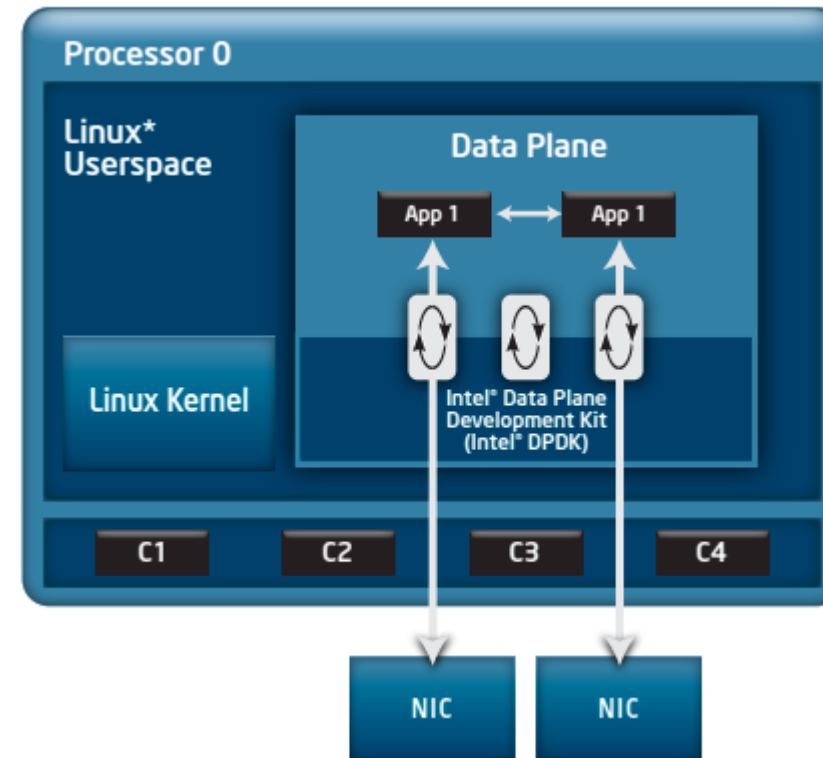
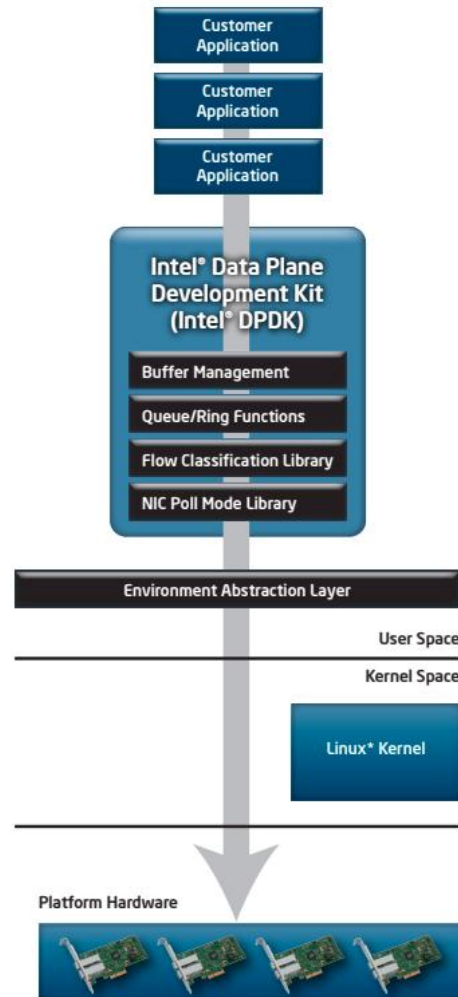
**Software ASIC**

FD.io VPP and FD.io logo are registered trademarks of The Fast Data Project, a series of LF Projects, LLC

### 3. FD.io VPP(1)

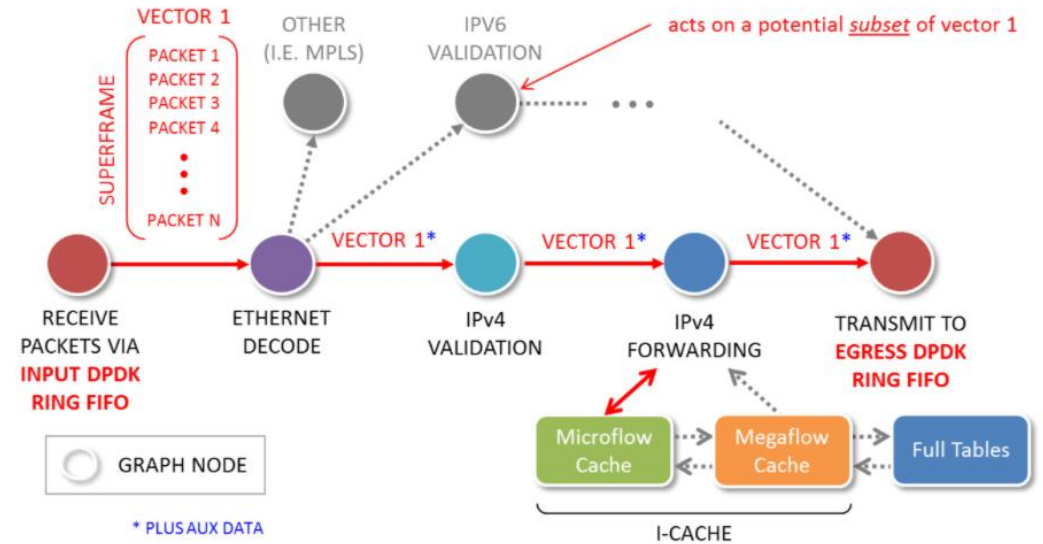
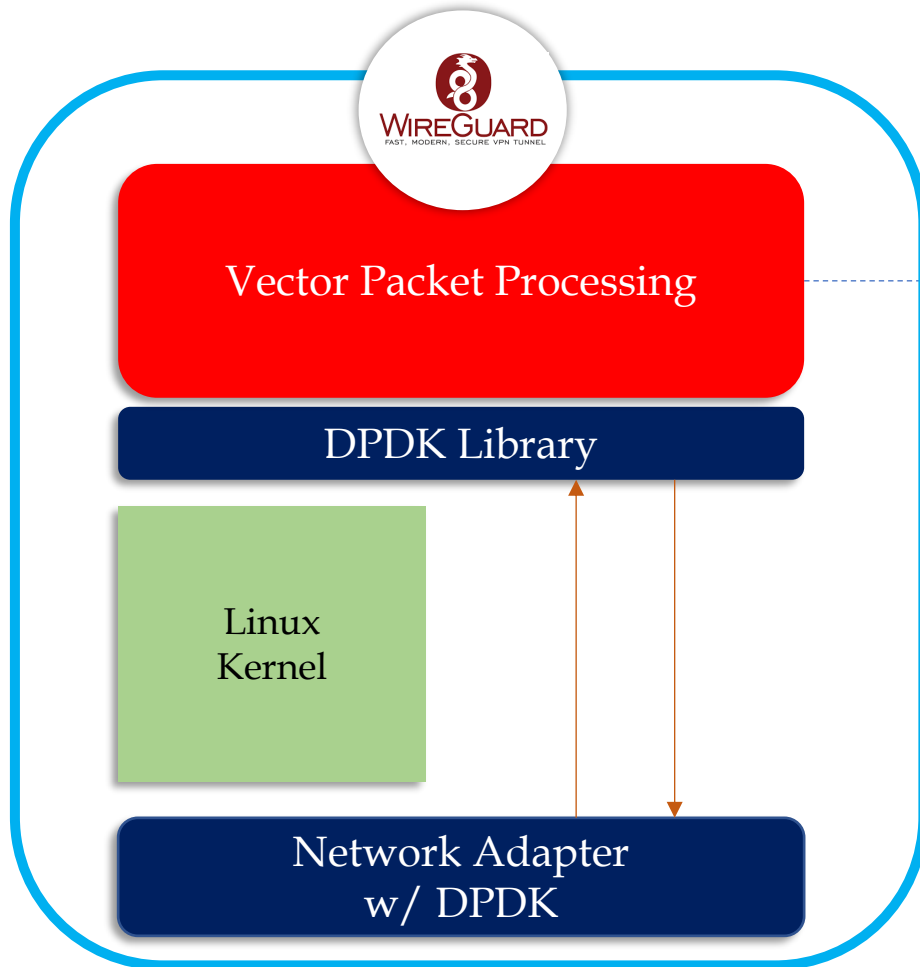


### 3. FD.io VPP(2)



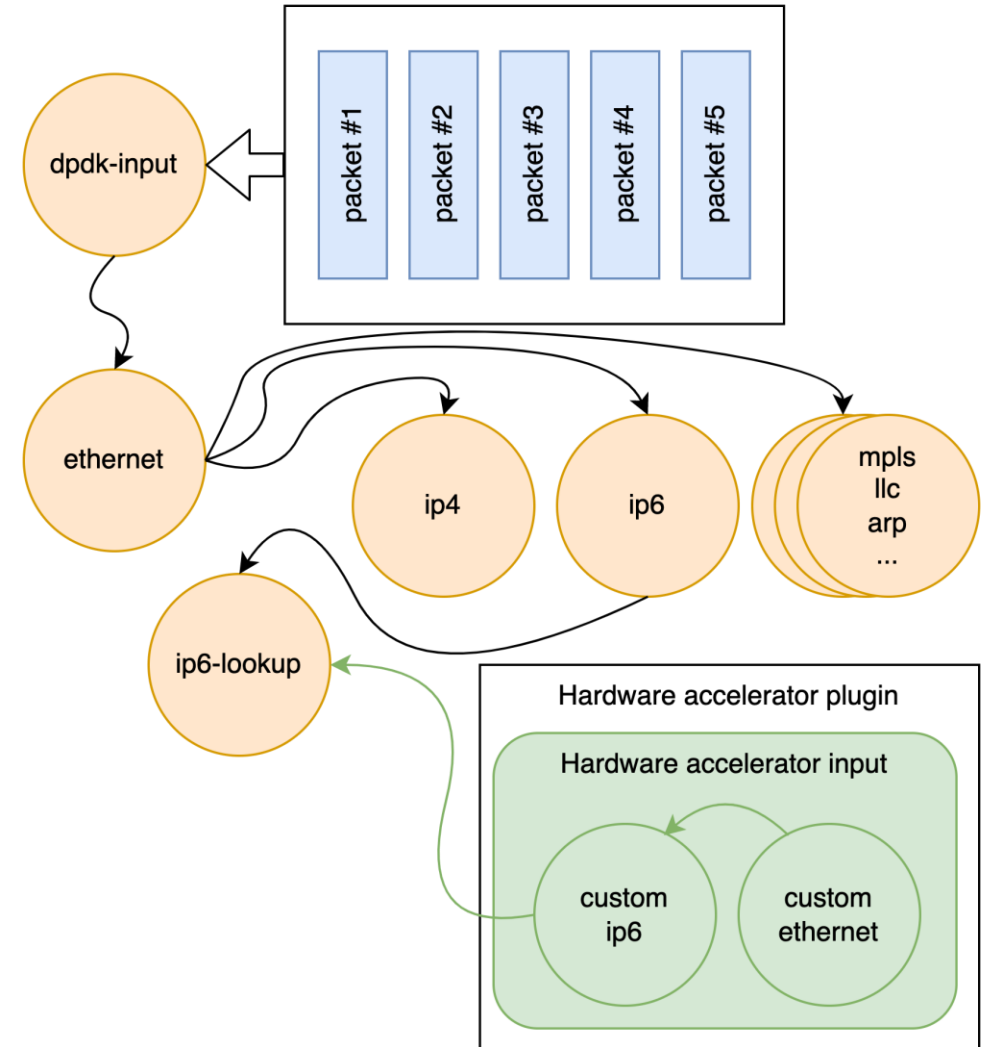
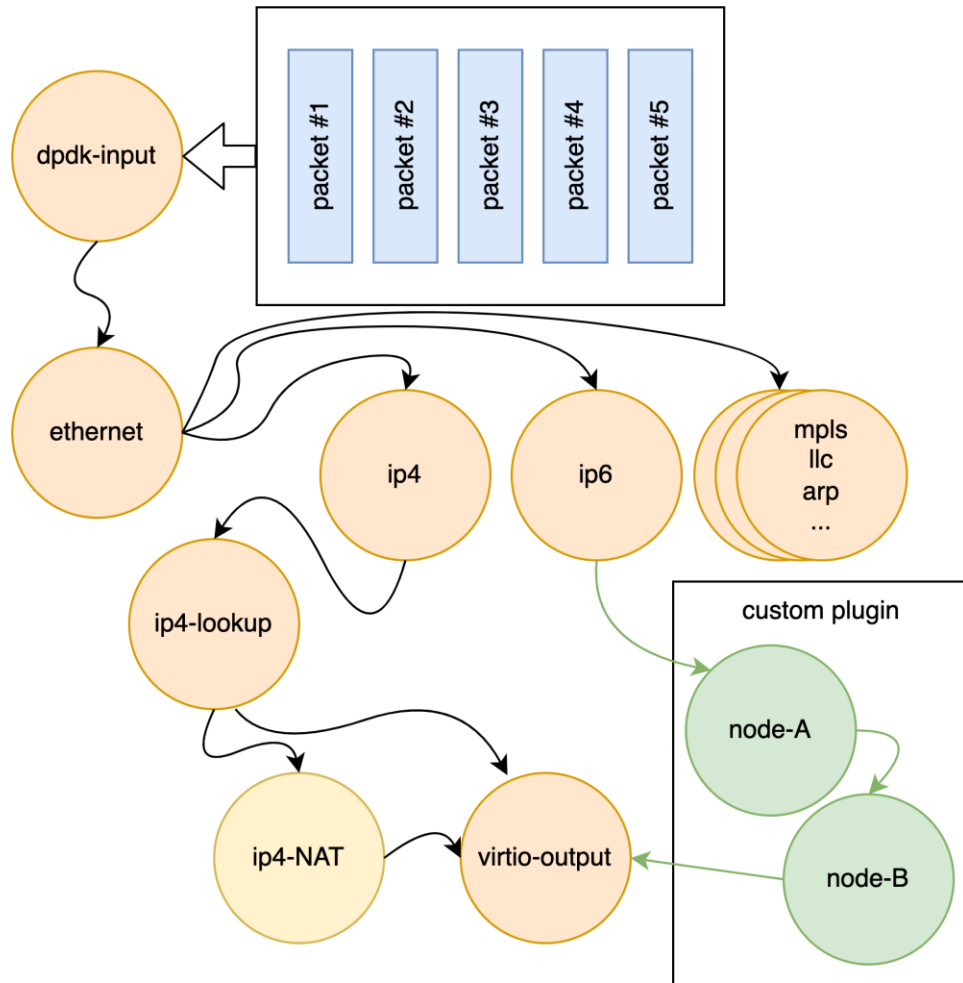
Ref) Impressive Packet Processing Performance Enables Greater Workload Consolidation

### 3. FD.io VPP(3)

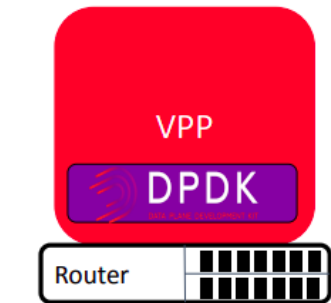


### 3. FD.io VPP(4)

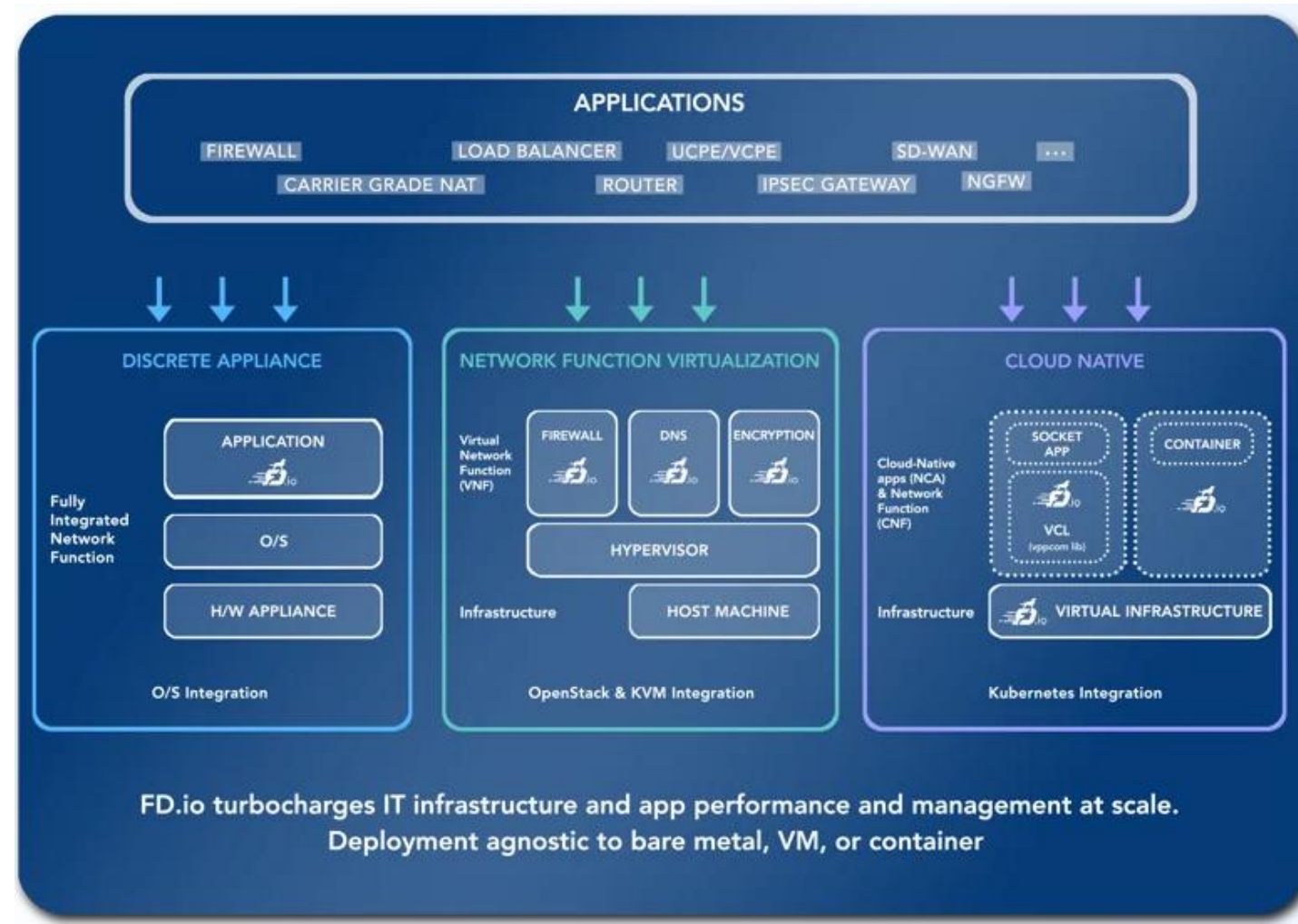
Ref <https://fd.io/technology/>



### 3. FD.io VPP(5)



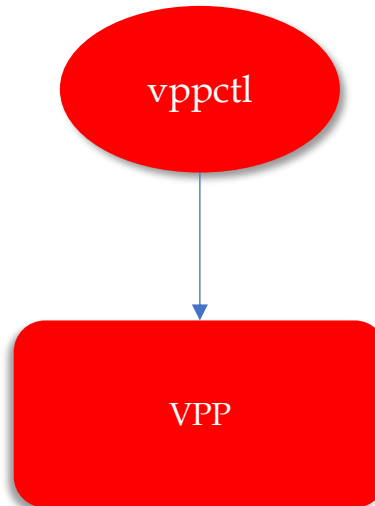
- L2 Switch
- L3 Router
- CGNAT
- L4 Switch
- WireGuard VPN
- IPSec VPN
- IPS
- VxLAN Gateway
- ...



Ref) <https://www.netgate.com/resources/articles-vector-packet-processing>



### 3. FD.io VPP(6)



```
/usr/bin/vpp -c /etc/vpp/startup.conf
```

```
P5 C:\Users\Michael> ssh -p 2241 ztnabox@192.168.1.57
ztnabox@192.168.1.57's password:
Last login: Fri Jan 5 08:34:57 2024 from 192.168.1.116

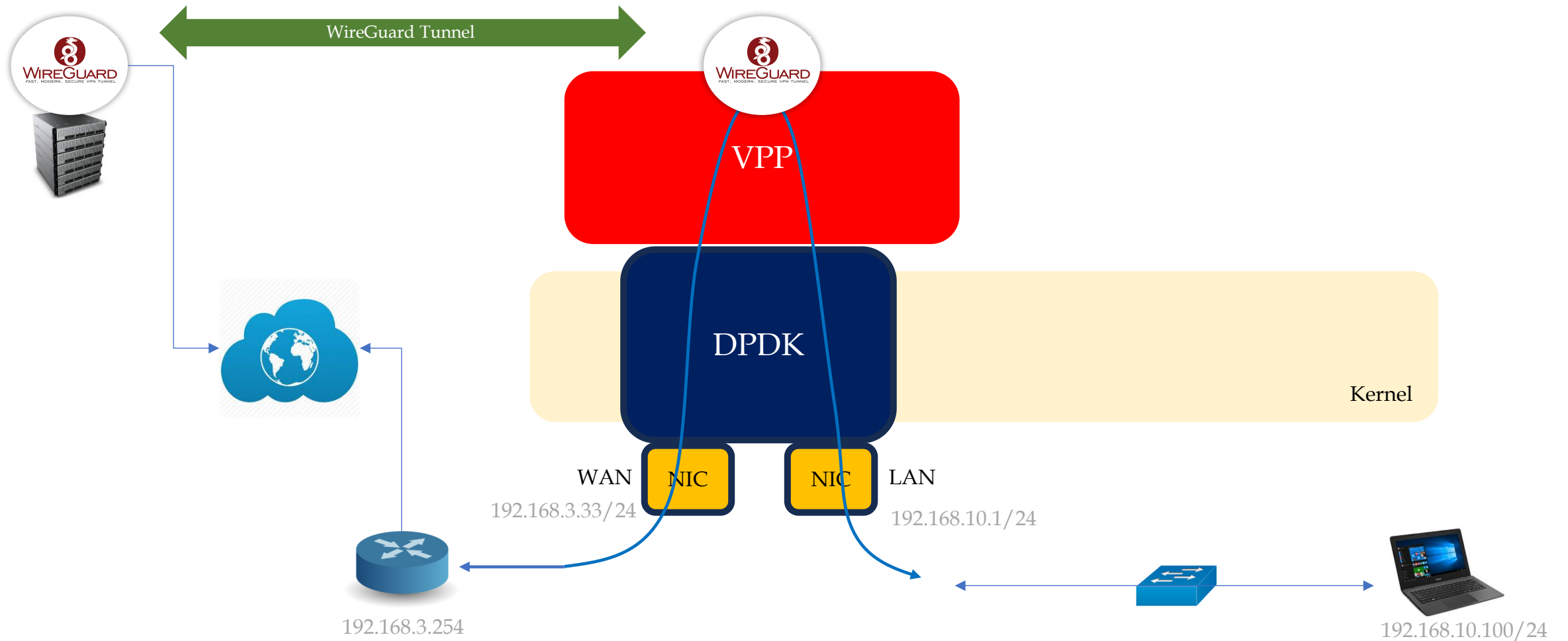
      _/_/_/_/_   \_  _/_/_/_/_
     /_/_/_/_/_   /_/_/_/_/_
    /_/_/_/_/_   /_/_/_/_/_
   /_/_/_/_/_   /_/_/_/_/_
  /_/_/_/_/_   /_/_/_/_/_
 /_/_/_/_/_   /_/_/_/_/_
/_/_/_/_/_   /_/_/_/_/_

vpp# show interface

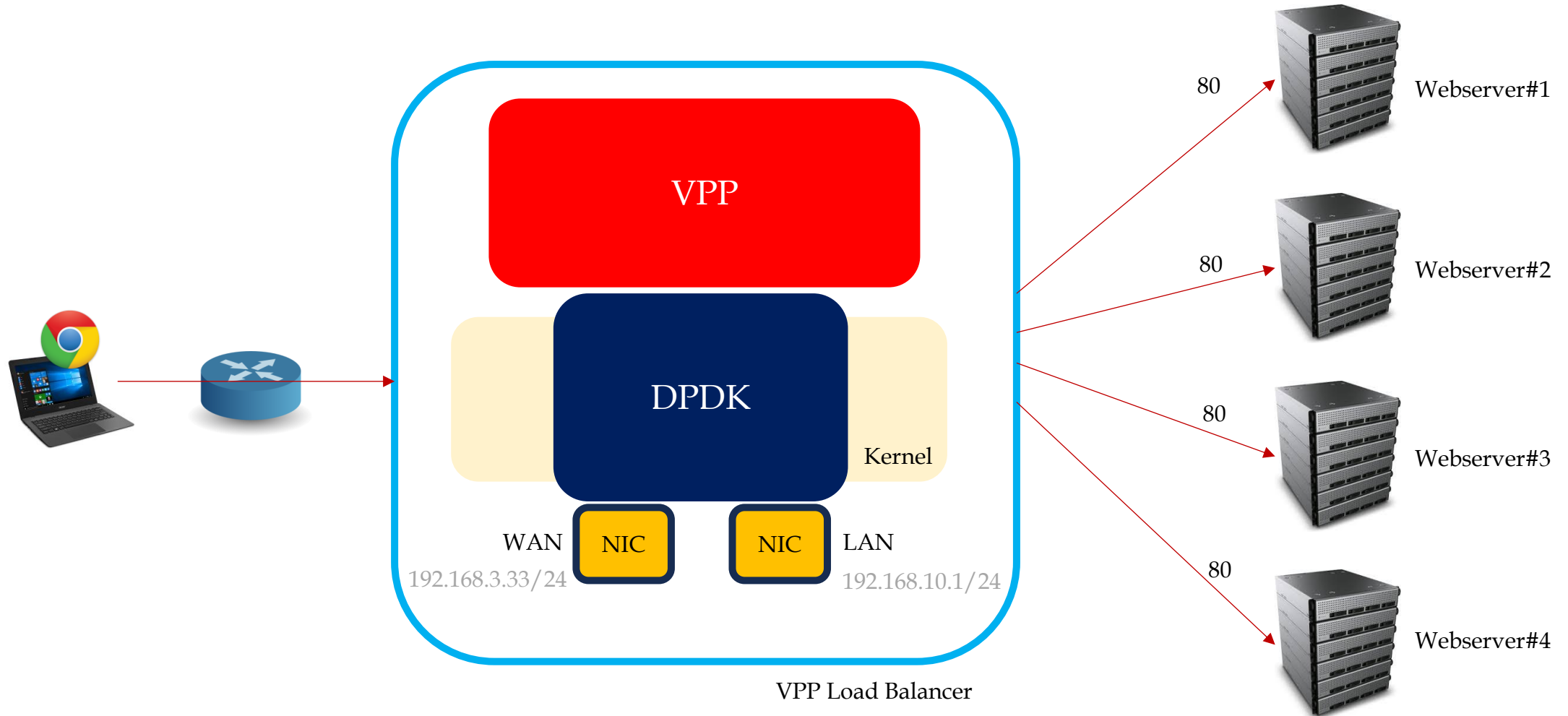
      Name                               Idx  State  MTU (L3/IP4/IP6/MPLS)  Counter                                Count
TenGigabitEthernet3b/0/0                1     up      9000/0/0/0             rx packets                  185665
                                           rx bytes                   15105688
                                           drops                      185665
                                           ip4                        180187
                                           ip6                         65
TenGigabitEthernet3b/0/1                 2    down      9000/0/0/0             rx packets                  219963
                                           rx bytes                   18226887
                                           tx packets                  6174
                                           tx bytes                   526266
                                           drops                      189314
                                           punt                       17411
                                           ip4                        198852
                                           ip6                         389
TenGigabitEthernet3b/0/3                 4    down      9000/0/0/0             rx packets                  16541
local0                                   0    down      0/0/0/0                 drops                       38
wg0                                       5     up       1420/0/0/0              rx packets                  14688
                                           rx bytes                   120
                                           tx packets                  18720
                                           tx bytes                    30
                                           drops                       153
                                           ip4

vpp#
```

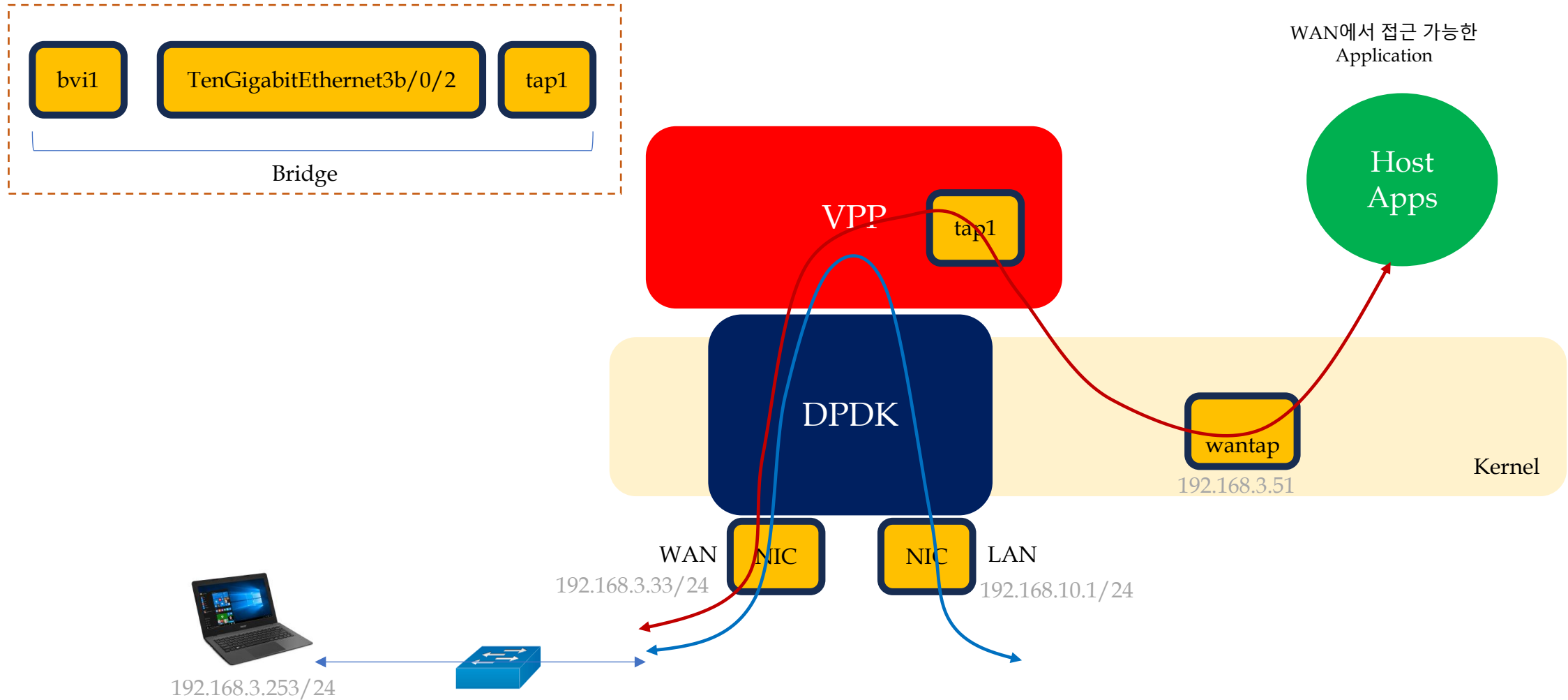
### 3. FD.io VPP(7) - WireGuard



### 3. FD.io VPP(8) – NAT & Load Balancing



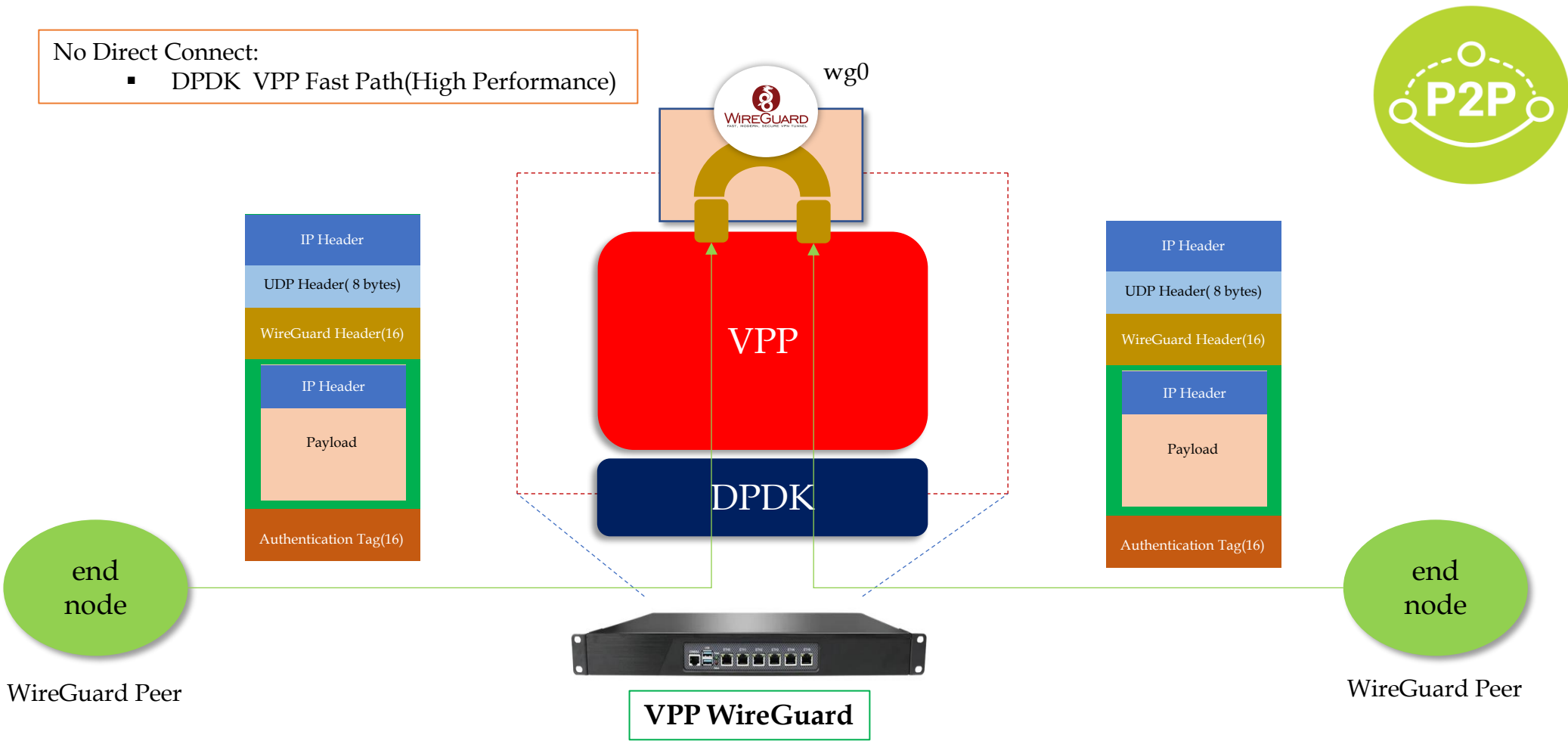
### 3. FD.io VPP(9) – Legacy App Access



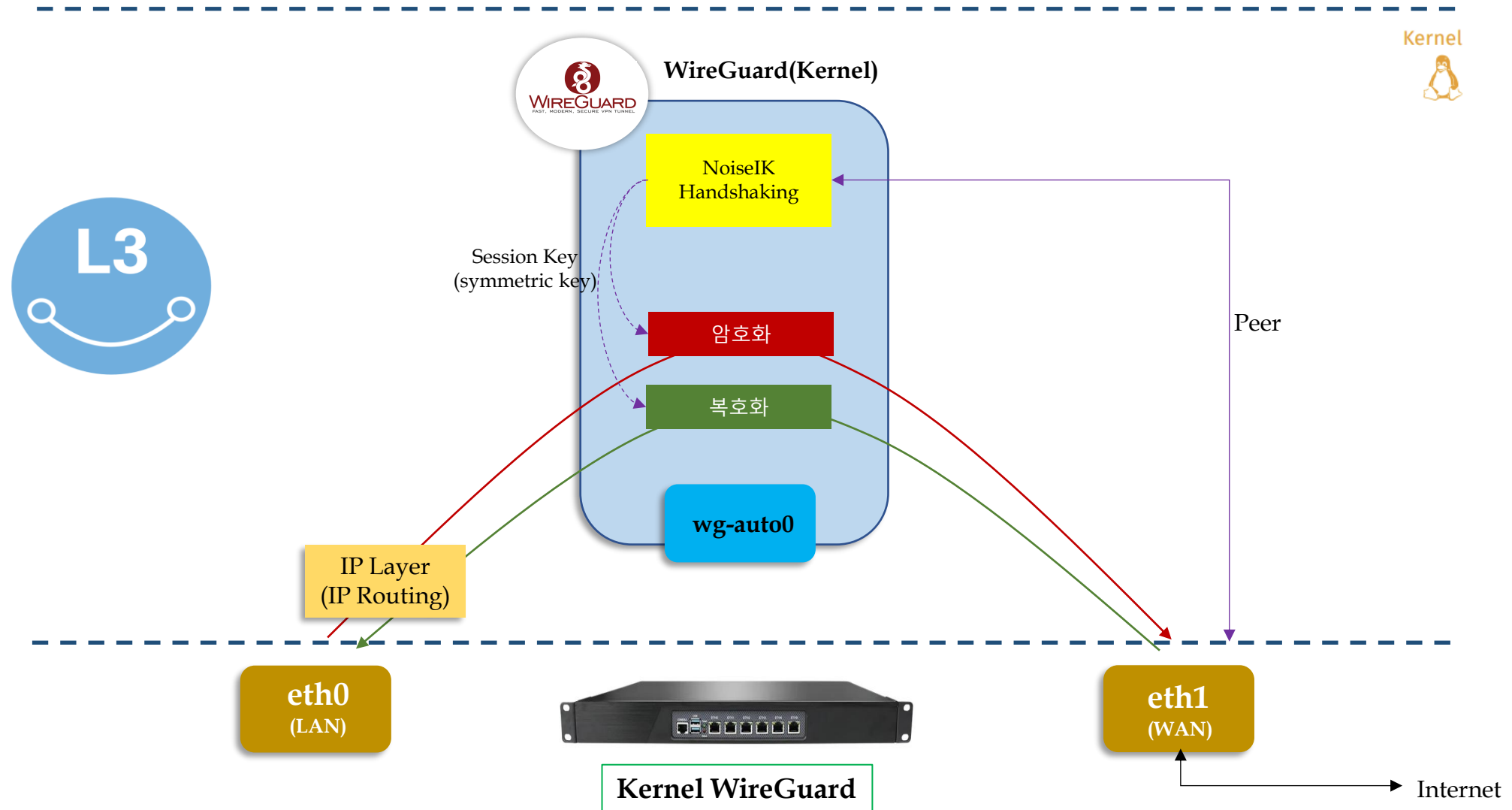
## 4. MightyConnect Security Box



# 4. MightyConnect Security Box(1) – WireGuard(1)



## 4. MightyConnect Security Box(1) – WireGuard(2-1)

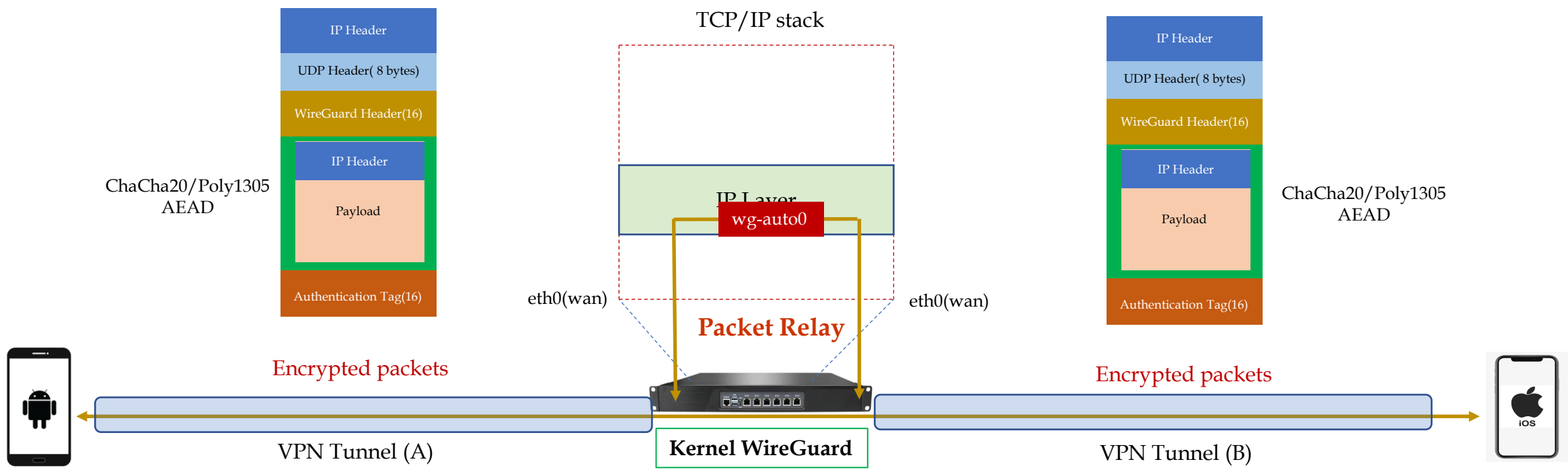


# 4. MightyConnect Security Box(1) – WireGuard(2-2)



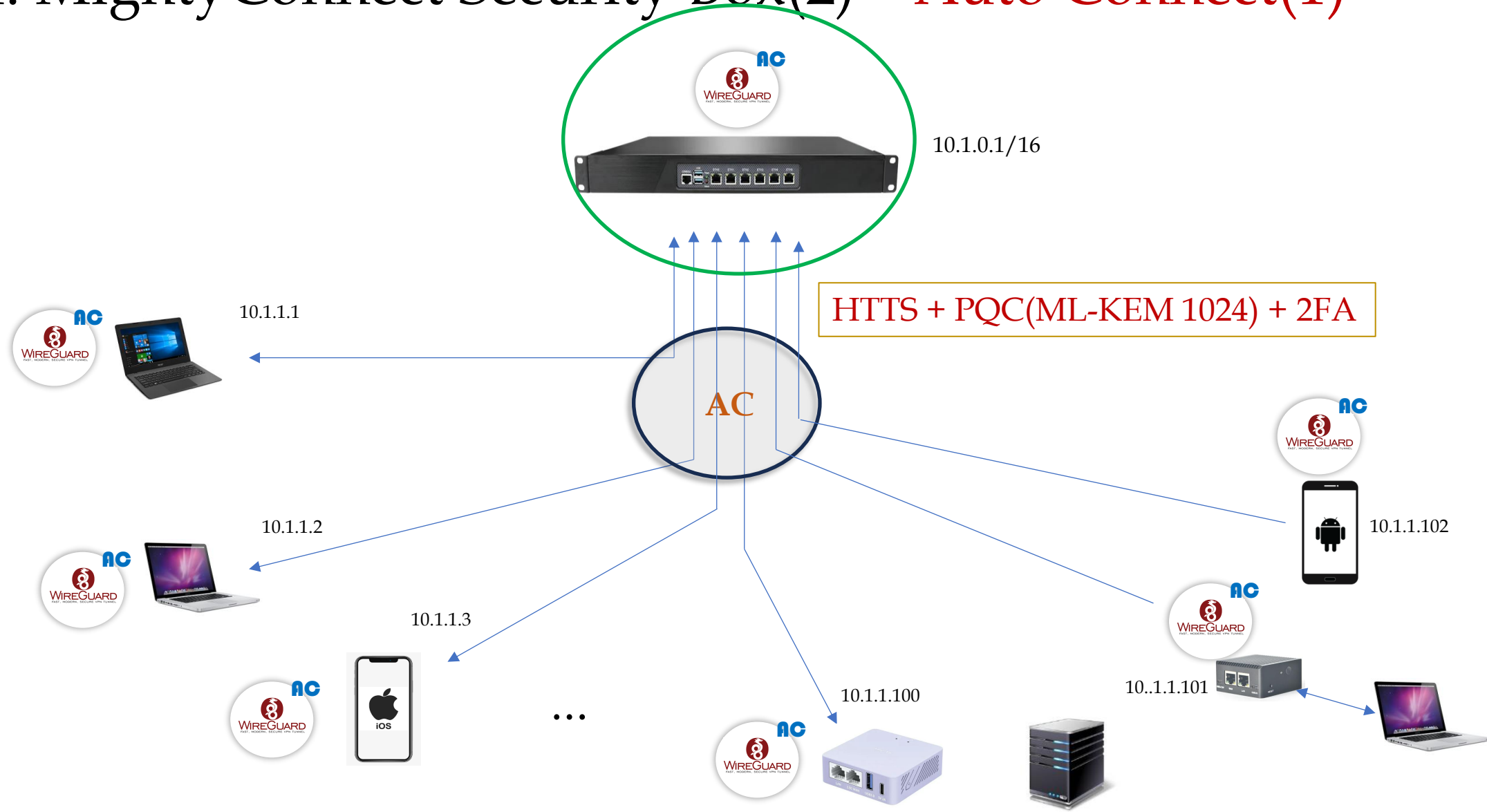
CryptoKey Routing

ID: Public Key::VPN IP

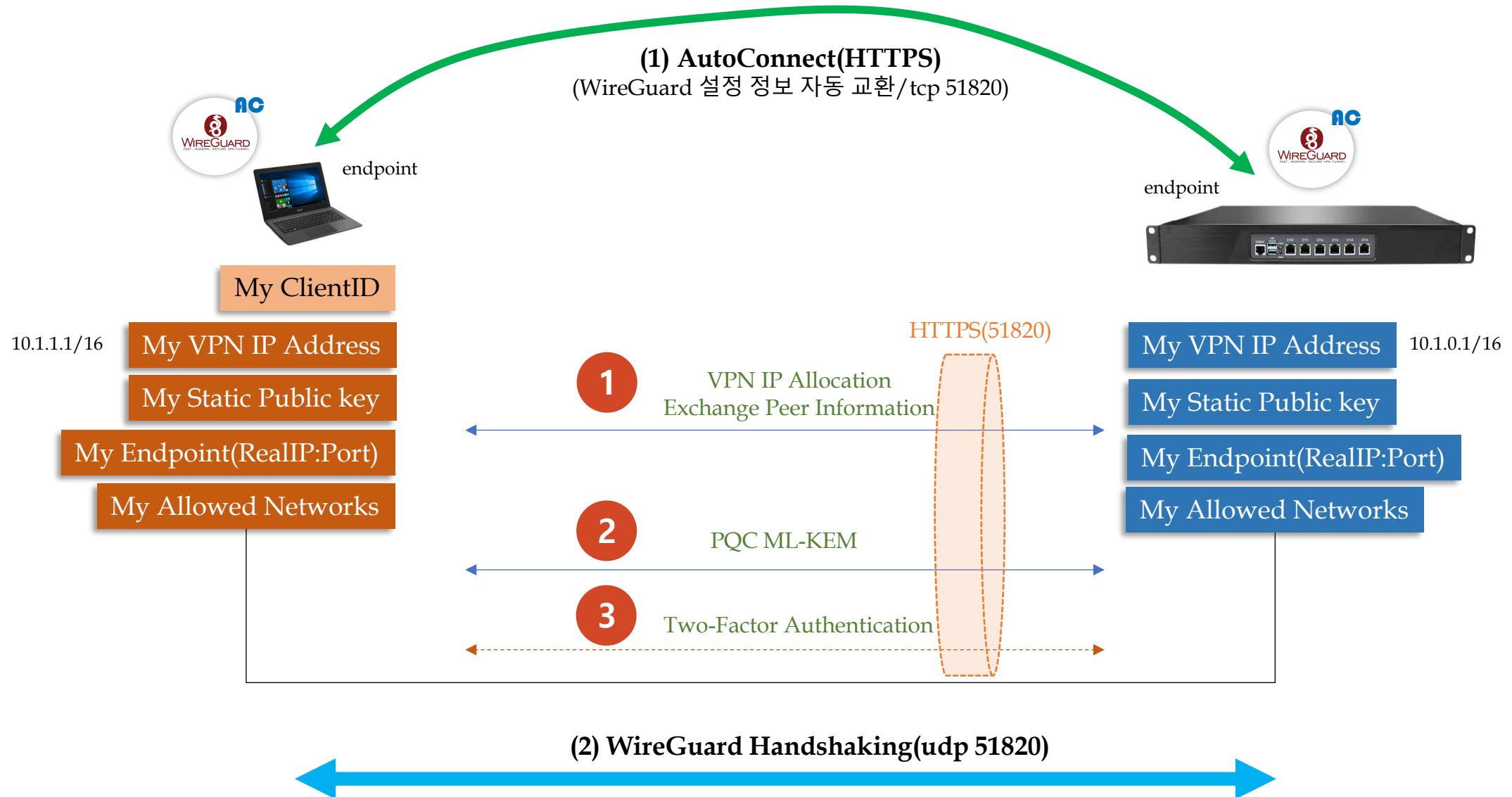




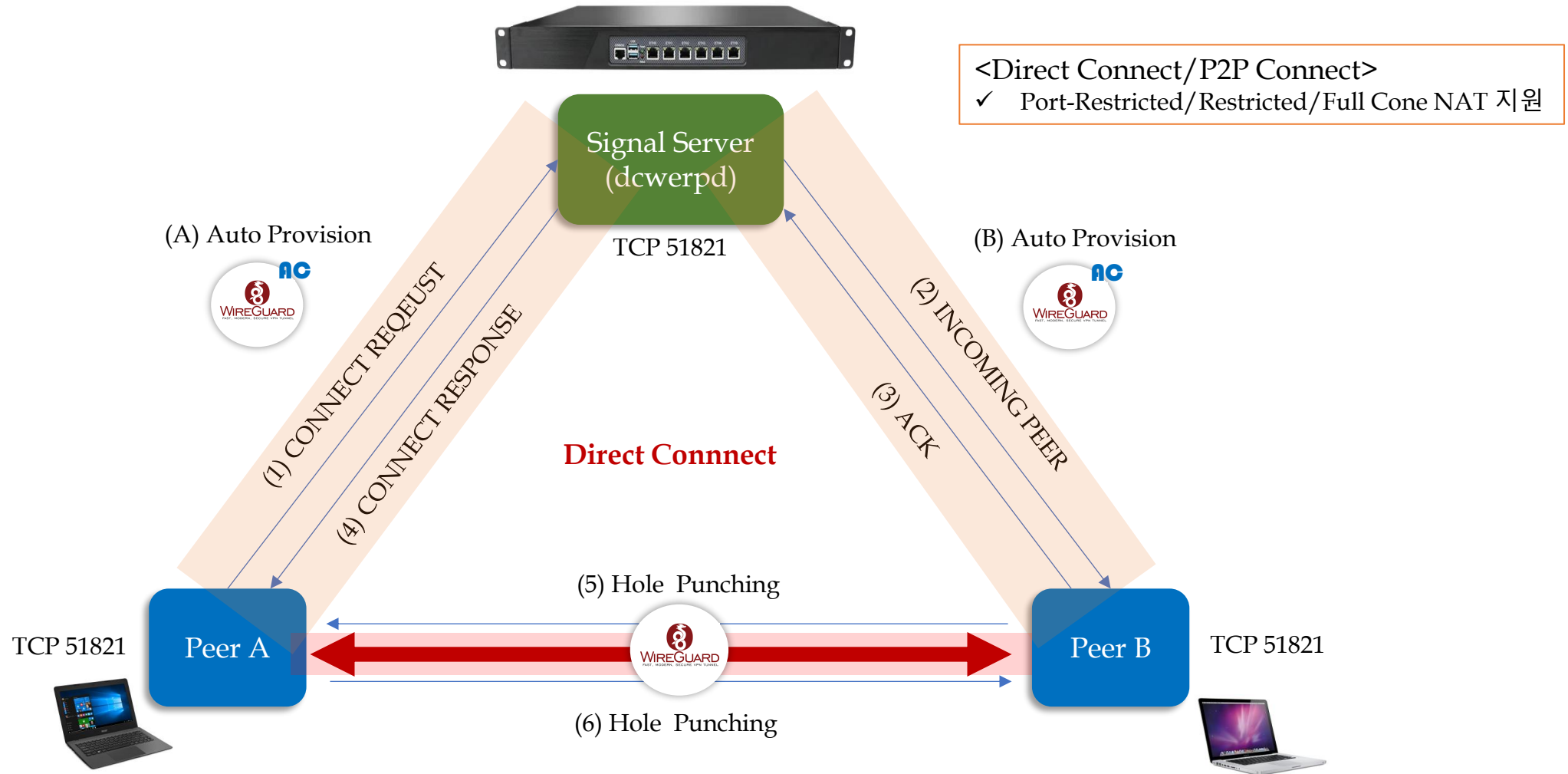
# 4. MightyConnect Security Box(2) – Auto Connect(1)



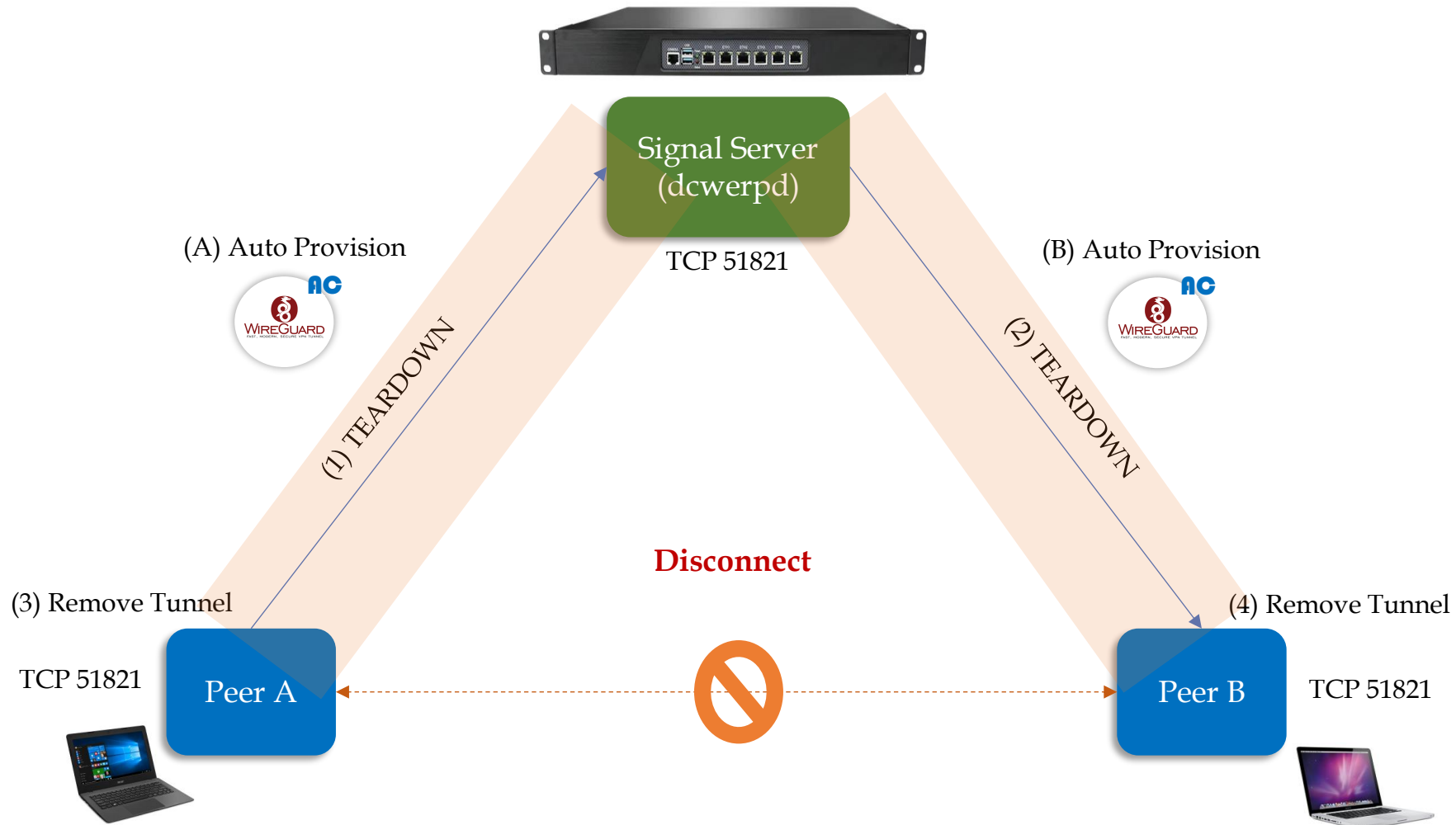
## 4. MightyConnect Security Box(2) – Auto Connect(2)



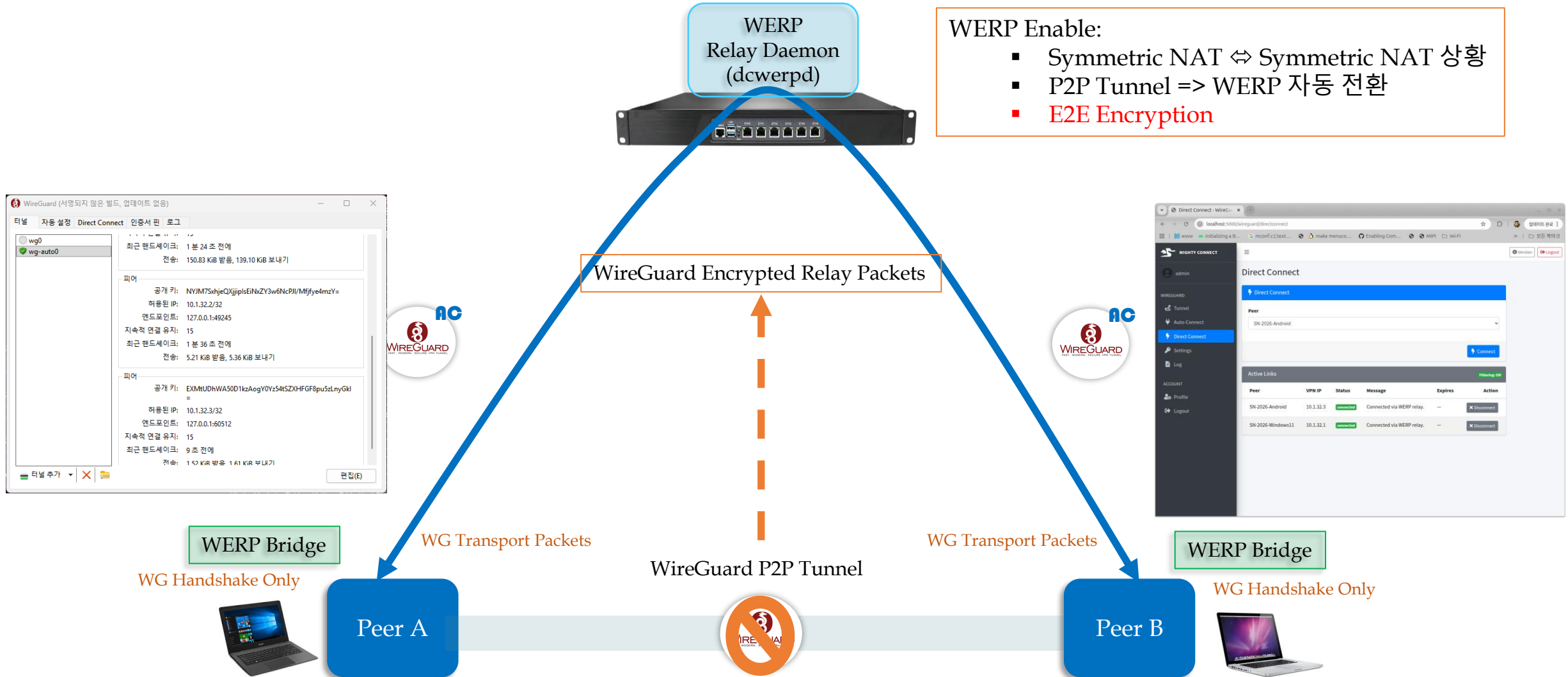
## 4. MightyConnect Security Box(3) – Direct Connect(1)



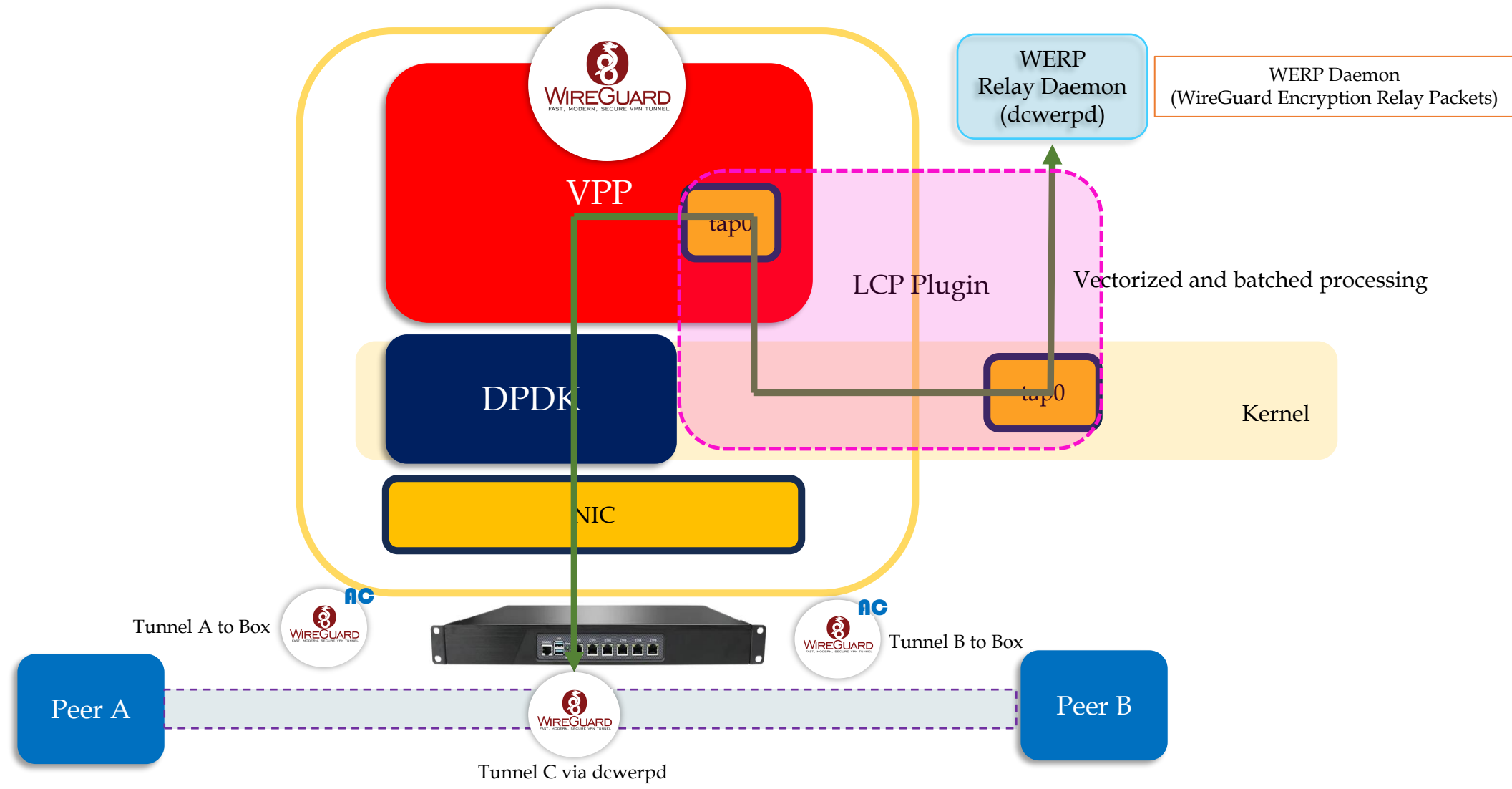
## 4. MightyConnect Security Box(3) – Direct Connect(2)



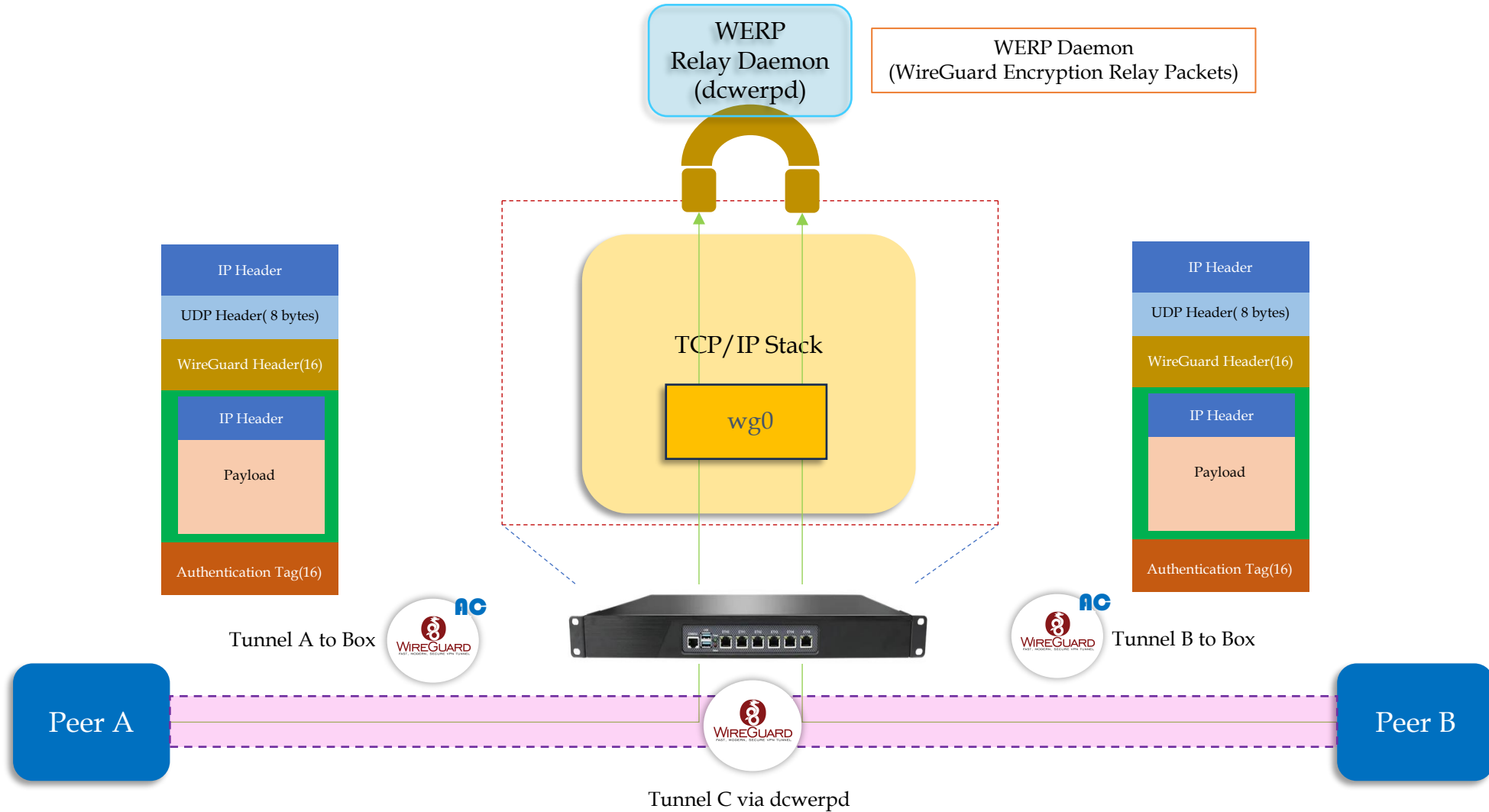
## 4. MightyConnect Security Box(3) – Direct Connect(3-1)



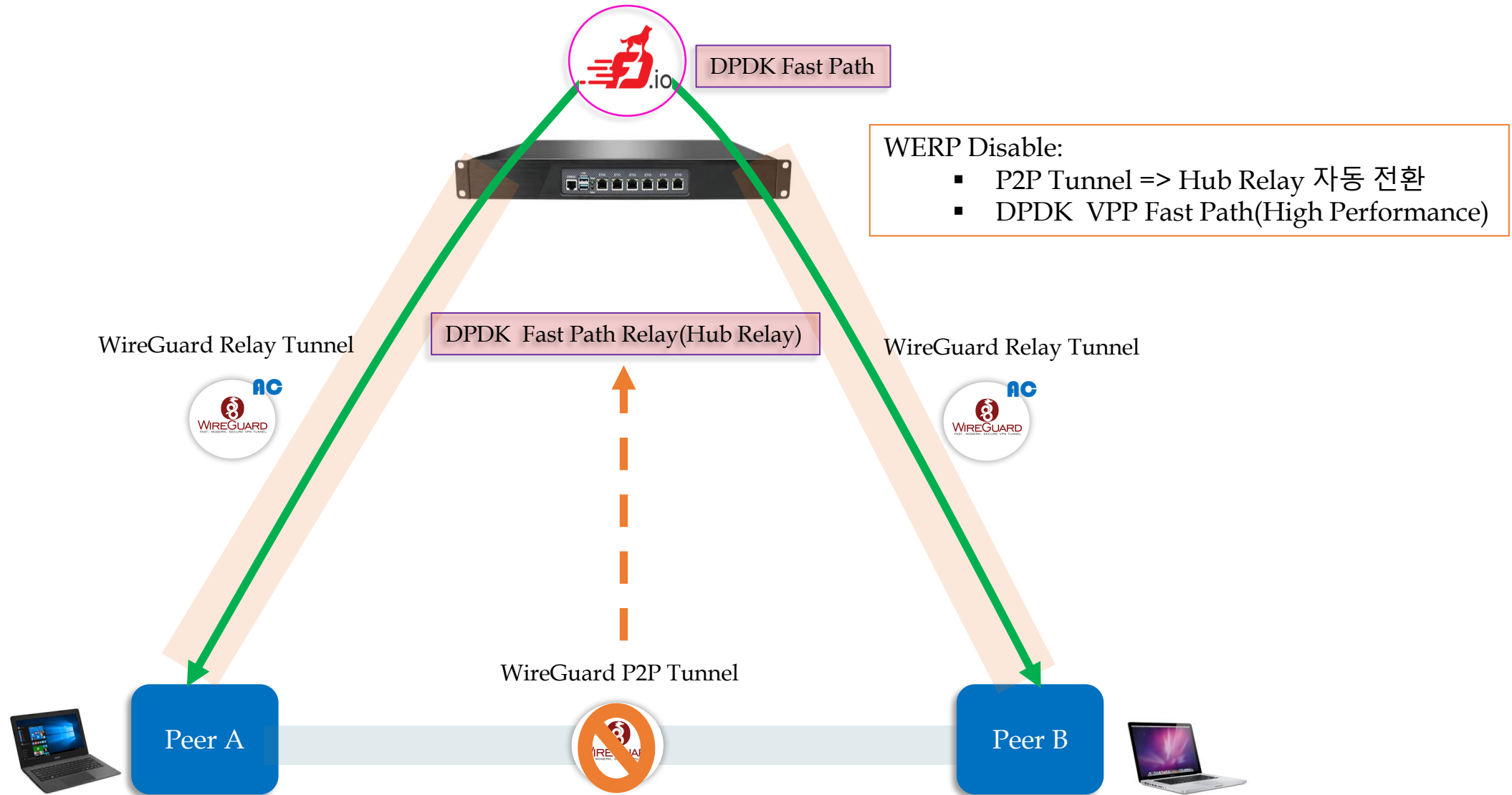
# 4. MightyConnect Security Box(3) – Direct Connect(3-2)



## 4. MightyConnect Security Box(3) – Direct Connect(3-3)

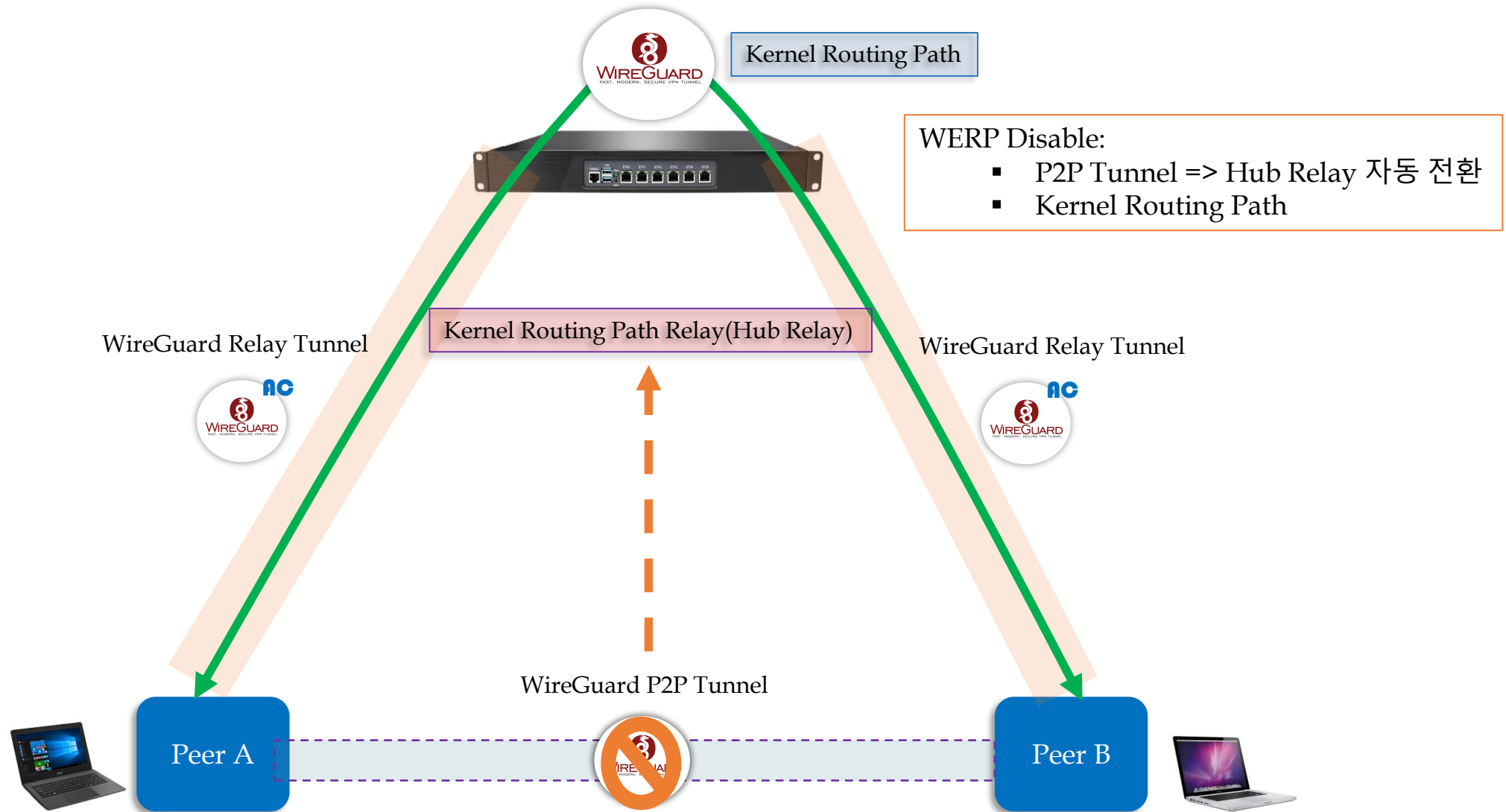


## 4. MightyConnect Security Box(4) – Fast Hub Relay(1)





## 4. MightyConnect Security Box(4) – Fast Hub Relay(2)



## 4. MightyConnect Security Box(5) – VPP ZTNA(1)

### Mighty ZTNA Filter (Micro Segmentation)

Allow Resources per not IP Address but Client ID

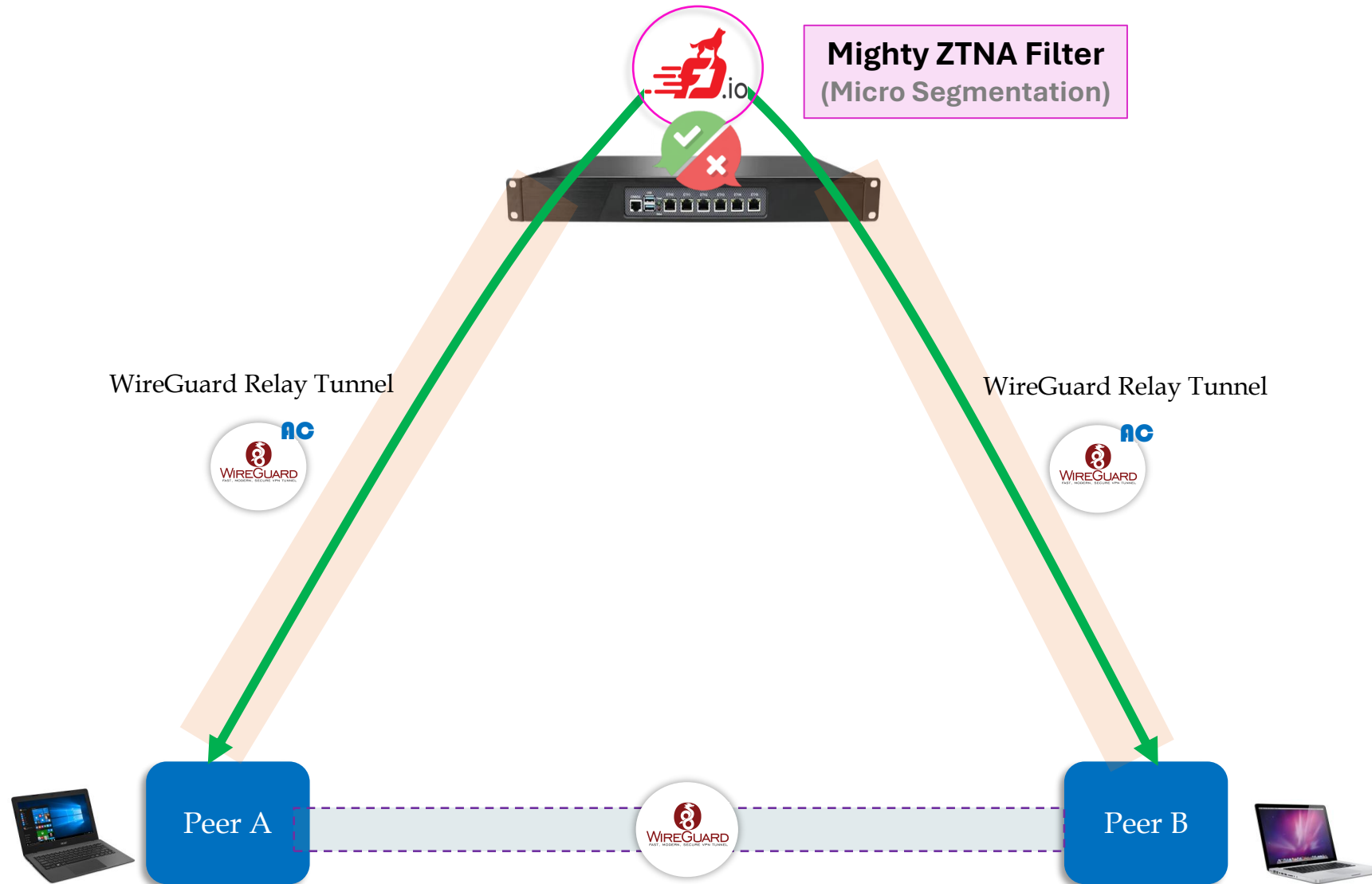


VPP ACL

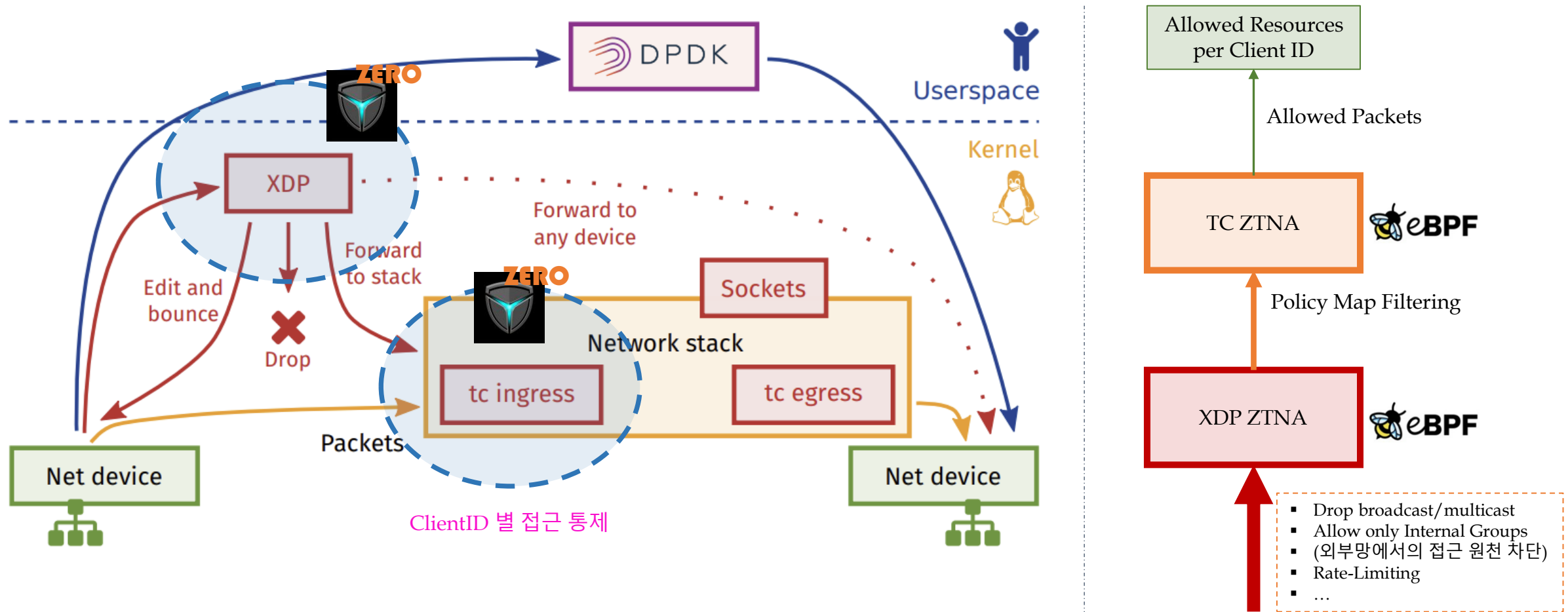
DPDK

```
VPP ACL state (show acl-plugin acl):
acl-index 0 count 9 tag {mightyconnect-acl-builtin-phys}
0: ipv4 permit+reflect src 192.168.5.0/24 dst 0.0.0.0/0 proto 0 sport 0-65535 dport 0-65535
1: ipv4 permit src 0.0.0.0/0 dst 0.0.0.0/0 proto 17 sport 0-65535 dport 51820
2: ipv4 permit src 0.0.0.0/0 dst 0.0.0.0/0 proto 6 sport 0-65535 dport 53
3: ipv4 permit src 0.0.0.0/0 dst 0.0.0.0/0 proto 6 sport 53 dport 0-65535
4: ipv4 permit src 0.0.0.0/0 dst 0.0.0.0/0 proto 17 sport 0-65535 dport 53
5: ipv4 permit src 0.0.0.0/0 dst 0.0.0.0/0 proto 17 sport 53 dport 0-65535
6: ipv4 permit src 0.0.0.0/0 dst 0.0.0.0/0 proto 6 sport 0-65535 dport 51820
7: ipv4 permit src 0.0.0.0/0 dst 0.0.0.0/0 proto 1 sport 0-65535 dport 0-65535
8: ipv4 permit src 0.0.0.0/0 dst 224.0.0.18/32 proto 112 sport 0-65535 dport 0-65535
  applied inbound on sw_if_index: 1, 2
  applied outbound on sw_if_index:
  used in lookup context index: 1, 3
acl-index 1 count 1 tag {mightyconnect-acl-egress-phys}
0: ipv4 permit+reflect src 0.0.0.0/0 dst 0.0.0.0/0 proto 0 sport 0-65535 dport 0-65535
  applied inbound on sw_if_index:
  applied outbound on sw_if_index: 1, 2
  used in lookup context index: 0, 2
acl-index 2 count 28 tag {mightyconnect-acl-ztna-wg0}
0: ipv4 permit+reflect src 10.1.1.3/32 dst 10.1.1.254/32 proto 6 sport 0-65535 dport 51821
1: ipv4 permit+reflect src 10.1.1.3/32 dst 10.1.1.0/24 proto 1 sport 0-65535 dport 0-65535
2: ipv4 permit+reflect src 10.1.1.3/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 22
3: ipv4 permit+reflect src 10.1.1.3/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 443
4: ipv4 permit+reflect src 10.1.1.1/32 dst 10.1.1.254/32 proto 6 sport 0-65535 dport 51821
5: ipv4 permit+reflect src 10.1.1.1/32 dst 10.1.1.0/24 proto 1 sport 0-65535 dport 0-65535
6: ipv4 permit+reflect src 10.1.1.1/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 22
7: ipv4 permit+reflect src 10.1.1.1/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 443
8: ipv4 permit+reflect src 10.1.1.1/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 80
9: ipv4 permit+reflect src 10.1.1.4/32 dst 10.1.1.254/32 proto 6 sport 0-65535 dport 51821
10: ipv4 permit+reflect src 10.1.1.4/32 dst 10.1.1.0/24 proto 1 sport 0-65535 dport 0-65535
11: ipv4 permit+reflect src 10.1.1.4/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 22
12: ipv4 permit+reflect src 10.1.1.4/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 443
13: ipv4 permit+reflect src 10.1.1.4/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 80
14: ipv4 permit+reflect src 10.1.1.4/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 5000
15: ipv4 permit+reflect src 10.1.1.5/32 dst 10.1.1.254/32 proto 6 sport 0-65535 dport 51821
16: ipv4 permit+reflect src 10.1.1.5/32 dst 10.1.1.0/24 proto 1 sport 0-65535 dport 0-65535
17: ipv4 permit+reflect src 10.1.1.5/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 22
18: ipv4 permit+reflect src 10.1.1.5/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 443
19: ipv4 permit+reflect src 10.1.1.5/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 80
20: ipv4 permit+reflect src 10.1.1.2/32 dst 10.1.1.254/32 proto 6 sport 0-65535 dport 51821
21: ipv4 permit+reflect src 10.1.1.2/32 dst 10.1.1.0/24 proto 1 sport 0-65535 dport 0-65535
22: ipv4 permit+reflect src 10.1.1.2/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 22
23: ipv4 permit+reflect src 10.1.1.2/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 443
24: ipv4 permit+reflect src 10.1.1.2/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 80
25: ipv4 permit+reflect src 10.1.1.2/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 5432
26: ipv4 permit+reflect src 10.1.1.2/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 5000
27: ipv4 permit+reflect src 10.1.1.2/32 dst 10.1.1.254/32 proto 6 sport 0-65535 dport 8080
  applied inbound on sw_if_index: 5
  applied outbound on sw_if_index:
  used in lookup context index: 5
acl-index 3 count 1 tag {mightyconnect-acl-egress-wg0}
0: ipv4 permit+reflect src 0.0.0.0/0 dst 0.0.0.0/0 proto 0 sport 0-65535 dport 0-65535
  applied outbound on sw_if_index: 5
```

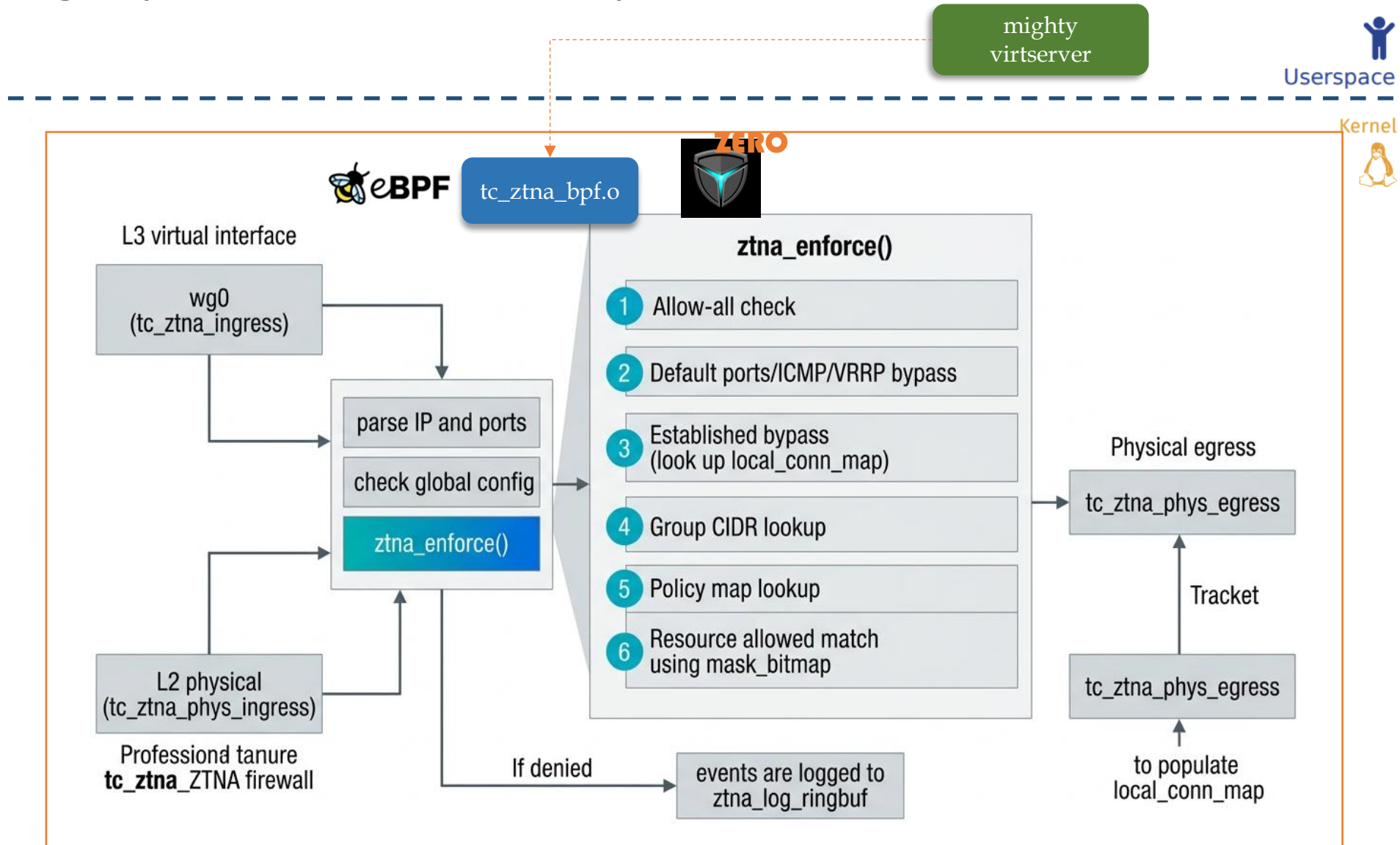
## 4. MightyConnect Security Box(5) – VPP ZTNA(2)



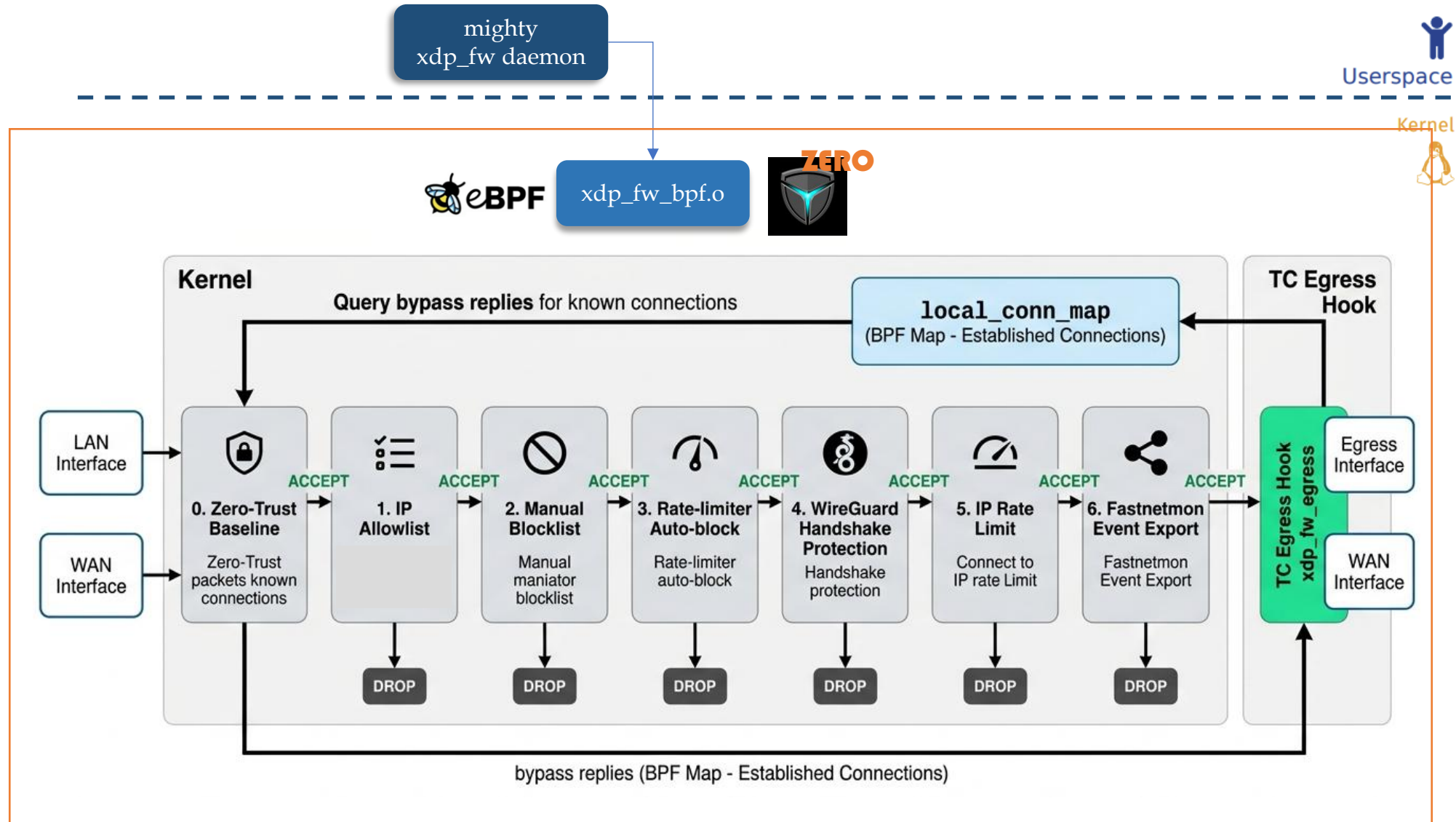
## 4. MightyConnect Security Box(6) – Kernel ZTNA(1)



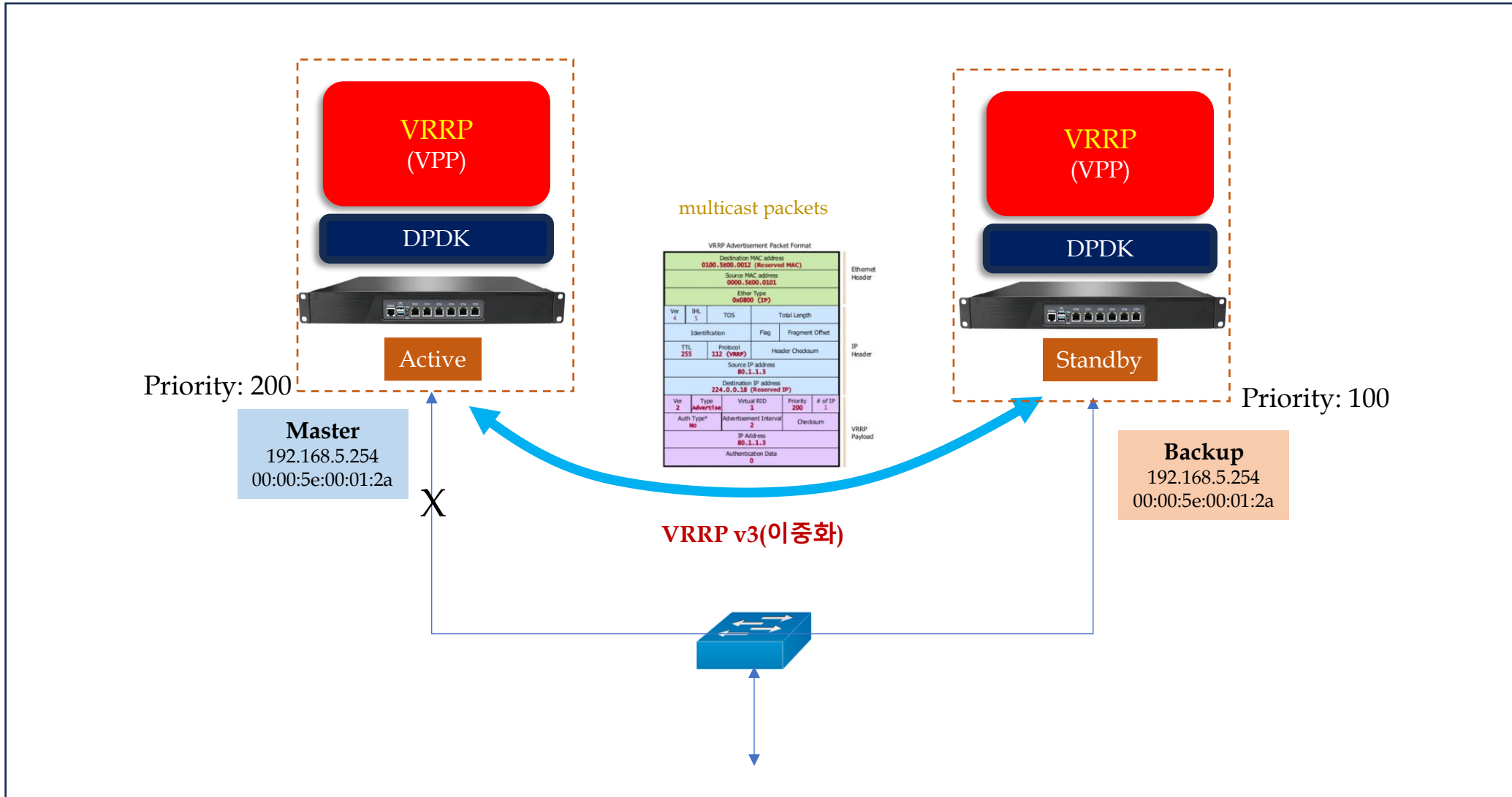
## 4. MightyConnect Security Box(6) – Kernel ZTNA(2)



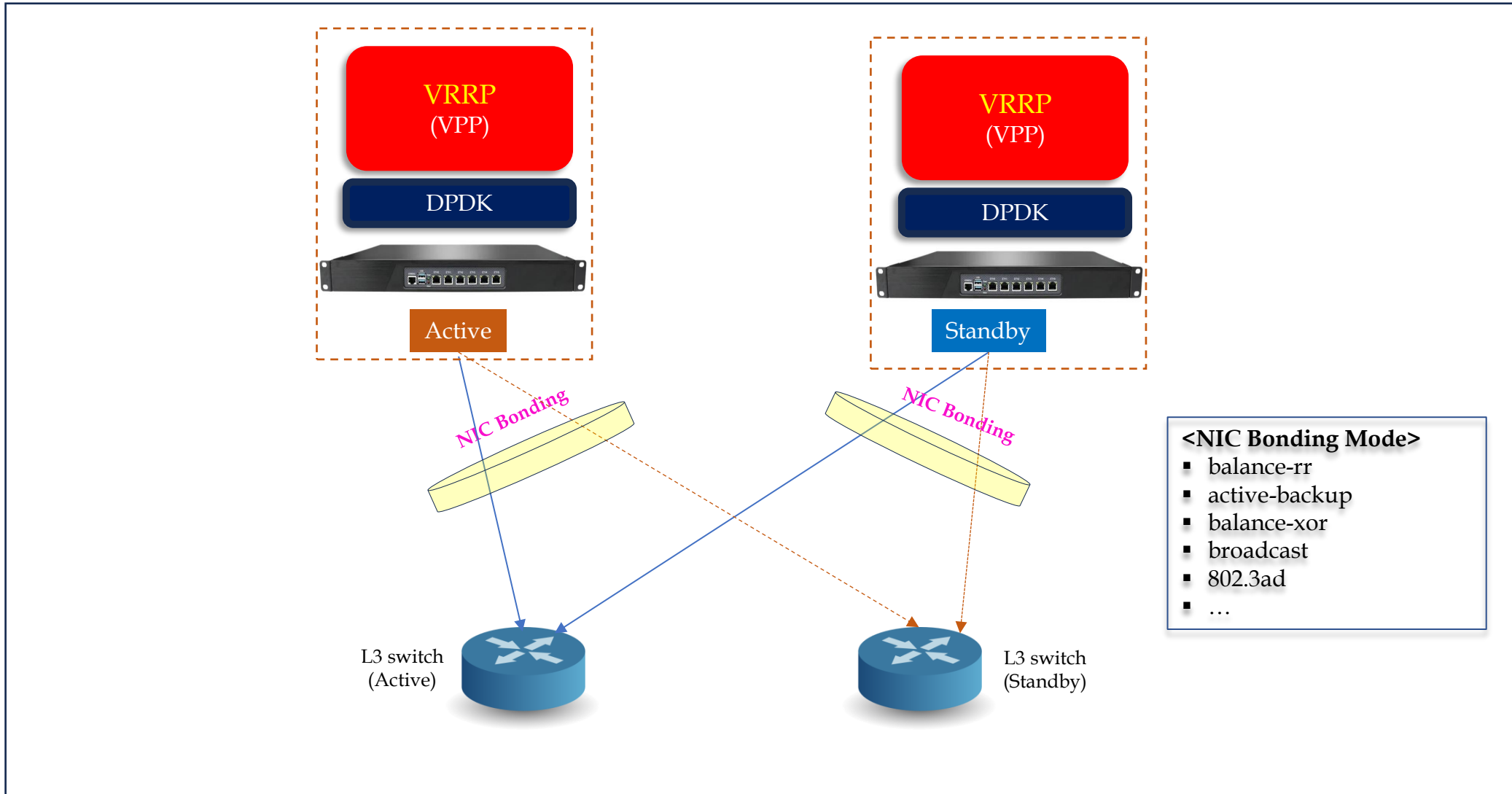
## 4. MightyConnect Security Box(6) – Kernel ZTNA(3)



## 4. MightyConnect Security Box(7) – Redundancy(1)

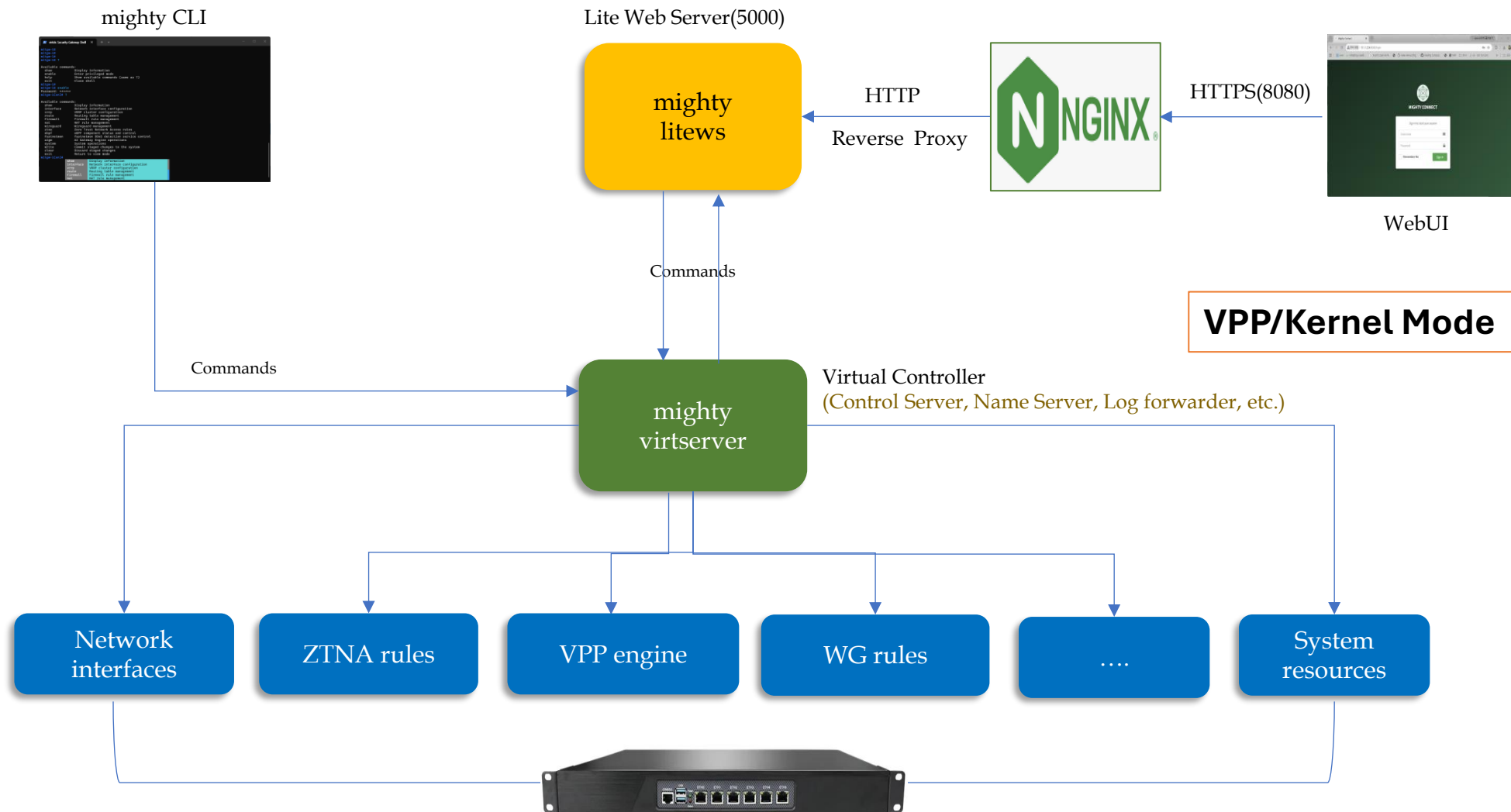


## 4. MightyConnect Security Box(7) – Redundancy(2)

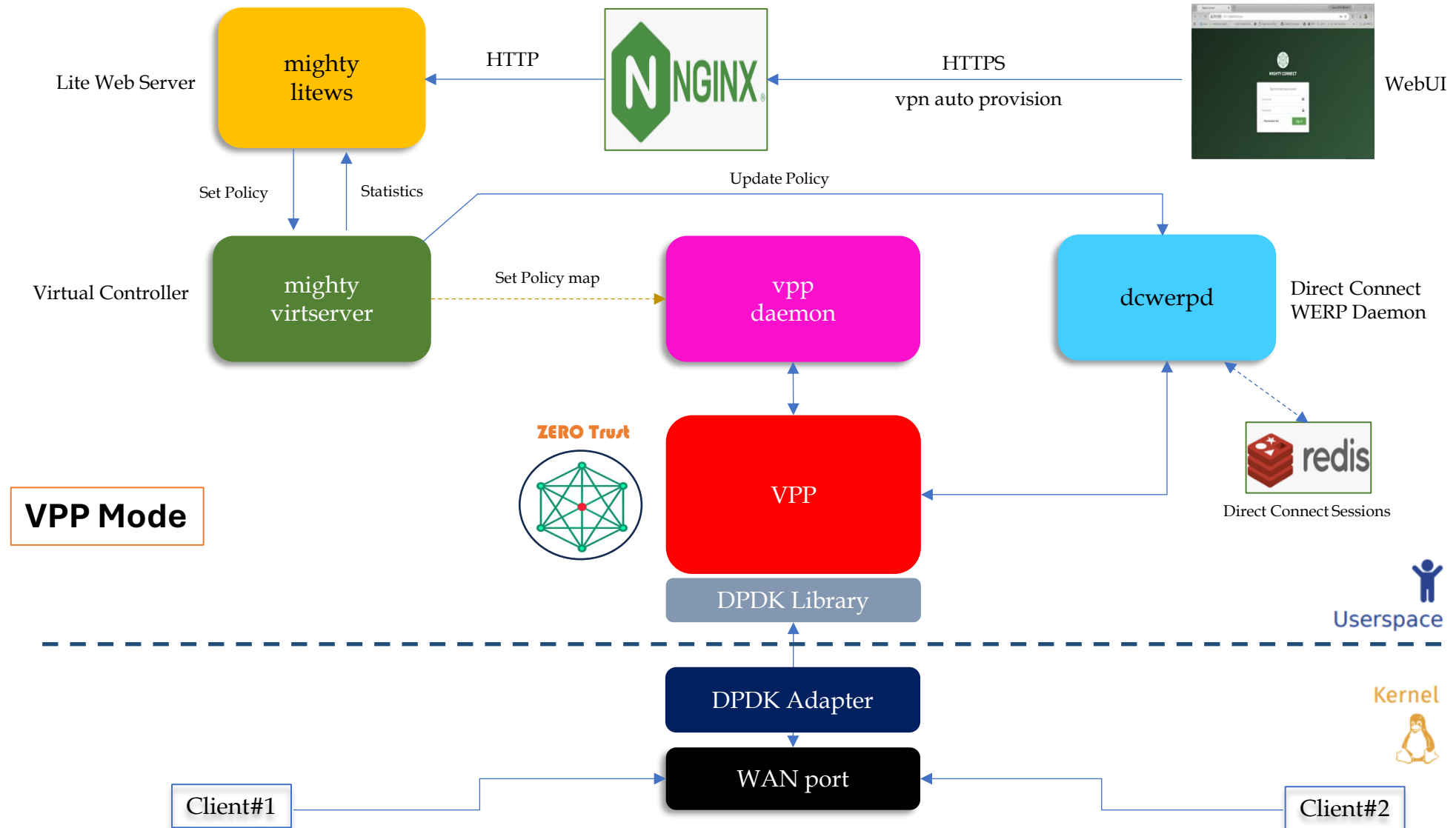




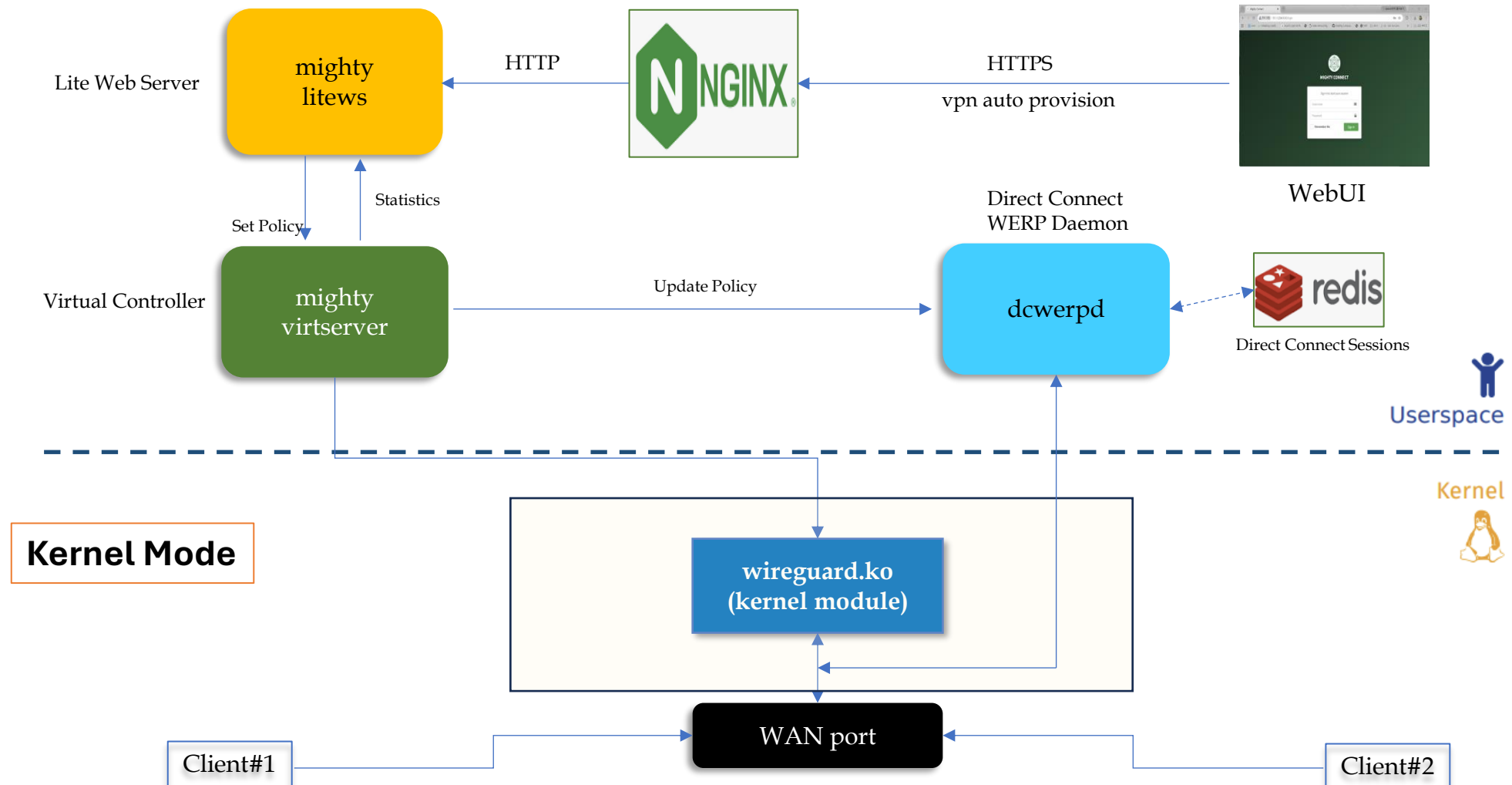
## 4. MightyConnect Security Box(8) – Mighty Engine(1)



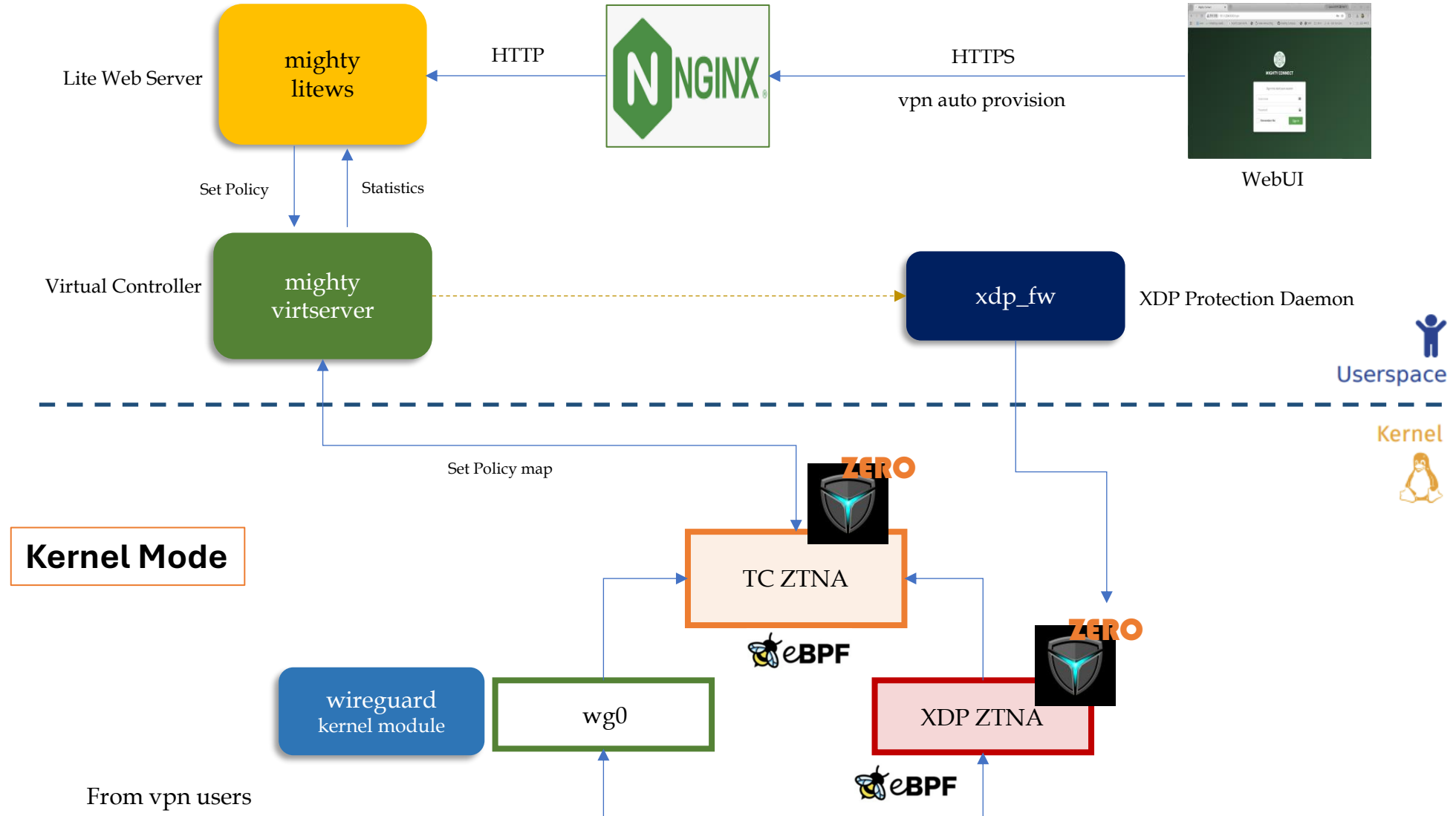
## 4. MightyConnect Security Box(8) – Mighty Engine(2)



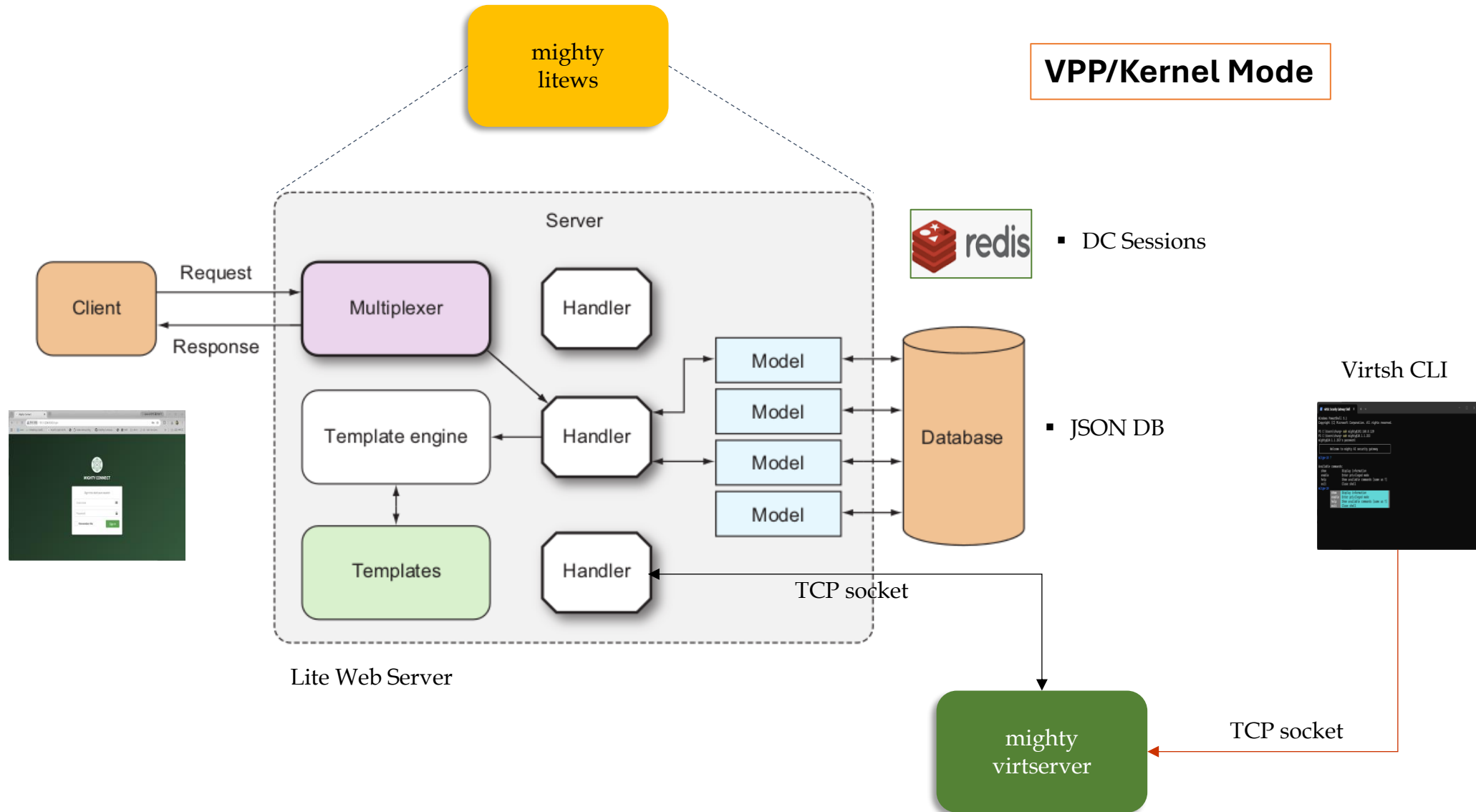
## 4. MightyConnect Security Box(8) – Mighty Engine(3-1)



## 4. MightyConnect Security Box(8) – Mighty Engine(3-2)

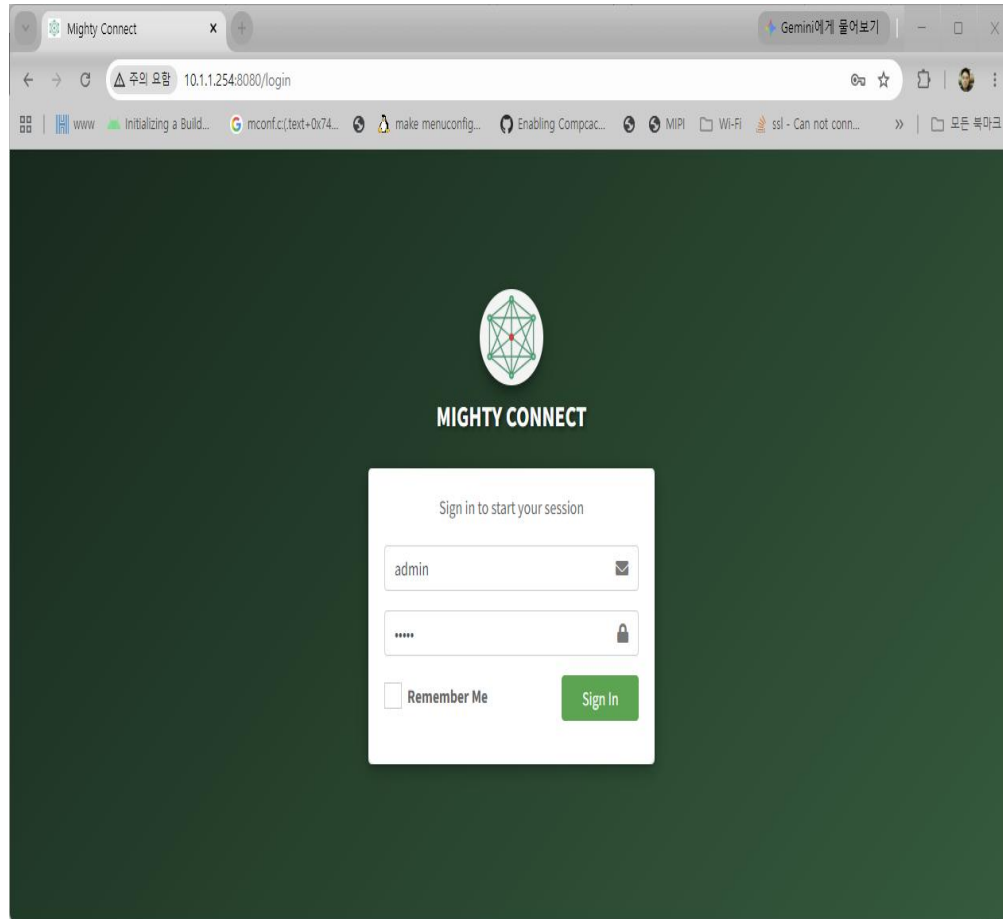


## 4. MightyConnect Security Box(8) – Mighty Engine(4)

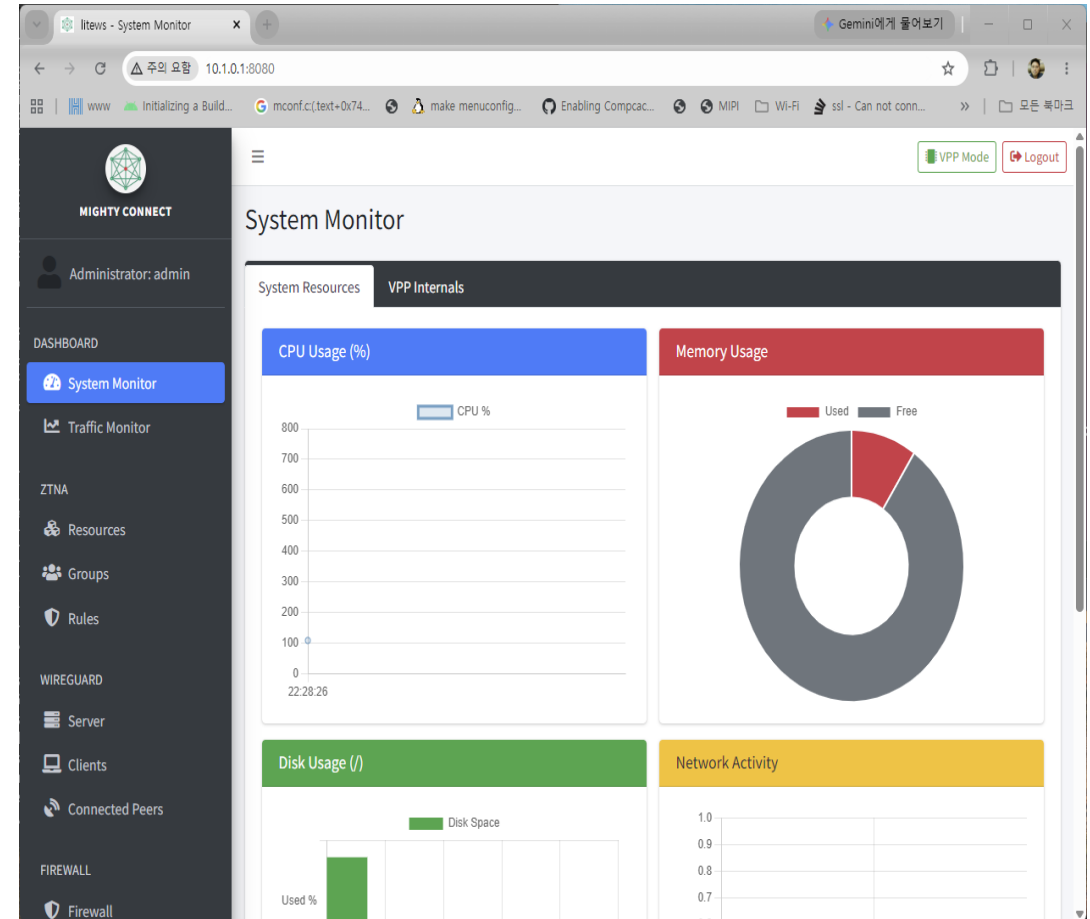


## 4. MightyConnect Security Box(9) – WebUI(1)

### ▪ WebUI Login Page



### ▪ Dashboard Page



## 4. MightyConnect Security Box(9) – WebUI(2)

### ▪ ZTNA VPN Groups Page

The screenshot shows the 'VPN Groups' page in the MightyConnect WebUI. The left sidebar contains navigation links: DASHBOARD, System Monitor, Traffic Monitor, ZTNA, Resources, Groups (selected), Rules, WIREGUARD, Server, Clients, Connected Peers, FIREWALL, and Firewall. The main content area has a 'VPN Groups' header with a 'VPP Mode' indicator and a 'Logout' button. Below the header is a form to 'Add New Group' with fields for 'Group Name' (e.g., Sales Team) and 'Expected Max Members' (e.g., 50), and a checkbox for 'Start with full internal mesh'. A table below lists existing groups:

| Name   | Dedicated Subnet                       | Members                                                                              | Created             | Removal  |
|--------|----------------------------------------|--------------------------------------------------------------------------------------|---------------------|----------|
| groupC | 10.1.33.0/25 (10.1.33.1 ~ 10.1.33.126) | 0 / 100                                                                              | 2026-08-11 15:10:08 | [Remove] |
| groupB | 10.1.34.0/23 (10.1.34.1 ~ 10.1.35.254) | 1 / 500<br>SN-2026-Windows11                                                         | 2026-08-11 14:27:25 | [Remove] |
| groupA | 10.1.32.0/24 (10.1.32.1 ~ 10.1.32.254) | 4 / 200<br>SN-2026-Windows11, SN-2026-EmbeddedLinux, SN-2026-Android, SN-2026-GI-Net | 2026-08-11 14:25:19 | [Remove] |

### ▪ ZTNA Access Rules Page

The screenshot shows the 'Access Rules' page in the MightyConnect WebUI. The left sidebar is identical to the previous page, with 'Rules' selected. The main content area has an 'Access Rules' header with a 'VPP Mode' indicator and a 'Logout' button. Below the header is a 'ZTNA Engine Status' section showing 'Bound (ACL Index 2 on wg0)'. A table displays engine statistics:

| Interface | Active Policies | Resource Rules |
|-----------|-----------------|----------------|
| wg0       | 4               | 7              |

Below this, a summary shows 'Allowed: 41' and 'Denied: 471'. A note states: 'Allowed/Denied are system-wide VPP ACL packet counts (every ACL-bound interface), not ZTNA-only.' The main section is 'Access Rules (Total: 4)' with '+ Add New Access Rule' and '+ Grant Resource Access' buttons. A table lists the rules:

| Source        | Destination   | Protocol      | Port                        | Enabled | Expires | Actions         |
|---------------|---------------|---------------|-----------------------------|---------|---------|-----------------|
| Group: groupC | Group: groupB | tcp           | 443,5000                    | [On]    | —       | [Edit] [Remove] |
| Group: groupC | Group: groupC | tcp           | 5432                        | [On]    | —       | [Edit] [Remove] |
| Group: groupB | Group: groupB | all           | any                         | [On]    | —       | [Edit] [Remove] |
| Group: groupA | Group: groupA | udp, tcp, tcp | 6000,7999, 22,80,5000, 8080 | [On]    | —       | [Edit] [Remove] |

## 4. MightyConnect Security Box(9) – WebUI(3)

### ■ WireGuard Server Page

The screenshot shows the 'WireGuard Server Settings' page. The left sidebar contains a navigation menu with 'Server' selected. The main content area has tabs for 'Interface/Keypair' and 'Settings'. The 'Interface' tab is active, showing fields for 'Server Interface Addresses' (10.1.0.1/16), 'Listen Port' (51820), and script fields for 'Post Up Script', 'Pre Down Script', and 'Post Down Script'. The 'Key Pair' tab is also visible, showing 'Private Key' and 'Public Key' fields with a 'Generate' button.

### ■ WireGuard Client Auto Provision Page

The screenshot shows the 'WireGuard Client Auto Provision' page. The left sidebar contains a navigation menu with 'Clients' selected. The main content area has a section for 'Enable ZTNA Access Control' with a toggle switch and a text input field. Below this is a table titled 'Authorized Devices' with columns for Device ID, Comment, Status, 2FA, Created At, and Action. The table lists several devices, including 'SN-2026-NanoP15', 'SN-2026-Android', 'SN-2026-G1-iNet', 'SN-2026-EmbeddedLinux', 'SN-2026-Windows11', 'SN-2026-X1022', and 'SN-2026-X1021'. Each device has a 'Registered' status and a 'Disabled' 2FA status. The 'Action' column contains a trash icon for each device.

| Device ID             | Comment                 | Status     | 2FA      | Created At       | Action |
|-----------------------|-------------------------|------------|----------|------------------|--------|
| SN-2026-NanoP15       | wireguard linux nanopi  | Registered | Disabled | 2026-08-06 11:28 |        |
| SN-2026-Android       | wireguard android       | Registered | Disabled | 2026-08-03 09:18 |        |
| SN-2026-G1-iNet       | wireguard linux gl-inet | Registered | Disabled | 2026-08-03 09:17 |        |
| SN-2026-EmbeddedLinux | wireguard linux         | Registered | Disabled | 2026-08-03 09:16 |        |
| SN-2026-Windows11     | wireguard windows       | Registered | Disabled | 2026-08-03 09:16 |        |
| SN-2026-X1022         | Bulk Imported           | Pending    | Disabled | 2026-08-03 09:00 |        |
| SN-2026-X1021         | Bulk Imported           | Pending    | Disabled | 2026-08-03 09:00 |        |



## 4. MightyConnect Security Box(9) – WebUI(4)

### ■ WireGuard Connected Peers Page

The screenshot shows the 'Connected Peers Status' page. The left sidebar contains the 'MIGHTY CONNECT' logo and a navigation menu with items: Administrator: admin, DASHBOARD, System Monitor, Traffic Monitor, ZTNA, Resources, Rules, WIREGUARD, Server, Clients, and Connected Peers (highlighted). The main content area has a 'Connected Peers Status' header with a refresh icon and 'Auto-refreshing every 10s'. Below this are tabs for 'Peer Status', 'Direct Connect Status', and 'Map View'. A blue box indicates 'Device: wg0'. A table lists connected peers with columns: #, Name, Allocated IPs, OS, Endpoint, Location, Received, Transmitted, and Status.

| # | Name              | Allocated IPs | OS                       | Endpoint            | Location         | Received  | Transmitted | Status    |
|---|-------------------|---------------|--------------------------|---------------------|------------------|-----------|-------------|-----------|
| 0 | SN-2026-GI-iNet   | 10.1.1.4/32   | OpenWrt 21.02-SNAPSHOT   | 192.168.8.239:51820 | Internal/Private | 794.93 KB | 662.31 KB   | Connected |
| 1 | SN-2026-Windows11 | 10.1.1.2/32   | Windows 11 (Build 26200) | 192.168.8.103:59719 | Internal/Private | 1.72 MB   | 20.18 MB    | Connected |

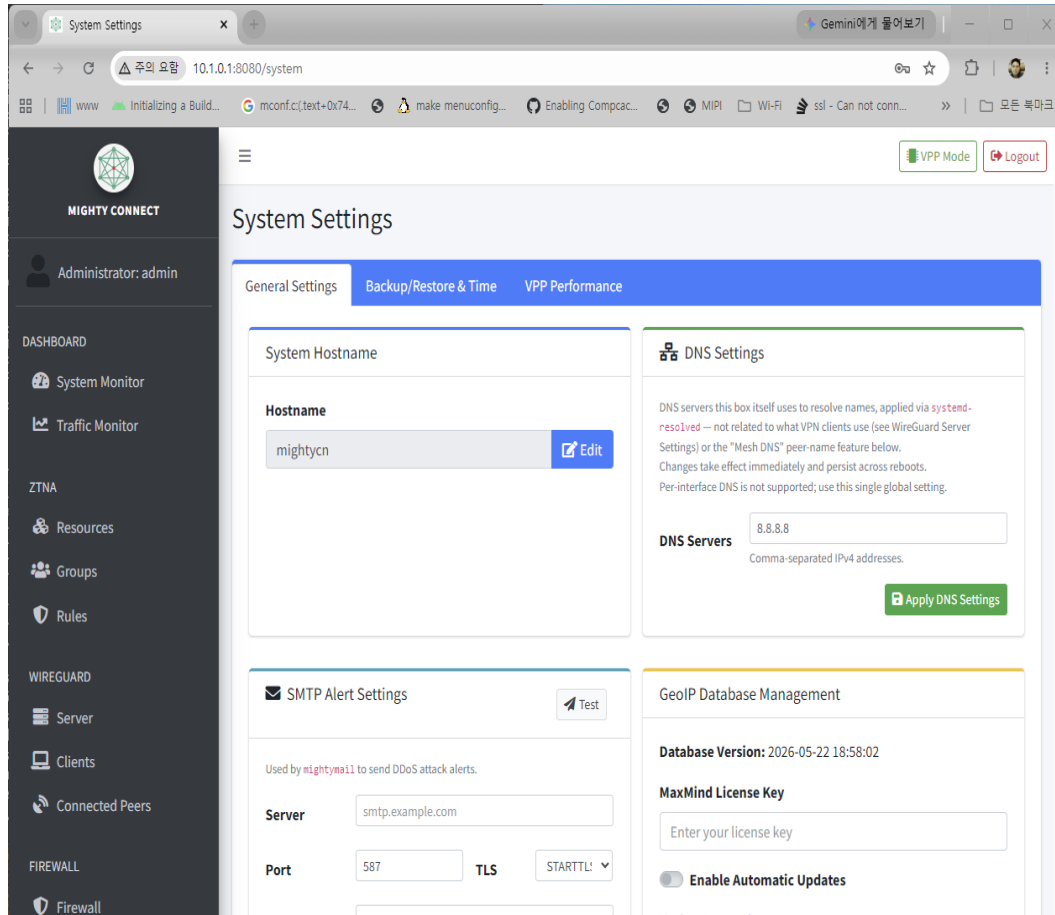
### ■ Network Interfaces Page

The screenshot shows the 'Network Interfaces' page. The left sidebar is identical to the previous page. The main content area has a 'Network Interfaces' header with a refresh icon. Below this are tabs for 'Interfaces' and 'VRRP'. A blue box indicates 'Ethernet Interfaces' with buttons for 'Restart Interfaces' and 'Create Bonding'. A table lists network interfaces with columns: Interface, Status, IP Address, Netmask, Role, DPDK, Link Info, and Action.

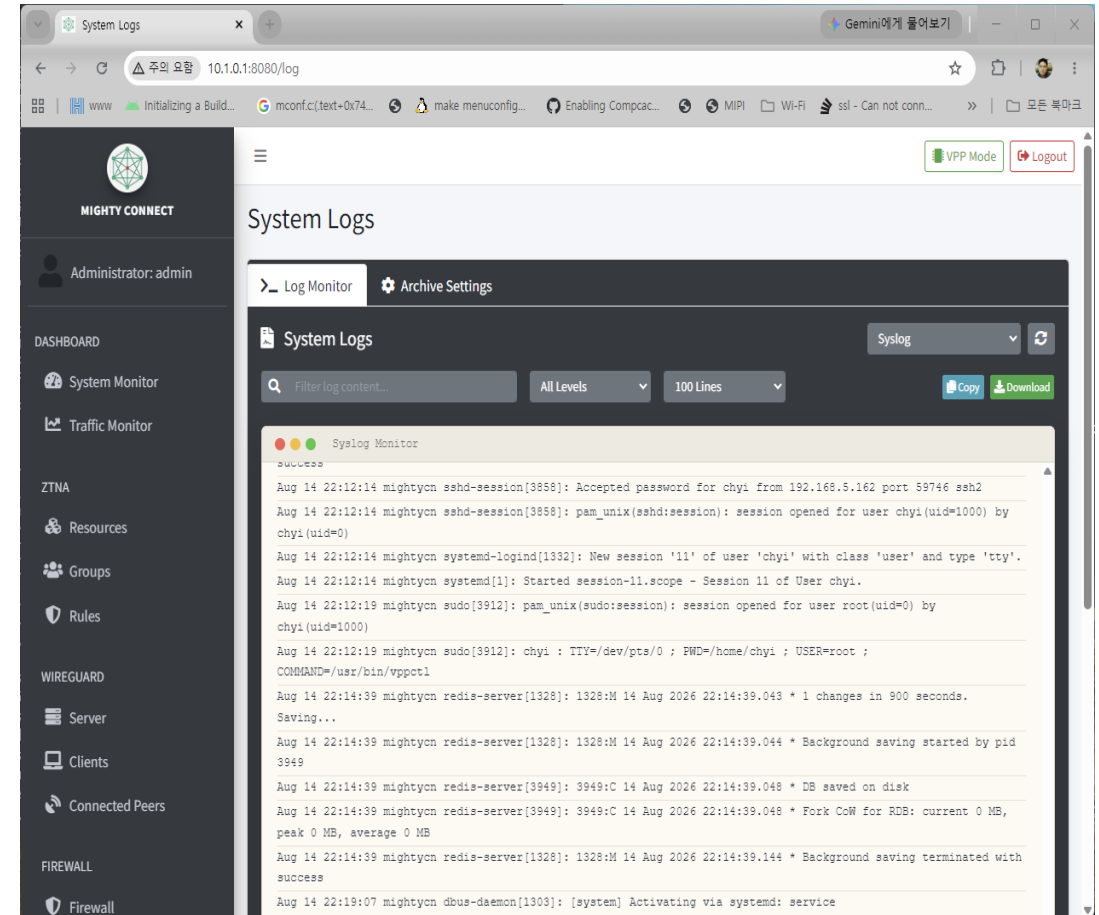
| Interface | Status | IP Address    | Netmask       | Role | DPDK | Link Info                                                             | Action |
|-----------|--------|---------------|---------------|------|------|-----------------------------------------------------------------------|--------|
| enp2s0    | UP     | 192.168.8.114 | 255.255.255.0 | LAN  | DPDK | Link 1000Mb/s Full                                                    | Edit   |
| enp6s0    | UP     | 192.168.10.1  | 255.255.255.0 | LAN2 | DPDK | —                                                                     | Edit   |
| enp1s0    | UP     | 192.168.5.1   | 255.255.255.0 | MGMT | —    | Link 2500Mb/s Full<br>Port: TP<br>Max: 2500baseT/Full<br>Auto-Neg: On | Edit   |
| enp3s0    | DOWN   | —             | —             | —    | —    | No Link<br>Port: TP<br>Max: 2500baseT/Full<br>Auto-Neg: On            | Edit   |
| enp4s0    | DOWN   | —             | —             | —    | —    | No Link<br>Port: TP<br>Max: 2500baseT/Full<br>Auto-Neg: On            | Edit   |

## 4. MightyConnect Security Box(9) – WebUI(5)

### System Settings Page



### Logs Page



## 4. MightyConnect Security Box(10) – CLI

```
PS C:\Users\chung> ssh mighty@10.1.1.254
mighty@10.1.1.254's password:

Welcome to mighty AI security gateway

mightycn# ?

Available commands:
show          Display information
enable       Enter privileged mode
help         Show available commands (same as ?)
exit         Close shell

mightycn# show interface
Interface Status IP Address DDPK
-----
enp2s0 UP 192.168.8.114/255.255.255.0 yes
enp6s0 UP 192.168.10.1/255.255.255.0 yes
tap4096 UP - yes
tap4097 UP - yes
tun4098 UP - yes
wg0 UP 10.1.1.254/255.255.255.0 yes
enp1s0 UP 192.168.5.1/255.255.255.0 -
enp3s0 DOWN -
enp4s0 DOWN -
enp5s0 DOWN 192.168.20.1/255.255.255.0 -

VPP interfaces (show interface):
Name Idx State MTU (L3/IP4/IP6/MPLS) Counter Count
enp2s0 1 up 9000/0/0/0 rx packets 83438
rx bytes 31067596
tx packets 96924
tx bytes 69994133
drops 1197
punt 2101
ip4 82166
ip6 286
tx-error 2
tx-error 1
drops 4
rx packets 547
rx bytes 163088
tx packets 862
tx bytes 191013
drops 16
ip4 459
ip6 17
enp6s0 2 up 9000/0/0/0 rx packets 41921
rx bytes 40105173
tx packets 27211
tx bytes 2196120
ip4 41903
ip6 18
tap4096 3 up 9000/0/0/0 rx packets 79471
tap4097 4 up 9000/0/0/0
tun4098 6 up 1420/0/0/0
wg0 5 up 1420/0/0/0
```

```
mightycn# show ztna
Network ZTNA Rules (L4 Access Control):
Client Name (Label) Client IP Authorized Resources Status
-----
SN-2026-Android 10.1.1.3/32 default-ping (icmp/10.1.1.0/24/any), ssh (tcp/10.1.1.0/24/22), https (tcp/10.1.1.0/24/443) Enabled
SN-2026-EmbeddedLinux 10.1.1.1/32 default-ping (icmp/10.1.1.0/24/any), ssh (tcp/10.1.1.0/24/22), https (tcp/10.1.1.0/24/443), http (tcp/10.1.1.0/24/80) Enabled
SN-2026-Gl-iNet 10.1.1.4/32 default-ping (icmp/10.1.1.0/24/any), ssh (tcp/10.1.1.0/24/22), https (tcp/10.1.1.0/24/443), http (tcp/10.1.1.0/24/80), litews (tcp/10.1.1.0/24/5000) Enabled
SN-2026-NanoPi5 10.1.1.5/32 default-ping (icmp/10.1.1.0/24/any), ssh (tcp/10.1.1.0/24/22), https (tcp/10.1.1.0/24/443), http (tcp/10.1.1.0/24/80) Enabled
SN-2026-Windows11 10.1.1.2/32 default-ping (icmp/10.1.1.0/24/any), ssh (tcp/10.1.1.0/24/22), https (tcp/10.1.1.0/24/443), http (tcp/10.1.1.0/24/80), postgresql (tcp/10.1.1.0/24/5432), litews (tcp/10.1.1.0/24/5000), boxlitews (tcp/10.1.1.254/32/8080) Enabled

Always Allowed (all clients, bypasses policy_map entirely): DNS tcp/53, DNS udp/53, ICMP

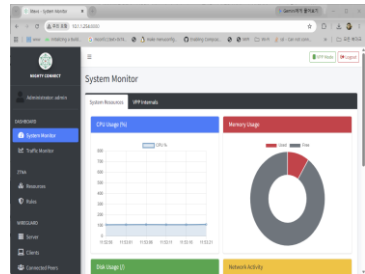
VPP ACL state (show acl-plugin acl):
acl-index 0 count 9 tag {mightyconnect-acl-builtin-phys}
0: ipv4 permit+reflect src 192.168.5.0/24 dst 0.0.0.0/0 proto 0 sport 0-65535 dport 0-65535
1: ipv4 permit src 0.0.0.0/0 dst 0.0.0.0/0 proto 17 sport 0-65535 dport 51820
2: ipv4 permit src 0.0.0.0/0 dst 0.0.0.0/0 proto 6 sport 0-65535 dport 53
3: ipv4 permit src 0.0.0.0/0 dst 0.0.0.0/0 proto 6 sport 53 dport 0-65535
4: ipv4 permit src 0.0.0.0/0 dst 0.0.0.0/0 proto 17 sport 0-65535 dport 53
5: ipv4 permit src 0.0.0.0/0 dst 0.0.0.0/0 proto 17 sport 53 dport 0-65535
6: ipv4 permit src 0.0.0.0/0 dst 0.0.0.0/0 proto 6 sport 0-65535 dport 51820
7: ipv4 permit src 0.0.0.0/0 dst 0.0.0.0/0 proto 1 sport 0-65535 dport 0-65535
8: ipv4 permit src 0.0.0.0/0 dst 224.0.0.18/32 proto 112 sport 0-65535 dport 0-65535
applied inbound on sw_if_index: 1, 2
applied outbound on sw_if_index:
used in lookup context index: 1, 3
acl-index 1 count 1 tag {mightyconnect-acl-egress-phys}
0: ipv4 permit+reflect src 0.0.0.0/0 dst 0.0.0.0/0 proto 0 sport 0-65535 dport 0-65535
applied inbound on sw_if_index:
applied outbound on sw_if_index: 1, 2
used in lookup context index: 0, 2
acl-index 2 count 28 tag {mightyconnect-acl-ztna-wg0}
0: ipv4 permit+reflect src 10.1.1.3/32 dst 10.1.1.254/32 proto 6 sport 0-65535 dport 51821
1: ipv4 permit+reflect src 10.1.1.3/32 dst 10.1.1.0/24 proto 1 sport 0-65535 dport 0-65535
2: ipv4 permit+reflect src 10.1.1.3/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 22
3: ipv4 permit+reflect src 10.1.1.3/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 443
4: ipv4 permit+reflect src 10.1.1.1/32 dst 10.1.1.254/32 proto 6 sport 0-65535 dport 51821
5: ipv4 permit+reflect src 10.1.1.1/32 dst 10.1.1.0/24 proto 1 sport 0-65535 dport 0-65535
6: ipv4 permit+reflect src 10.1.1.1/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 22
7: ipv4 permit+reflect src 10.1.1.1/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 443
8: ipv4 permit+reflect src 10.1.1.1/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 80
9: ipv4 permit+reflect src 10.1.1.4/32 dst 10.1.1.254/32 proto 6 sport 0-65535 dport 51821
10: ipv4 permit+reflect src 10.1.1.4/32 dst 10.1.1.0/24 proto 1 sport 0-65535 dport 0-65535
11: ipv4 permit+reflect src 10.1.1.4/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 22
12: ipv4 permit+reflect src 10.1.1.4/32 dst 10.1.1.0/24 proto 6 sport 0-65535 dport 443

--More--
```

## 5. MightyConnect Devices

**MightyConnect Devices**

# 5. MightyConnect Devices



MightyConnect™ Box



MightyConnect™ Mini



MightyConnect™ Windows



MightyConnect™ Ubuntu



MightyConnect™ macOS/iOS



MightyConnect™ Android

## 5.1 MightyConnect Mini

### MightyConnect Mini Security Gateway



GL-iNet Series  
<https://www.gl-inet.com/>

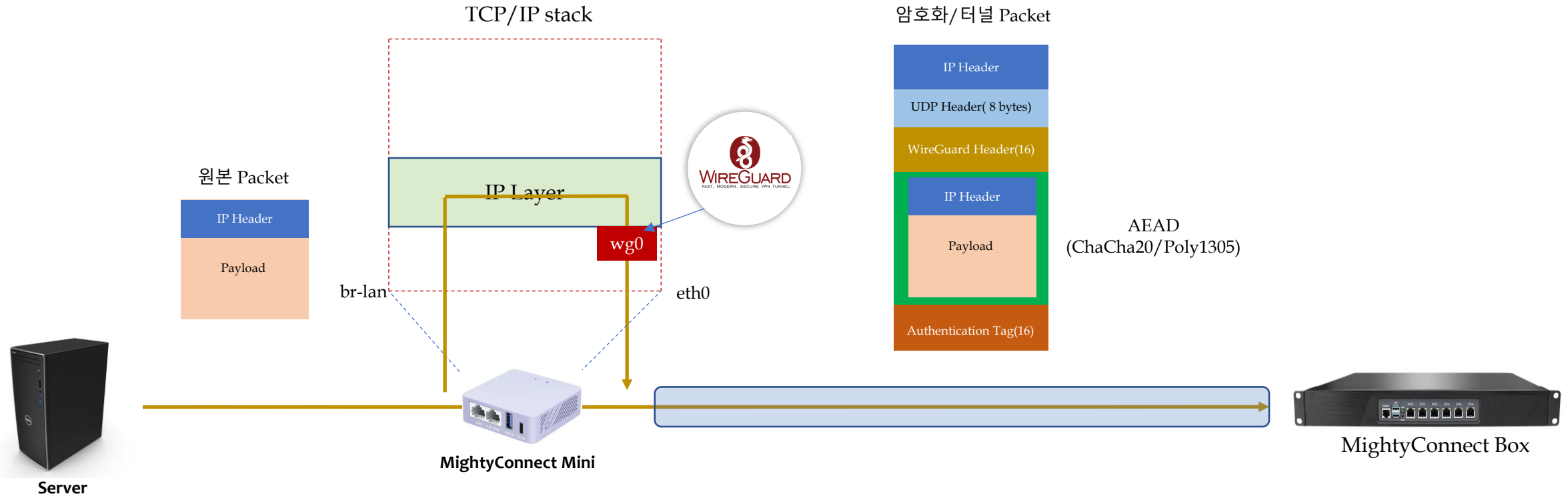


NanoPi Series  
<https://www.friendlyelec.com/>

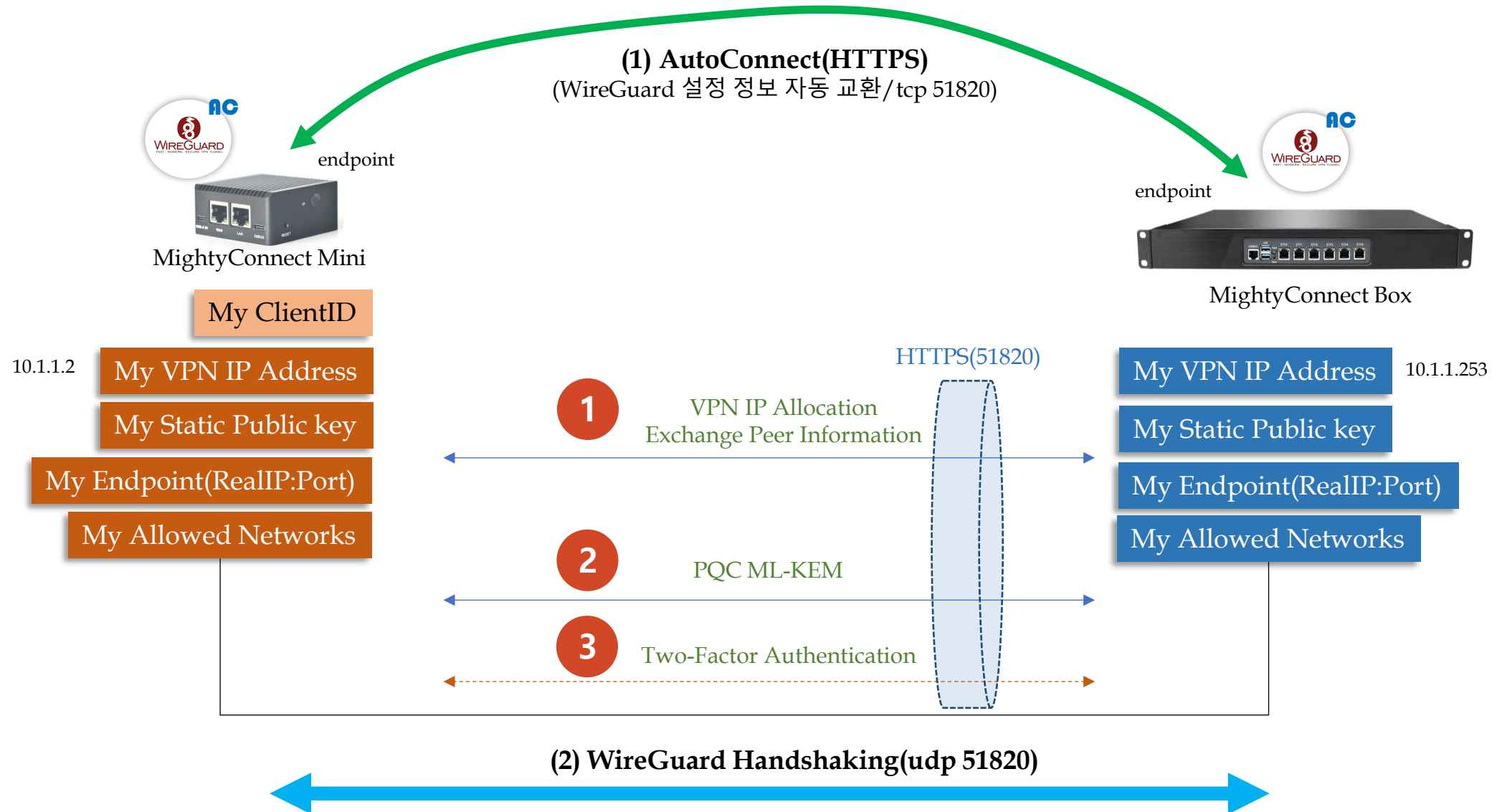
# 5.1 MightyConnect Mini(1) – Overview

## CryptoKey Routing

ID: Public Key::VPN IP

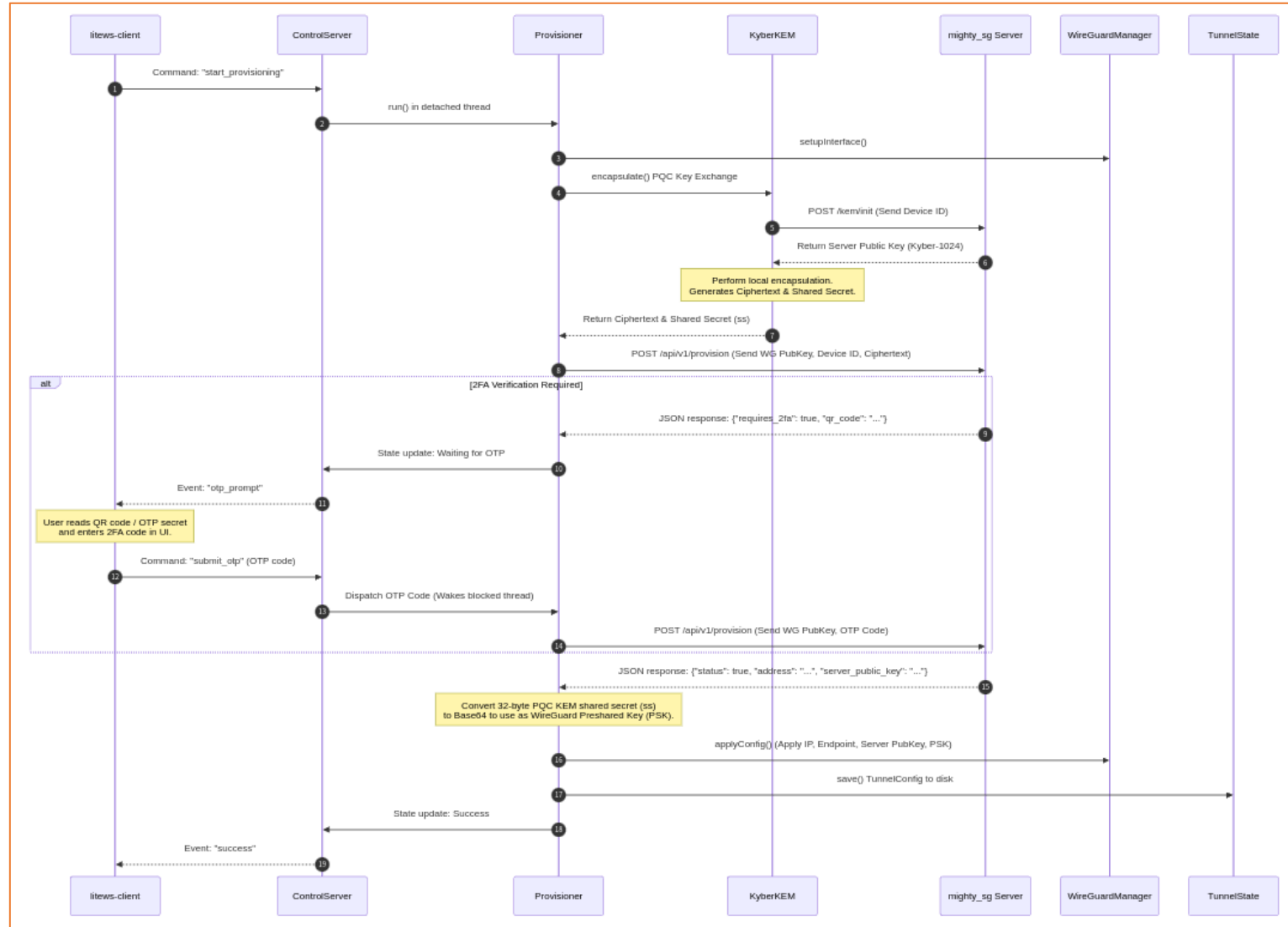


# 5.1 MightyConnect Mini(2) – Auto Connection



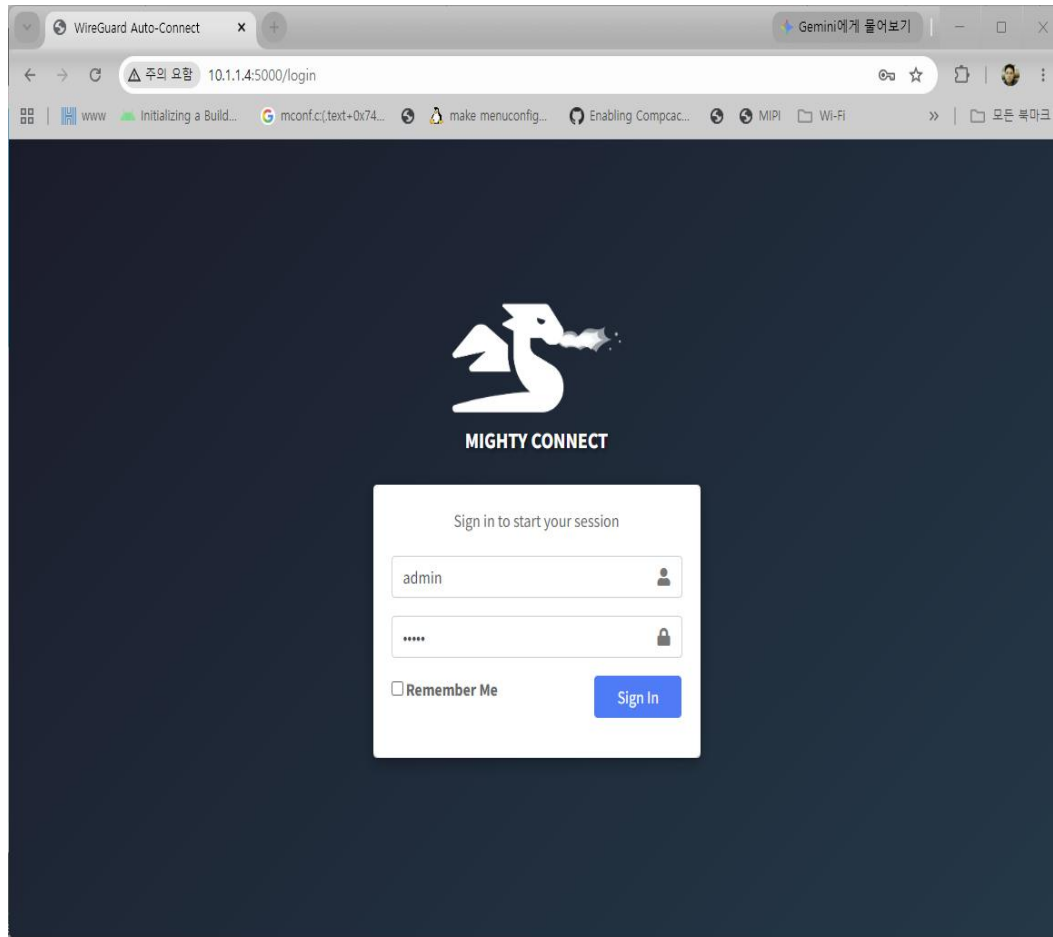


## 5.1 MightyConnect Mini(3) – Architecture



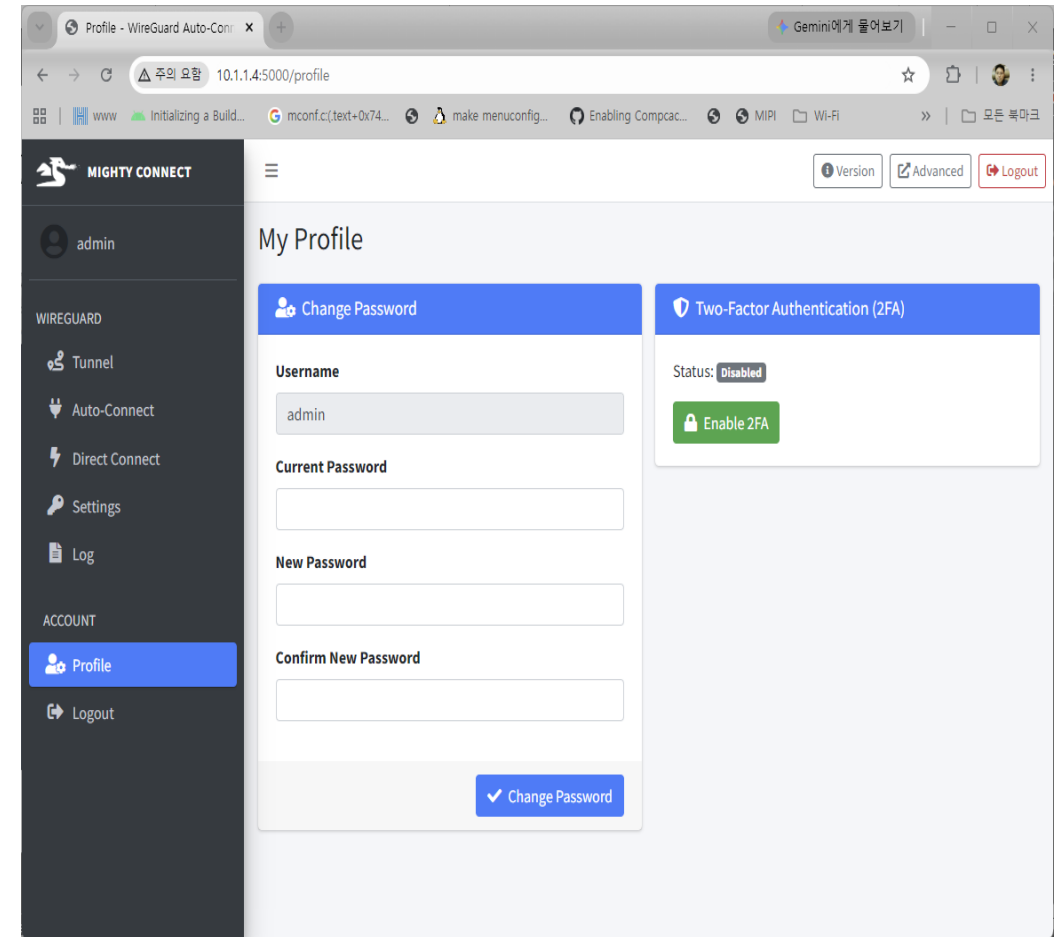
## 5.1 MightyConnect Mini(4) – WebUI(1)

### ■ Login Page



The screenshot shows the login page of the MightyConnect Mini WebUI. The browser address bar displays '10.1.14:5000/login'. The page features a dark blue background with the 'MIGHTY CONNECT' logo at the top center. Below the logo is a white login form with the text 'Sign in to start your session'. The form contains a username field with 'admin' entered, a password field with masked characters '\*\*\*\*\*', a 'Remember Me' checkbox, and a blue 'Sign In' button.

### ■ Admin User Page



The screenshot shows the admin user profile page of the MightyConnect Mini WebUI. The browser address bar displays '10.1.14:5000/profile'. The page has a dark sidebar on the left with the 'MIGHTY CONNECT' logo and a menu including 'Tunnel', 'Auto-Connect', 'Direct Connect', 'Settings', 'Log', 'ACCOUNT', 'Profile' (highlighted), and 'Logout'. The main content area is light blue and titled 'My Profile'. It includes a 'Change Password' section with fields for 'Username' (admin), 'Current Password', 'New Password', and 'Confirm New Password', along with a 'Change Password' button. To the right is a 'Two-Factor Authentication (2FA)' section showing a 'Status: Disabled' and an 'Enable 2FA' button. At the top right of the main area are links for 'Version', 'Advanced', and 'Logout'.

# 5.1 MightyConnect Mini(4) – WebUI(2)

## ▪ Tunnel Status Page

The screenshot shows the 'Tunnel Status' page of the WG Auto-Connect web interface. The left sidebar contains a menu with 'Tunnel' selected. The main content area displays the status of the 'wg-auto0' interface. A 'Restart Daemon' button is visible in the top right corner of the status section.

| Tunnel Status                                      |                                             |
|----------------------------------------------------|---------------------------------------------|
| interface: wg-auto0                                |                                             |
| daemon status                                      | running                                     |
| public key                                         | wac8svbrogFLtgZ1H3IXZ0qXLxA07XwyfKI+KGqJlc= |
| private key                                        | (hidden)                                    |
| listening port                                     | 51820                                       |
| assigned ip                                        | 10.1.1.12/24                                |
| peer: 93Pq7SGTfy7o+8P1SXaf40J4wJ8Kq19o5CPRc8jJbg4= |                                             |
| preshared key                                      | (hidden)                                    |
| endpoint                                           | 192.168.8.129:51820                         |
| allowed ips                                        | 10.1.1.0/24                                 |
| latest handshake                                   | 46 seconds ago                              |
| transfer                                           | 2.1 MiB received, 2.11 MiB sent             |
| persistent keepalive                               | every 15 seconds                            |

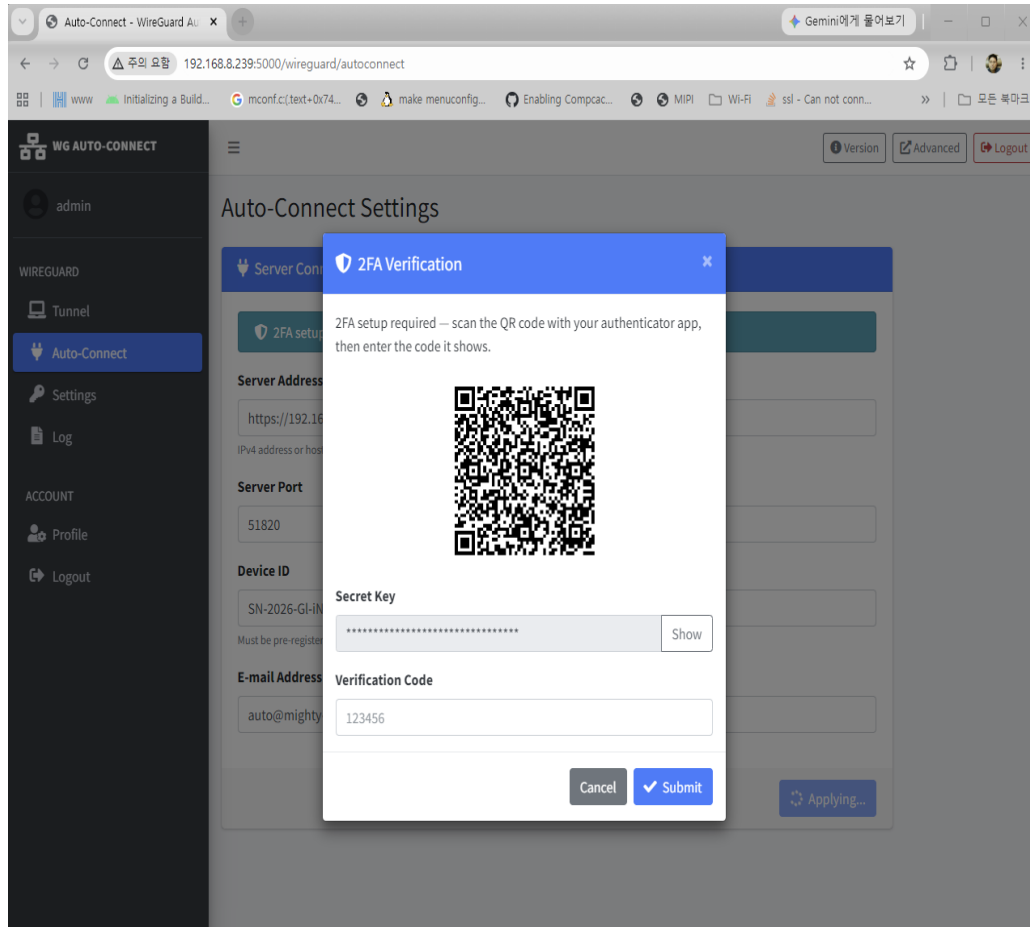
## ▪ Auto Connection Page

The screenshot shows the 'Auto-Connect Settings' page of the WG Auto-Connect web interface. The left sidebar contains a menu with 'Auto-Connect' selected. The main content area displays the configuration for the auto-connect server. A green 'Connected.' status bar is visible at the top of the settings section. The 'Apply' button is at the bottom right.

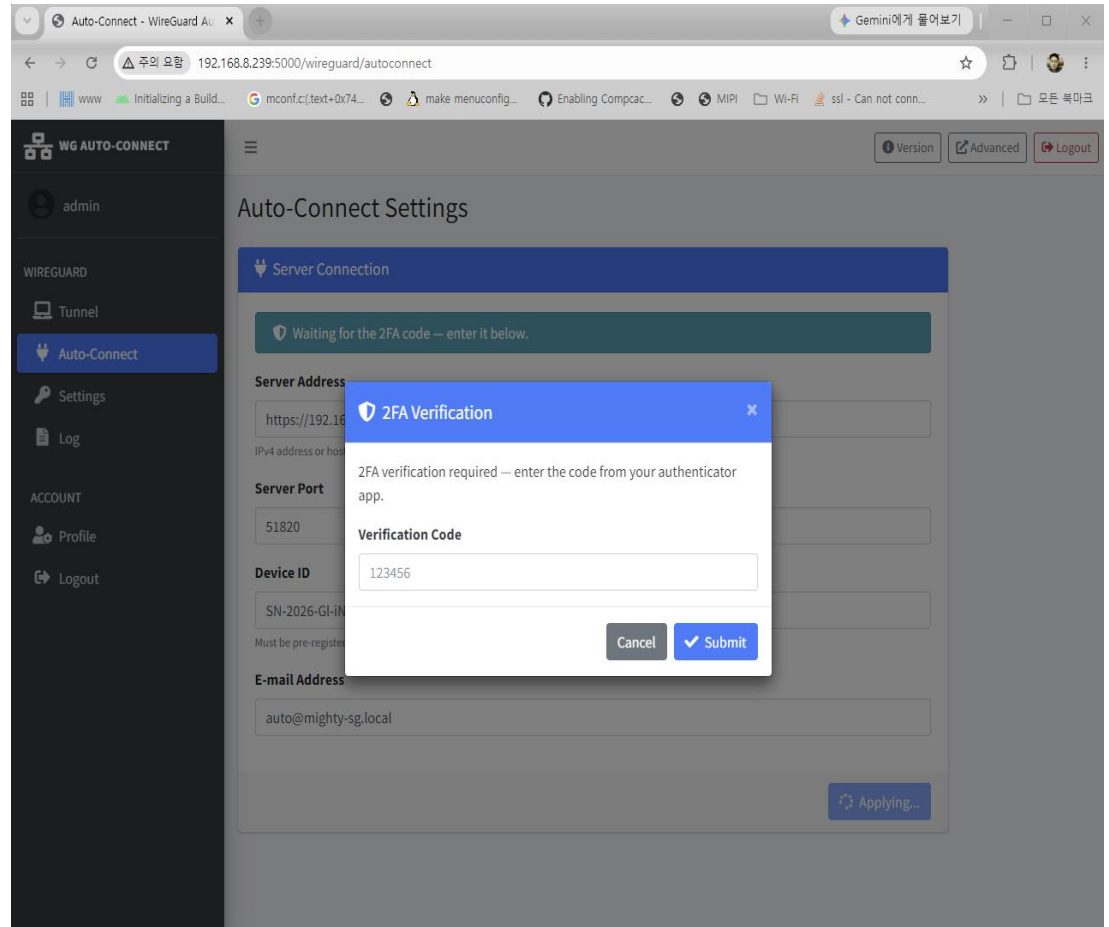
| Auto-Connect Settings                                                                  |                       |
|----------------------------------------------------------------------------------------|-----------------------|
| Server Connection                                                                      |                       |
| Connected.                                                                             |                       |
| Server Address                                                                         | https://192.168.8.129 |
| IPv4 address or hostname of the auto-connect server. "https://" is assumed if omitted. |                       |
| Server Port                                                                            | 51820                 |
| Device ID                                                                              | SN-2026-GI-iNet       |
| Must be pre-registered in MightyBox.                                                   |                       |
| E-mail Address                                                                         | auto@mighty-sg.local  |
| Apply                                                                                  |                       |

# 5.1 MightyConnect Mini(4) – WebUI(3)

- Auto Connection 2FA Step1 Page



- Auto Connection 2FA Step2 Page



# 5.1 MightyConnect Mini(4) – WebUI(4)

## ▪ Direct Connect Page

Direct Connect

Peer: SN-2026-Android

Connect

Active Links

| Peer              | VPN IP   | Status    | Message             | Expires | Action     |
|-------------------|----------|-----------|---------------------|---------|------------|
| SN-2026-Windows11 | 10.1.1.2 | connected | Connected directly. | —       | Disconnect |

## ▪ Direct Connect Filter Rules

Direct Connect

Peer: SN-2026-Android

Connect

Active Links

Outbound traffic to the VPN is restricted to these services (plus the box itself):

ICMP TCP/22 TCP/443 TCP/80 TCP/5000

Other direct-connected peers may reach these services on this device:

TCP/5432 TCP/5000 ICMP TCP/22 TCP/443 TCP/80

| Peer              | VPN IP   | Status    | Message             | Expires | Action     |
|-------------------|----------|-----------|---------------------|---------|------------|
| SN-2026-Windows11 | 10.1.1.2 | connected | Connected directly. | —       | Disconnect |

# 5.1 MightyConnect Mini(4) – WebUI(5)

## ■ Settings Page

Settings - WireGuard Auto-Connect

10.1.1.4:5000/wireguard/settings

MIGHTY CONNECT

admin

WIREGUARD

Tunnel

Auto-Connect

Direct Connect

Settings

Log

ACCOUNT

Profile

Logout

Auto-Connect Certificate Pin

TLS Certificate Pin

Pinned Certificate (SPKI SHA-256)

sha256//cuKwPCvgk04PNk7xHw4ao7UOD5+jCf2NXikMjdPZqY=

Must match the auto-connect server's certificate pin exactly, in curl's CURLOPT\_PINNEDPUBLICKEY format (sha256//<base64>). Get the current value from the gateway's System page or scripts/generate\_autoconnect\_cert.sh output. Leave empty to disable pinning.

Apply

Internal PC IP (DMZ)

Traffic reaching this gateway's own VPN IP — whether relayed through the box or reached via a Direct Connect link — is redirected to the internal PC below. Leave empty to disable.

Internal PC IP

192.168.3.139

Save

## ■ Log Page

Log - WireGuard Auto-Connect

10.1.1.4:5000/wireguard/log

MIGHTY CONNECT

admin

WIREGUARD

Tunnel

Auto-Connect

Direct Connect

Settings

Log

ACCOUNT

Profile

Logout

Daemon Log

Log Viewer

Log Size Limit

Auto-Refresh

wg-auto-daemon.log

200 lines

Log

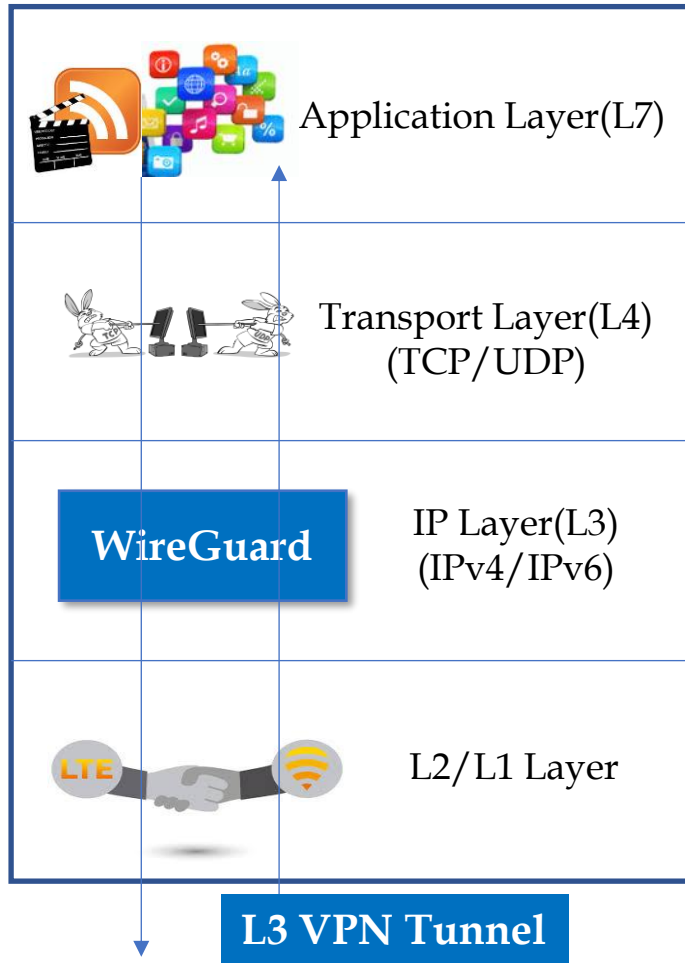
```
[2026-08-08 15:47:01] [DEBUG] Executing: wg set wg-auto0 listen-port 51820 private-key /usr/local/mightysg/etc/wg-auto-private.key peer s5BT+Cd1Vbjz0n9f3fY2VP2Ux6W7jFj1Q18GEBKpAE= preshared-key /tmp/wg_psk_wg-auto0 endpoint 192.168.8.114:51820 allowed-ips 10.1.1.0/24 persistent-keepalive 15
[2026-08-08 15:47:01] [INFO] WireGuard tunnel wg-auto0 is UP and configured.
[2026-08-08 15:47:01] [INFO] Tunnel restored successfully.
[2026-08-08 15:47:01] [INFO] DirectConnectClient: signaling listener up on 10.1.1.4:51821
[2026-08-08 15:47:02] [INFO] DirectConnectFilter: applied outbound filter to wg-auto0 (5 allowed, 6 exposed service(s))
[2026-08-08 15:47:03] [INFO] OpenWrtForwardFilter: applied (5 allowed, 6 exposed service(s)); wgauto->lan forwarding restricted to these, zone default REJECT backstops the rest
[2026-08-08 15:47:03] [INFO] OpenWrtDnat: applied - traffic to 10.1.1.4/32 (this gateway's VPN IP) redirected to 192.168.3.139, except ports reserved for this gateway itself (Direct Connect signaling 51821, litews UI 5000)
[2026-08-08 15:47:03] [INFO] ControlServer: listening on 127.0.0.1:51821
[2026-08-08 16:03:24] [INFO] Direct-connect: adding peer Pn17WjIG1+gc5f4KVkBDWTPghBQ5ctGLXFDorCjh908= (10.1.1.2/32 via 192.168.8.103:50855)
[2026-08-08 16:03:24] [DEBUG] Executing: wg set wg-auto0 peer Pn17WjIG1+gc5f4KVkBDWTPghBQ5ctGLXFDorCjh908= endpoint 192.168.8.103:50855 allowed-ips 10.1.1.2/32 persistent-keepalive 15
[2026-08-08 16:03:24] [DEBUG] NatPuncher: probing 10.1.1.2 via wg-auto0
[2026-08-08 16:03:26] [DEBUG] NatPuncher: probing 10.1.1.2 via wg-auto0
[2026-08-08 16:03:27] [DEBUG] NatPuncher: probing 10.1.1.2 via wg-auto0
```

## 5.2 MightyConnect Ubuntu



## 5.2 MightyConnect Ubuntu(1) – WireGuard

### Cryptokey Routing

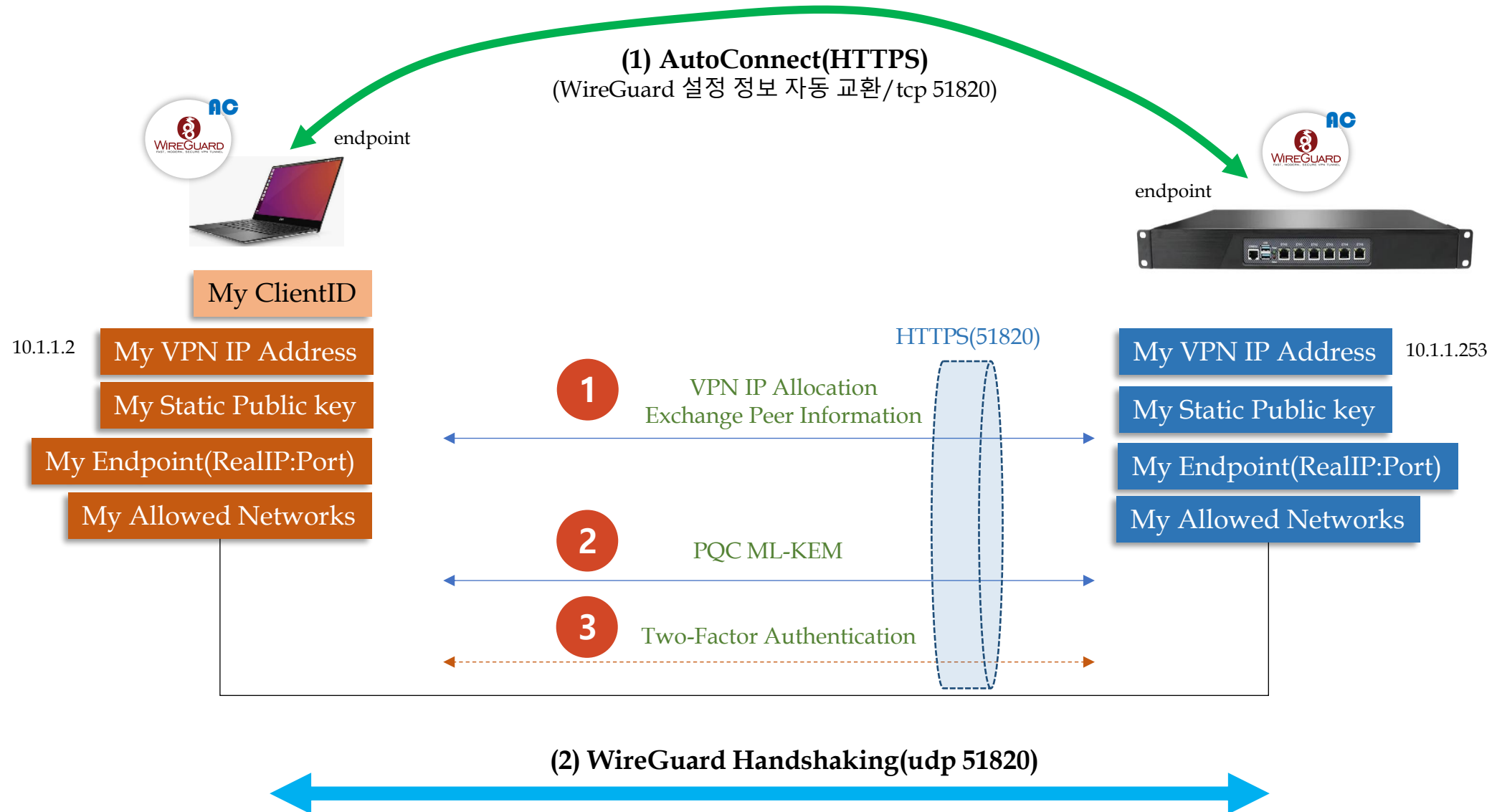


```
chyi@earth-new:/mnt/hdd/workspace/mighty_sg_prj/07162026$ sudo wg show
interface: wg-auto0
  public key: NYJM7SxhjeQXjjipIsEiNxZY3w6NcPJl/Mfjfye4mzY=
  private key: (hidden)
  listening port: 51820

peer: 9JPq7SGTFy7o+8PLSXaf40J4wjBKql9o5CPRc8jJbg4=
  preshared key: (hidden)
  endpoint: 192.168.8.129:51820
  allowed ips: 10.1.1.0/24
  latest handshake: 1 minute, 40 seconds ago
  transfer: 316 B received, 180 B sent
  persistent keepalive: every 15 seconds
chyi@earth-new:/mnt/hdd/workspace/mighty_sg_prj/07162026$ ping 10.1.1.253
PING 10.1.1.253 (10.1.1.253) 56(84) bytes of data:
64 bytes from 10.1.1.253: icmp_seq=1 ttl=64 time=2.70 ms
64 bytes from 10.1.1.253: icmp_seq=2 ttl=64 time=2.94 ms
^C
--- 10.1.1.253 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 2.698/2.817/2.936/0.119 ms
```

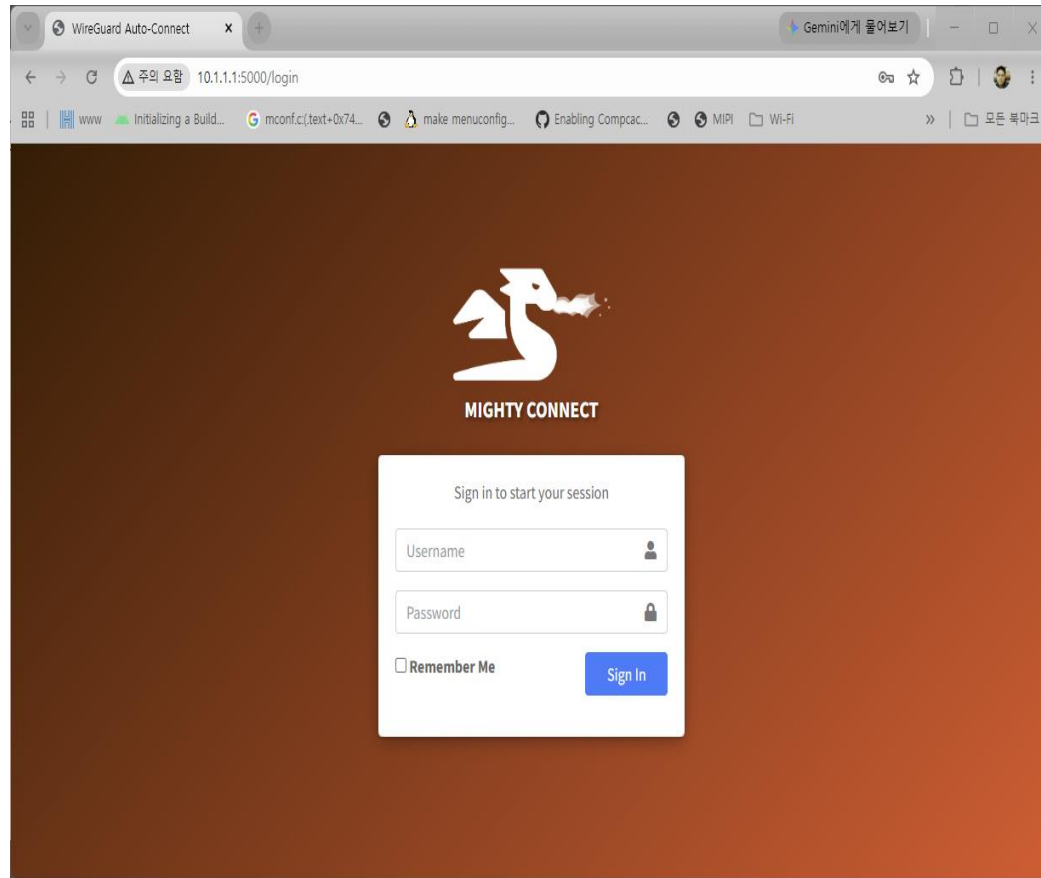


## 5.2 MightyConnect Ubuntu(2) – Auto Connection

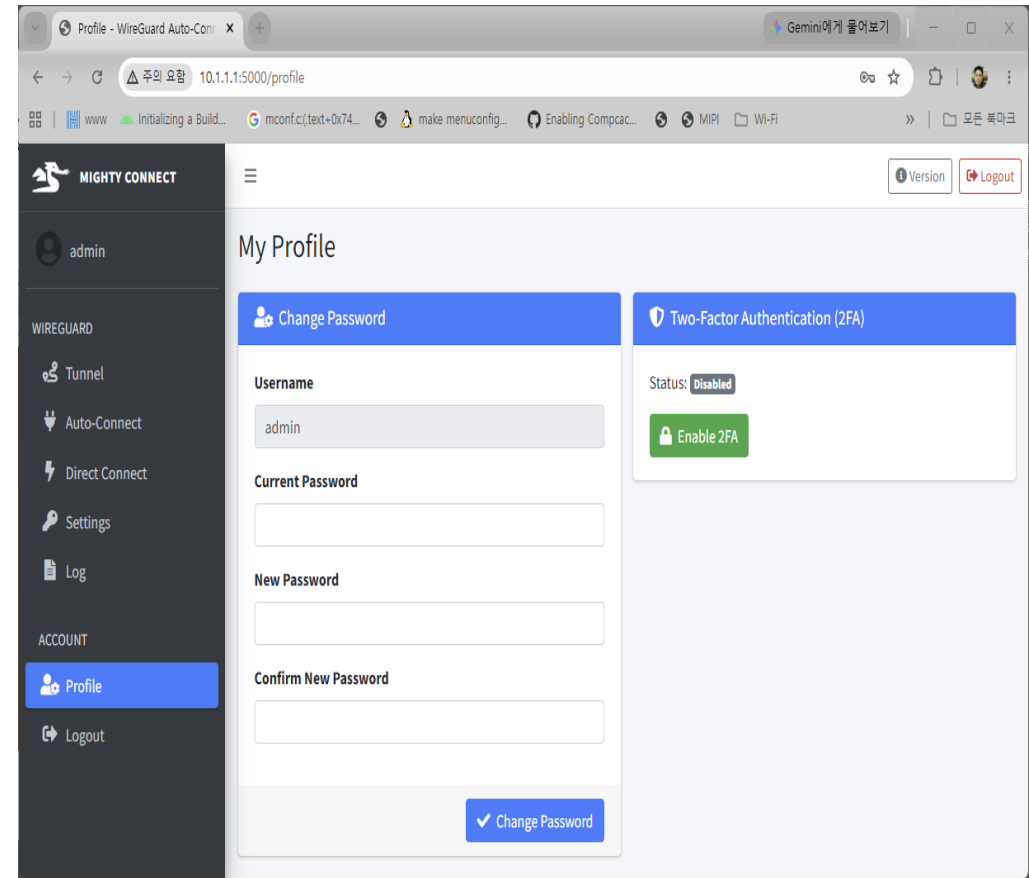


## 5.2 MightyConnect Ubuntu(3) – WebUI(1)

### ■ Login Page

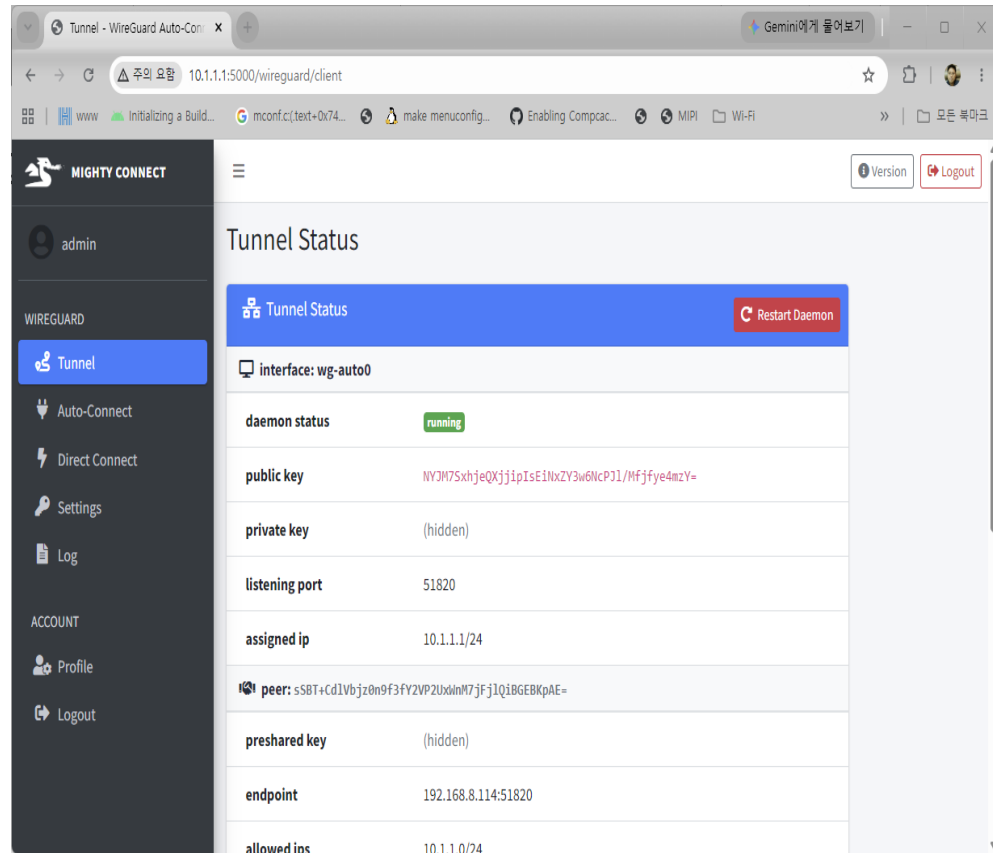


### ■ Admin User Page

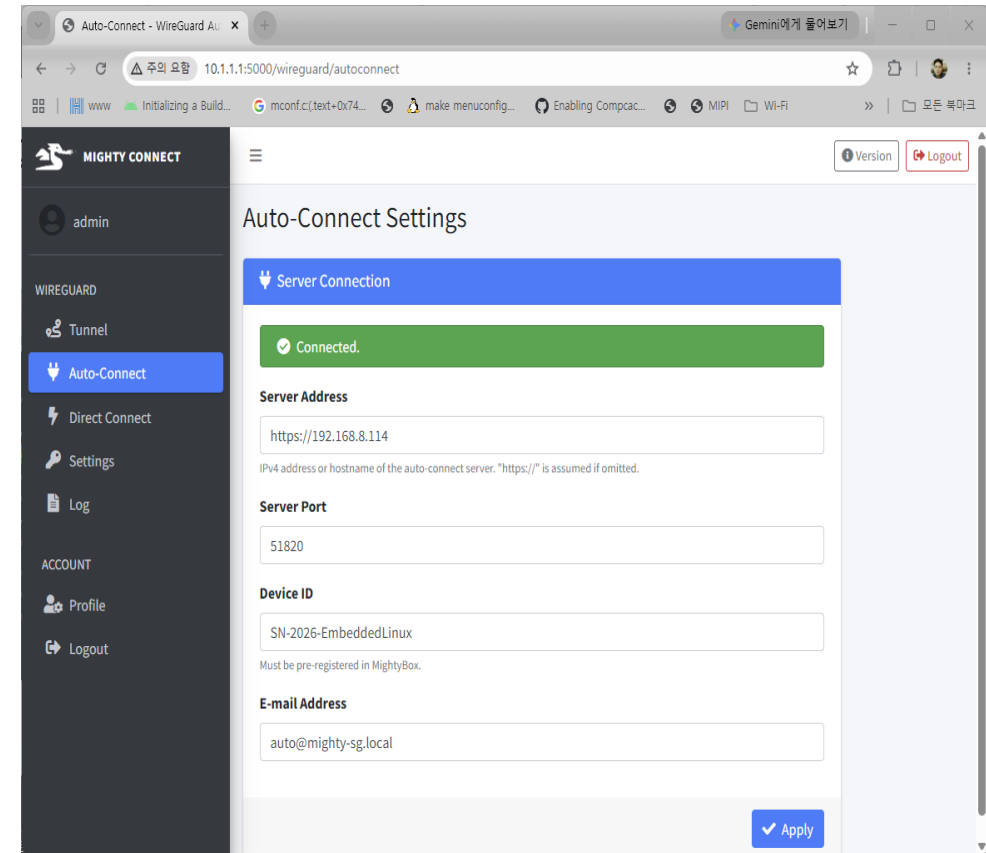


## 5.2 MightyConnect Ubuntu(3) – WebUI(2)

### ■ Tunnel Status Page

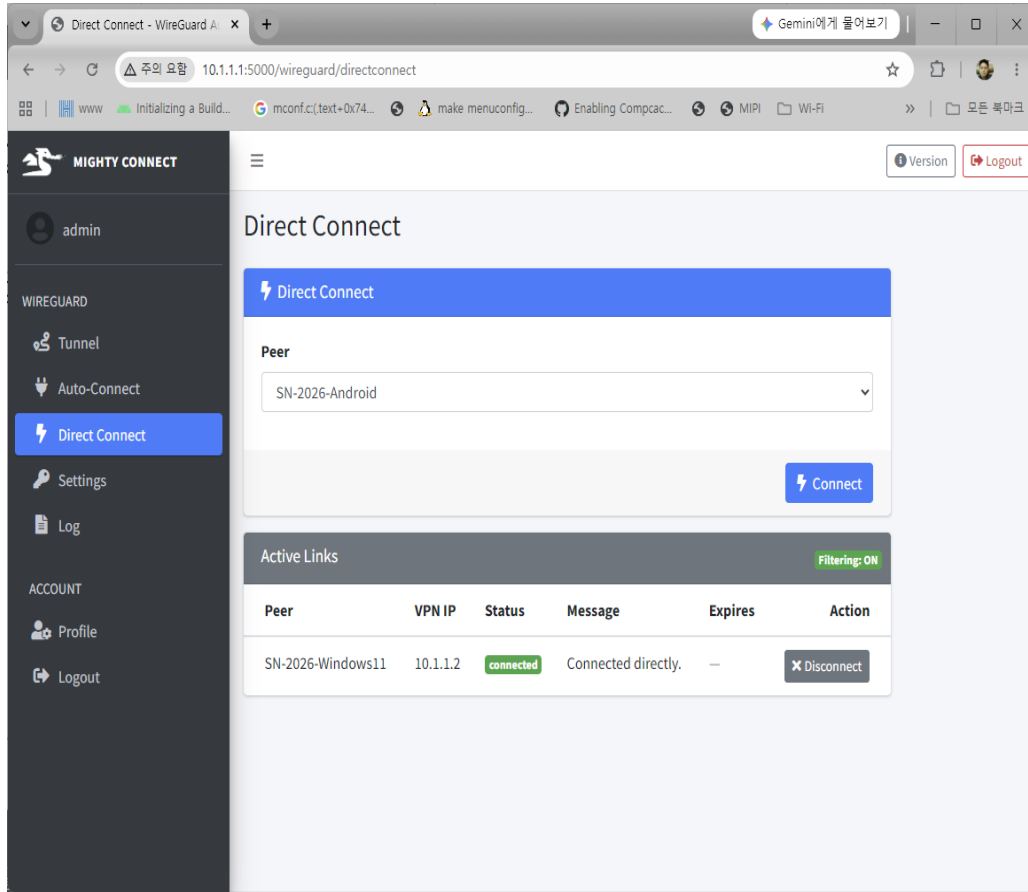


### ■ Auto Connection Page



## 5.2 MightyConnect Ubuntu(3) – WebUI(3)

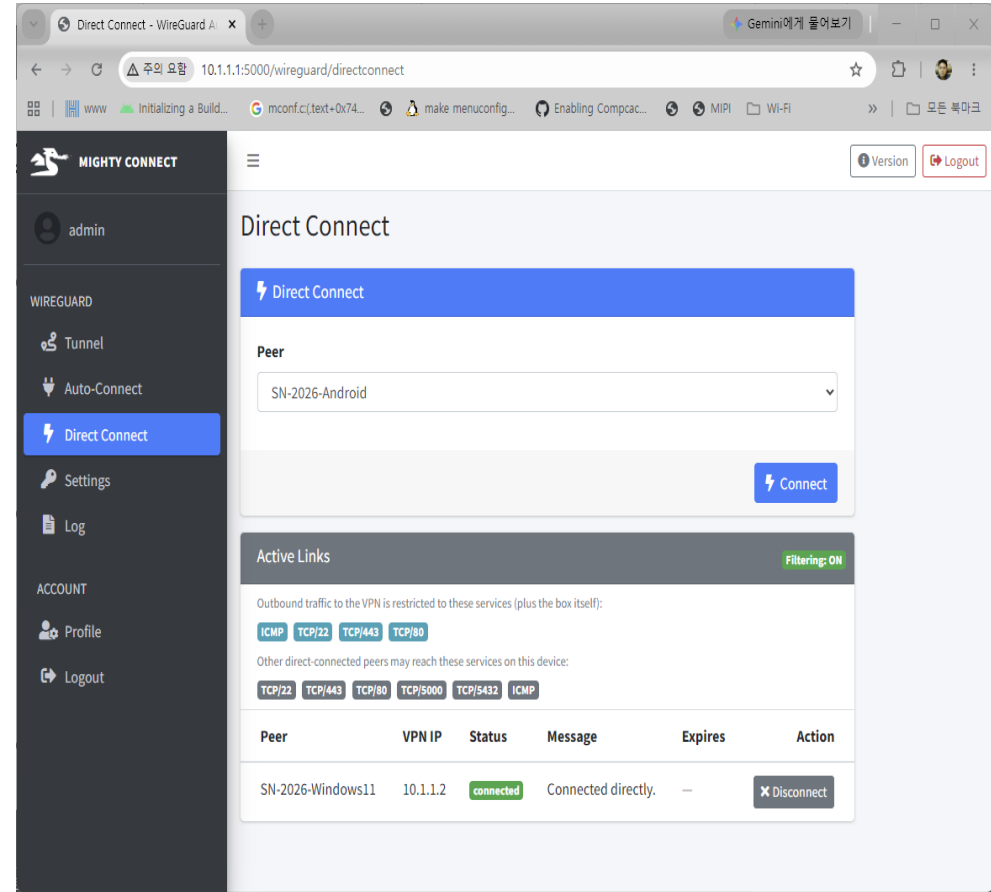
### ▪ Direct Connect Page



The screenshot shows the 'Direct Connect' page in the MightyConnect WebUI. The left sidebar contains navigation links for 'admin', 'WIREGUARD', 'Tunnel', 'Auto-Connect', 'Direct Connect' (highlighted), 'Settings', 'Log', 'ACCOUNT', 'Profile', and 'Logout'. The main content area has a 'Direct Connect' header and a 'Peer' dropdown menu set to 'SN-2026-Android'. A 'Connect' button is visible. Below the 'Connect' button is an 'Active Links' section with a table showing one active connection.

| Peer              | VPN IP   | Status    | Message             | Expires | Action     |
|-------------------|----------|-----------|---------------------|---------|------------|
| SN-2026-Windows11 | 10.1.1.2 | connected | Connected directly. | —       | Disconnect |

### ▪ Direct Connect Filtering



The screenshot shows the 'Direct Connect' page in the MightyConnect WebUI, specifically the 'Direct Connect Filtering' view. The left sidebar is identical to the previous screenshot. The main content area has a 'Direct Connect' header and a 'Peer' dropdown menu set to 'SN-2026-Android'. A 'Connect' button is visible. Below the 'Connect' button is an 'Active Links' section with a table showing one active connection. Above the table, there is a section for 'Outbound traffic to the VPN is restricted to these services (plus the box itself):' with a list of services: ICMP, TCP/22, TCP/443, TCP/80. Below this, there is a section for 'Other direct-connected peers may reach these services on this device:' with a list of services: TCP/22, TCP/443, TCP/80, TCP/5000, TCP/5432, ICMP.

| Peer              | VPN IP   | Status    | Message             | Expires | Action     |
|-------------------|----------|-----------|---------------------|---------|------------|
| SN-2026-Windows11 | 10.1.1.2 | connected | Connected directly. | —       | Disconnect |

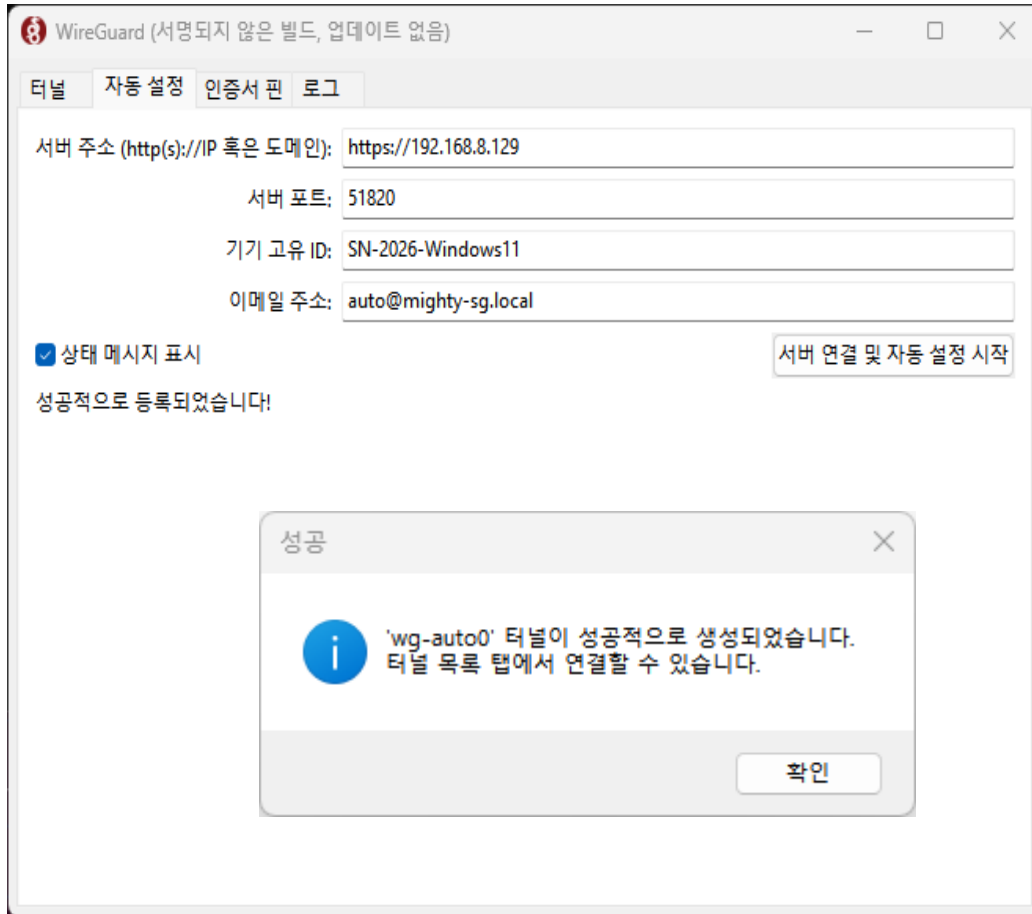
## 5.3 MightyConnect Windows

### MightyConnect Windows Agent

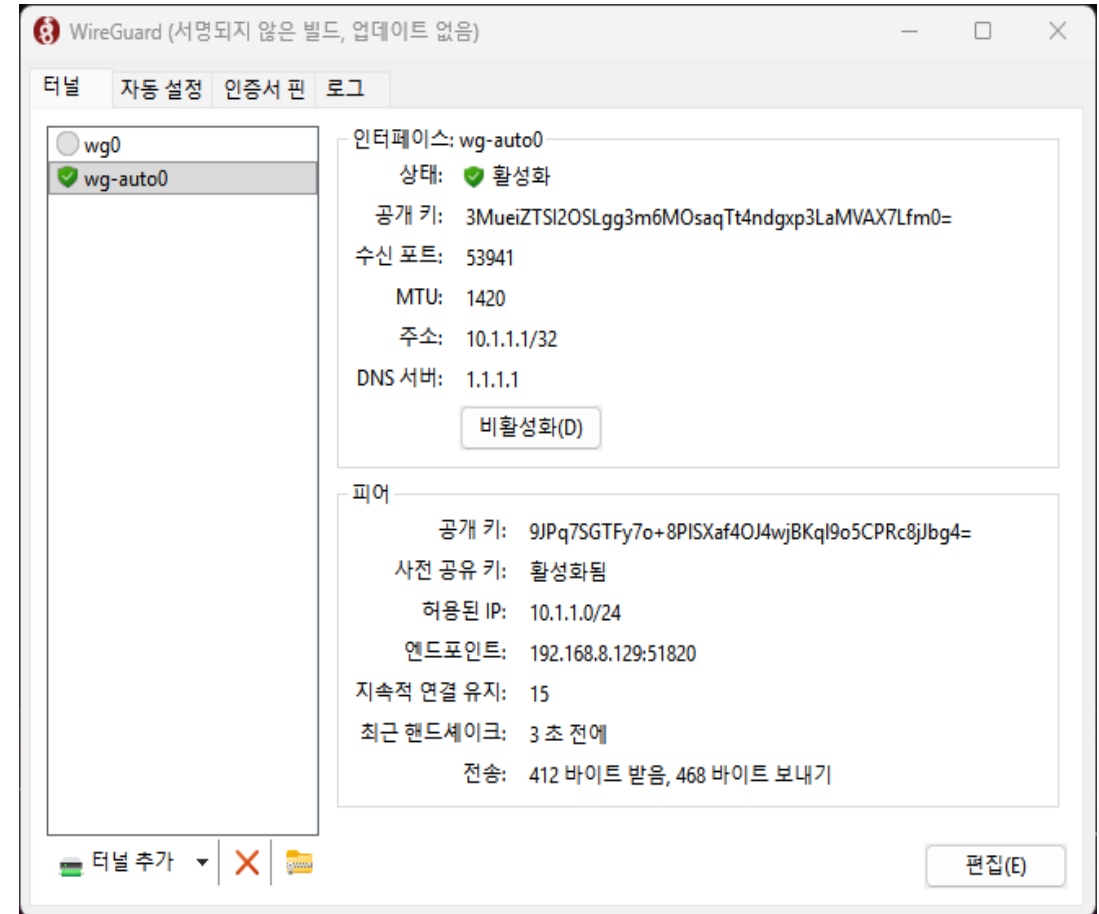


## 5.3 MightyConnect Windows(1) – User Interface(1)

- Auto Connection Page

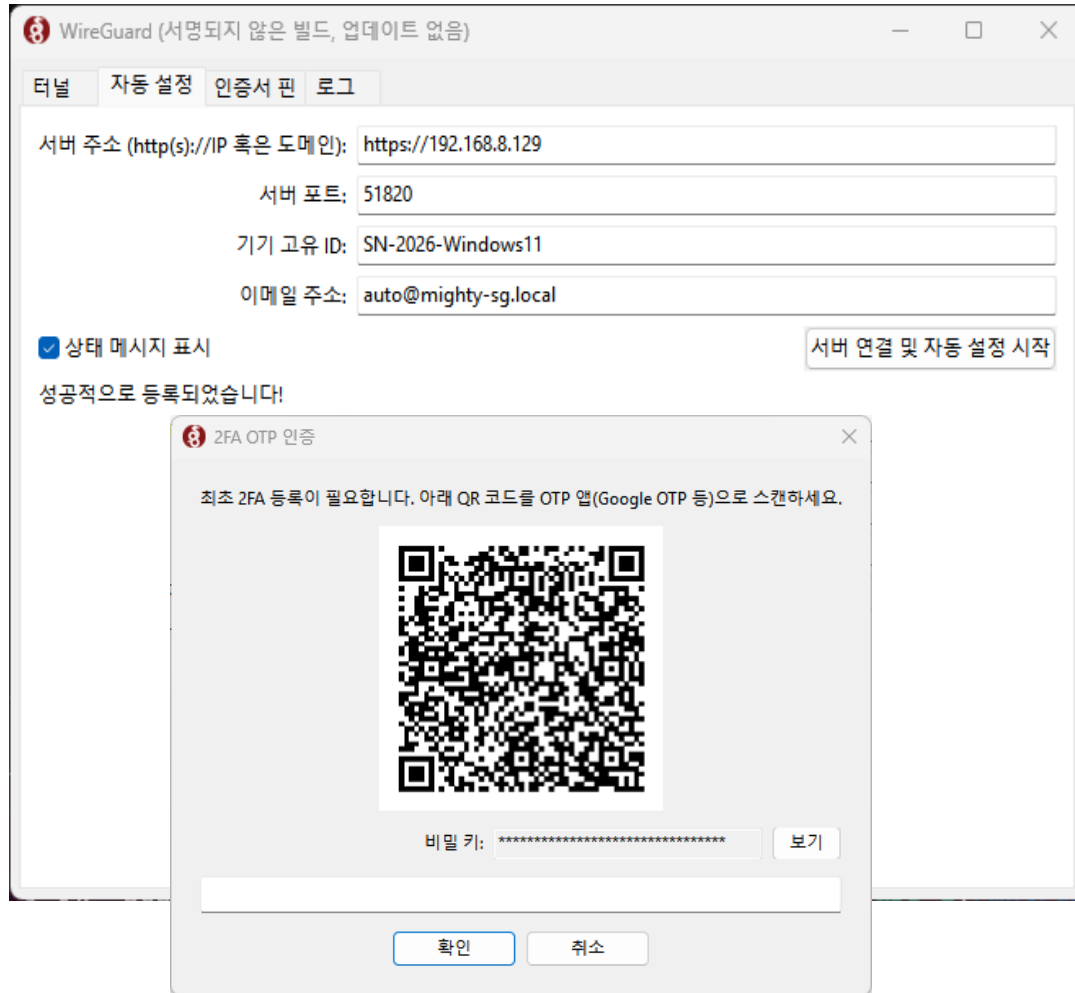


- wg-auto0 interface 생성 및 터널 활성화 Page

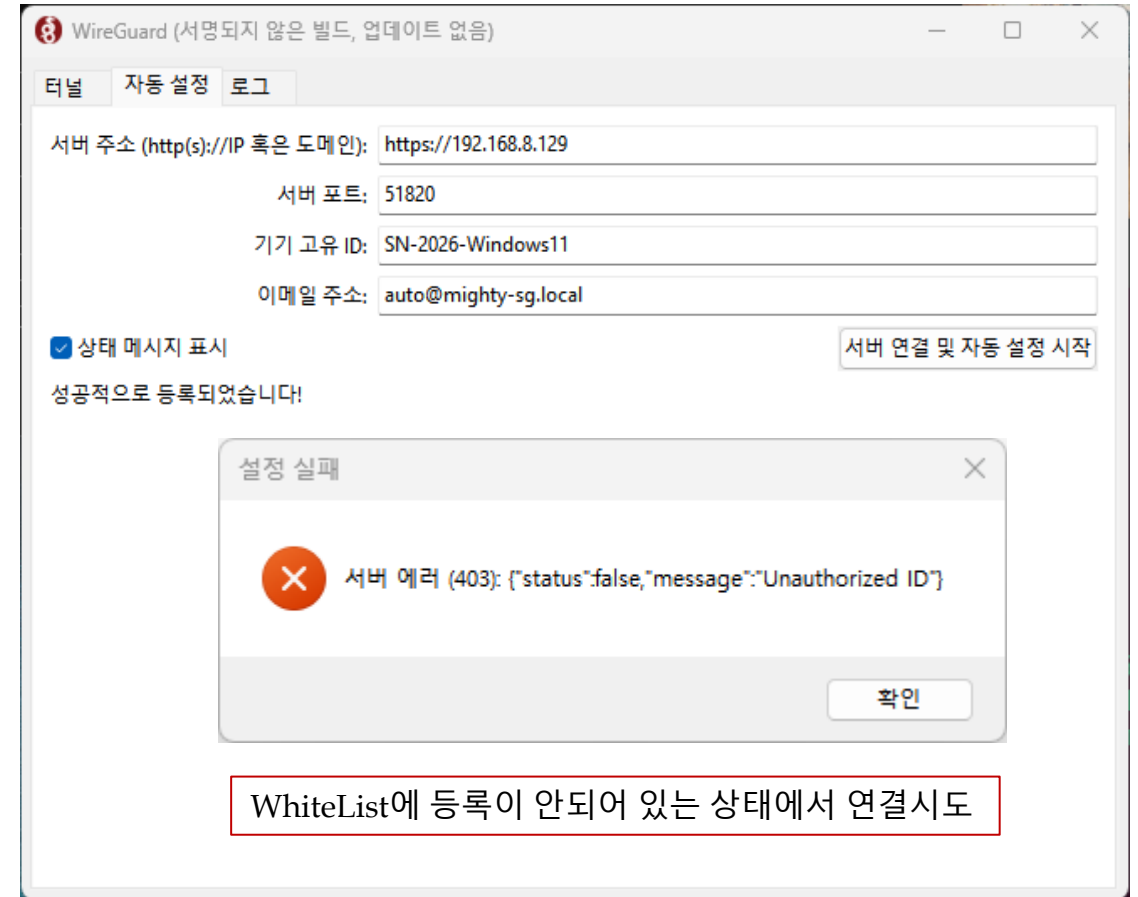


## 5.3 MightyConnect Windows(1) – User Interface(2)

- Auto Connection – 2FA 인증 Page

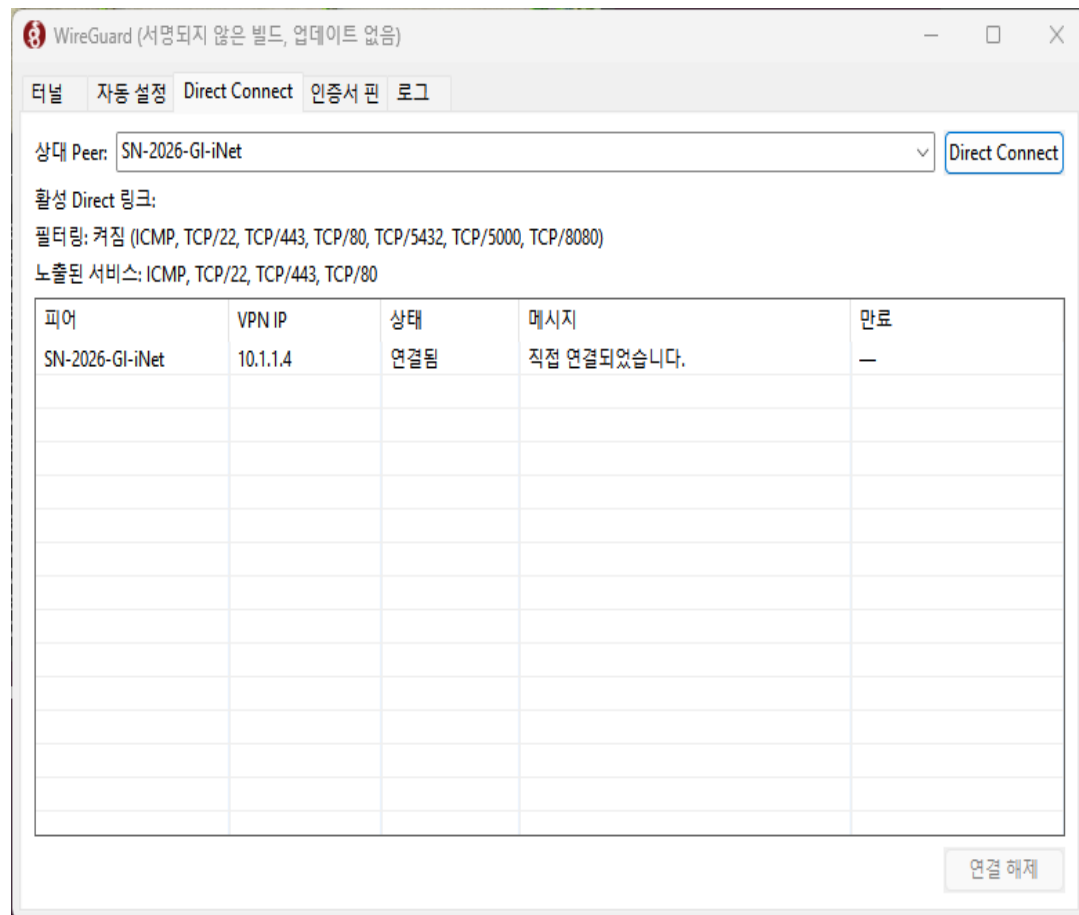


- Auto Connection – 인증 실패 예시

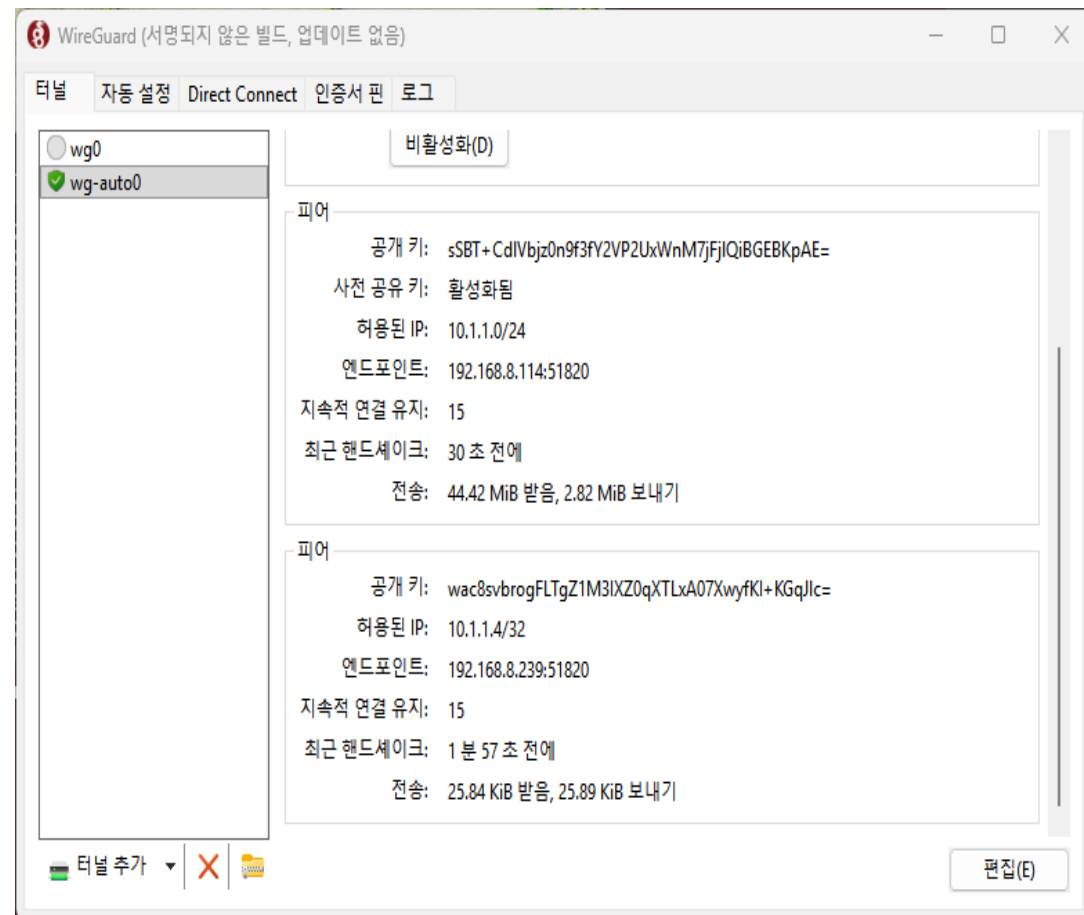


## 5.3 MightyConnect Windows(1) – User Interface(3)

- Direct Connect Page



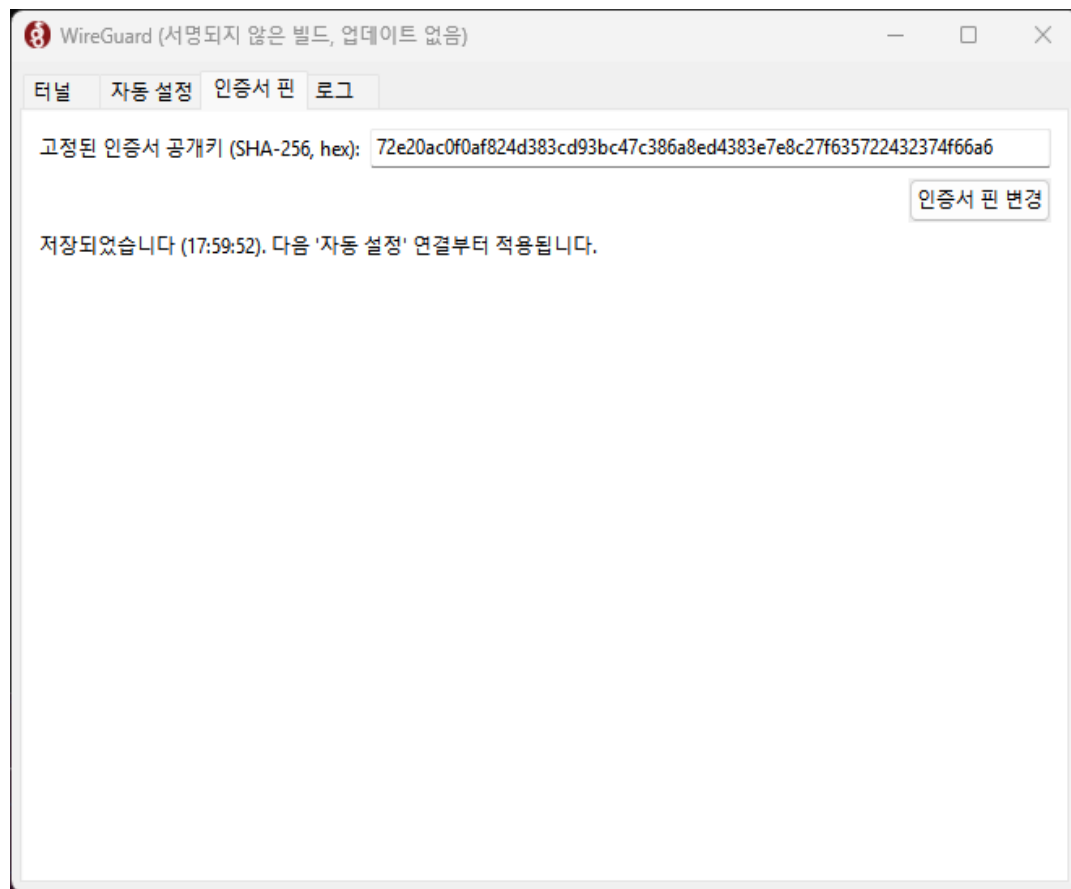
- Direct Connect Peer 확인



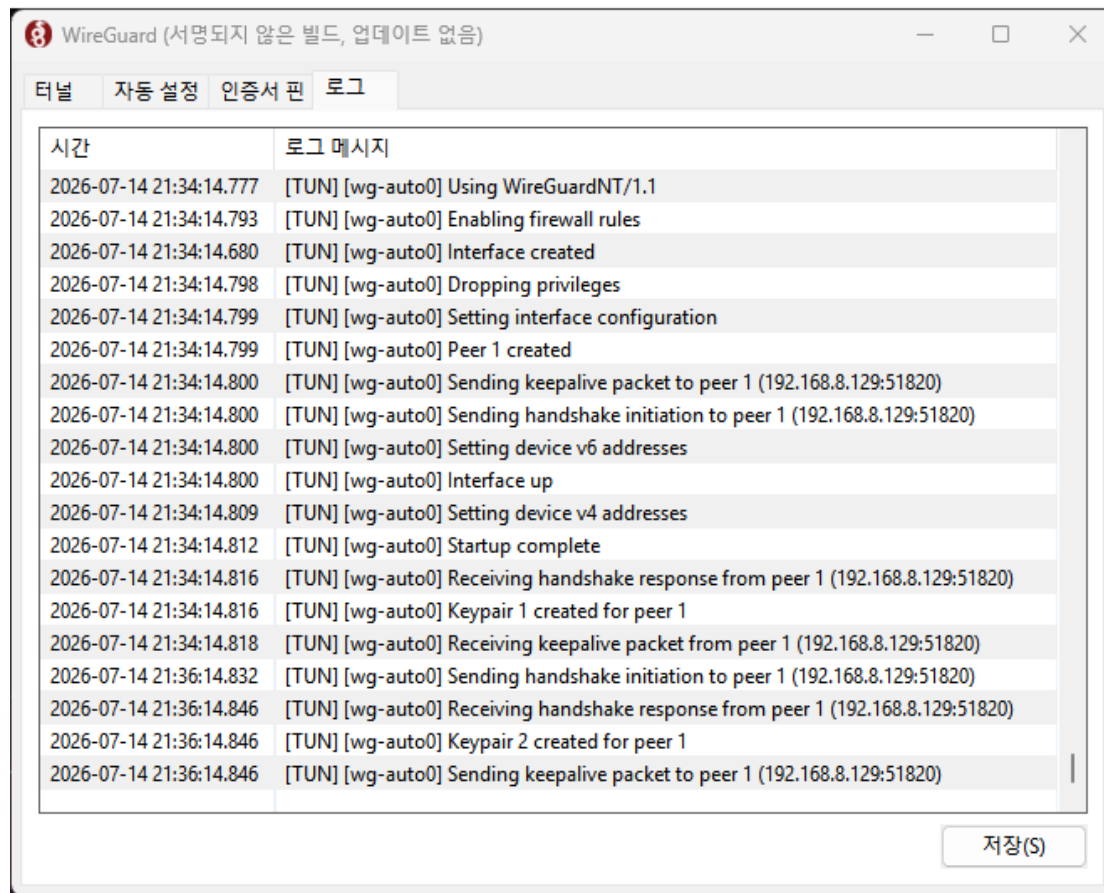


## 5.3 MightyConnect Windows(1) – User Interface(4)

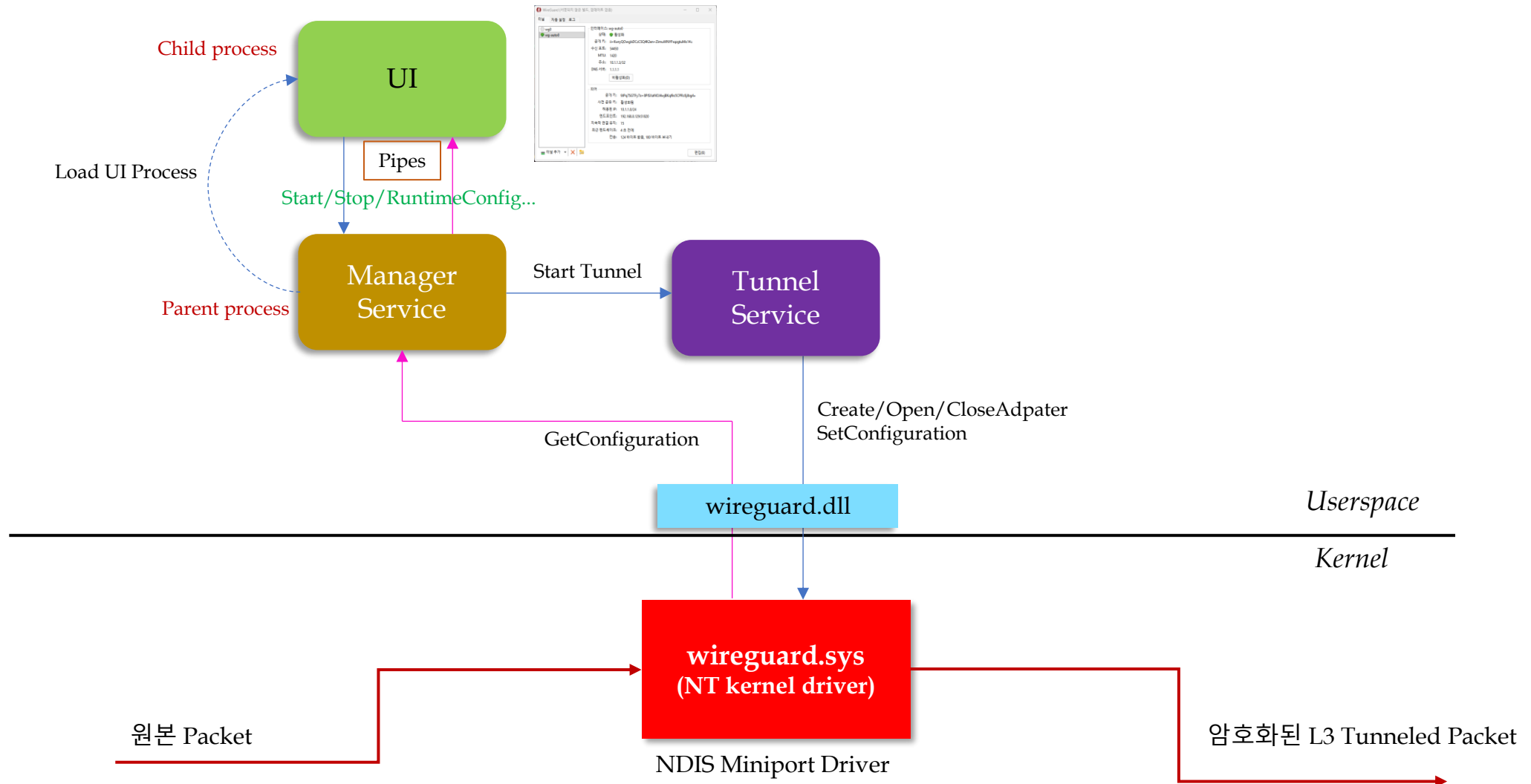
- X.509 인증서 Pin 값 변경 Page



- Log Page



## 5.3 MightyConnect Windows(2) – Architecture

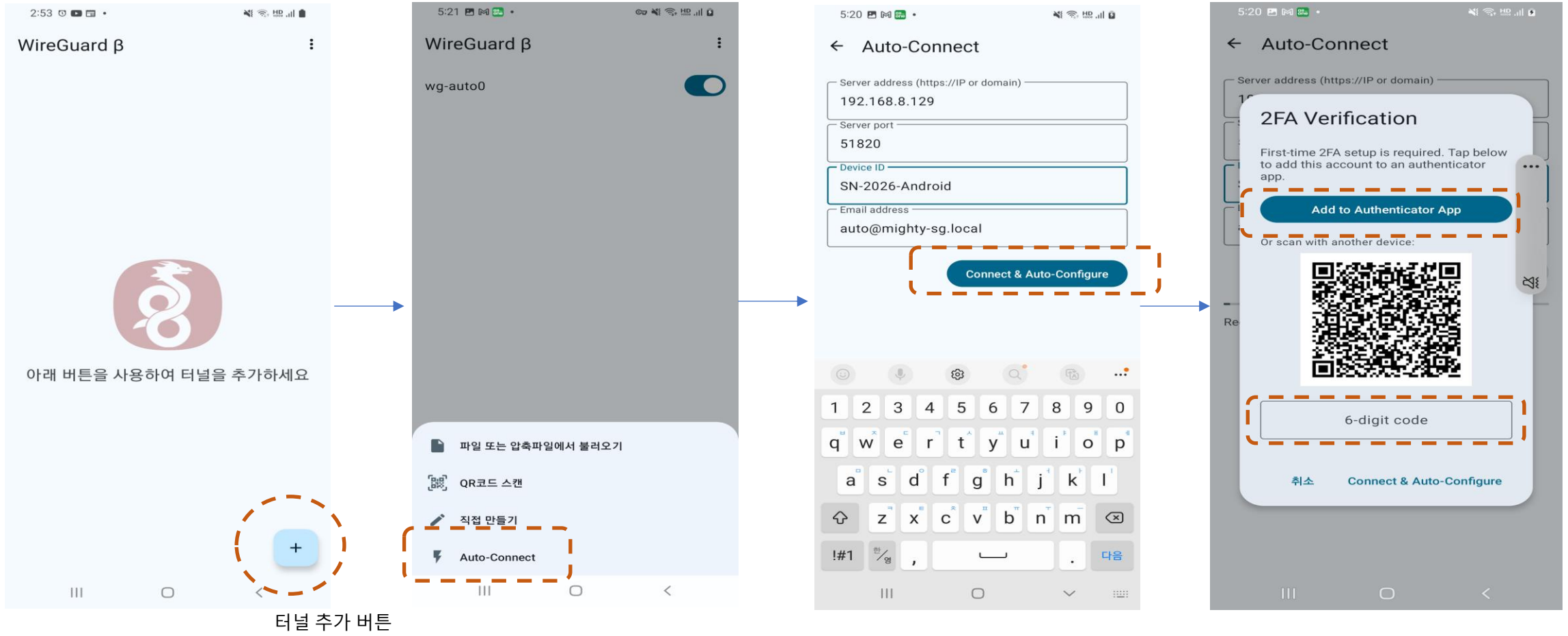


## 5.4 MightyConnect Android

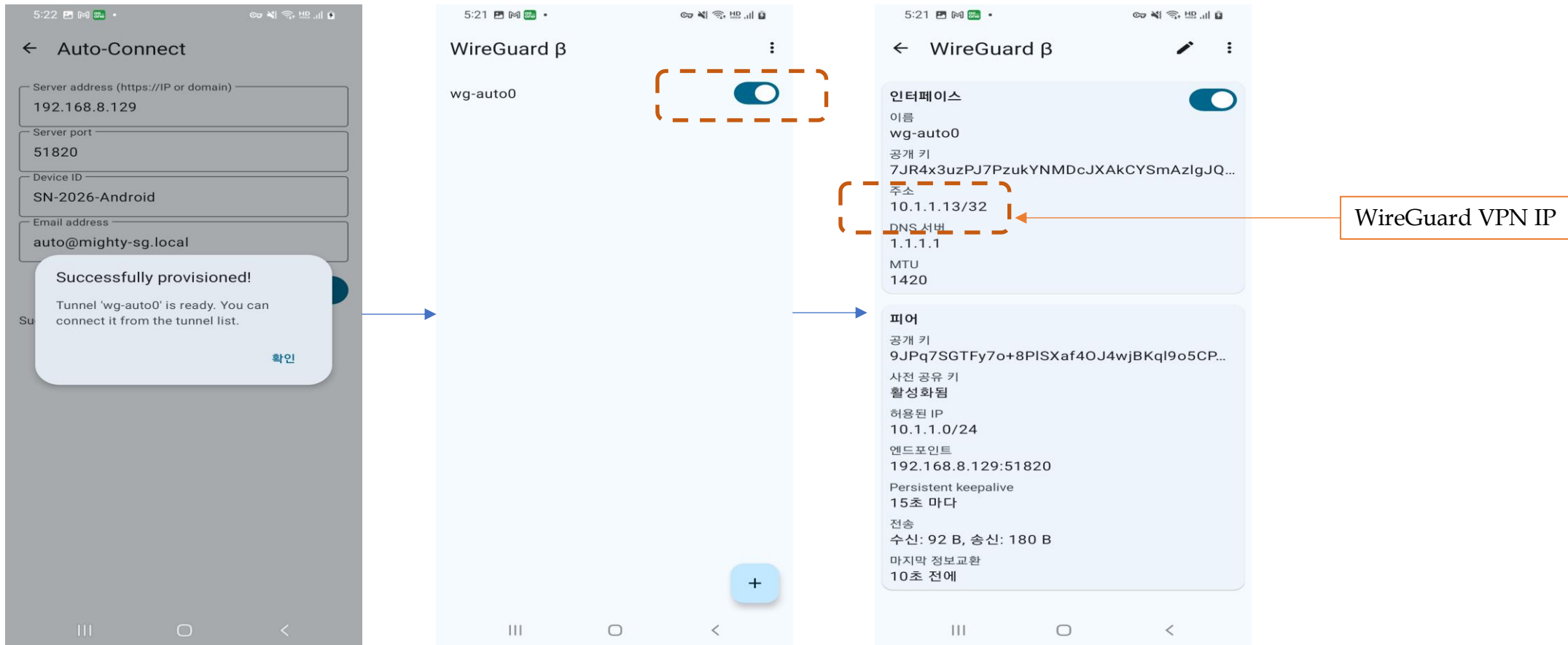
### Android App



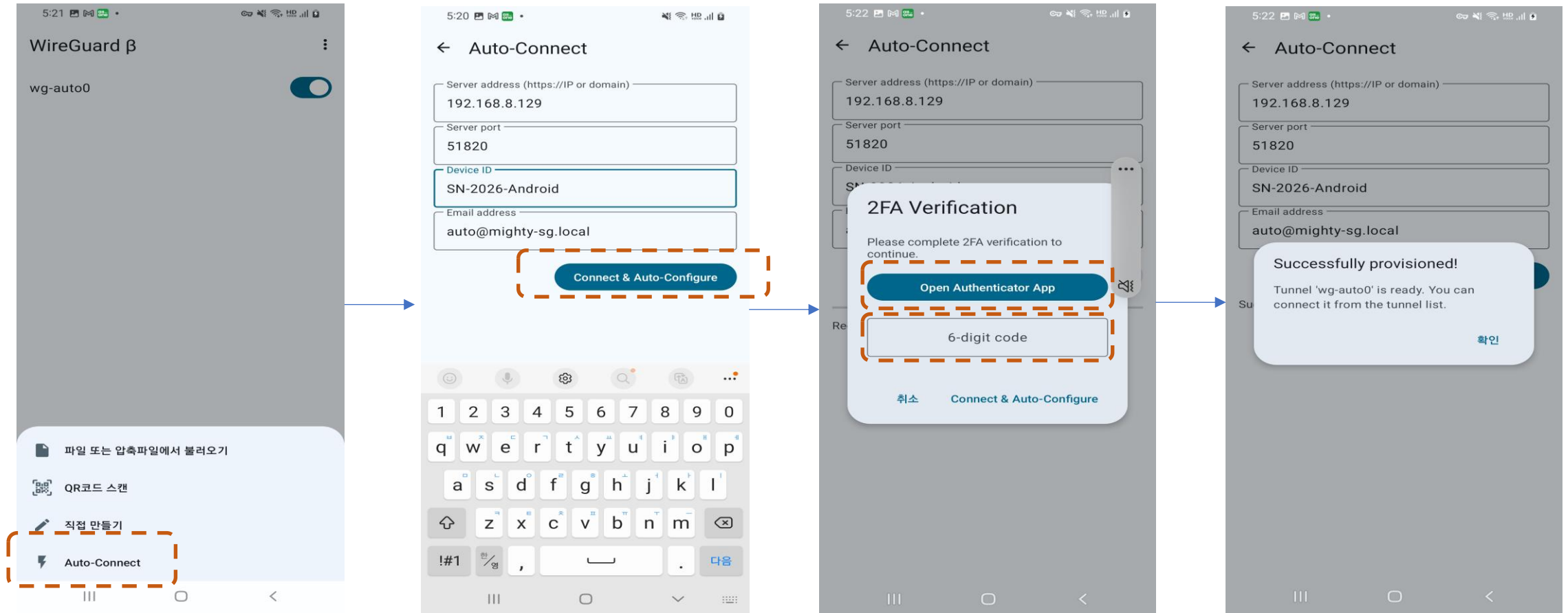
## 5.4 MightyConnect Android(1) – Auto Connect(1)



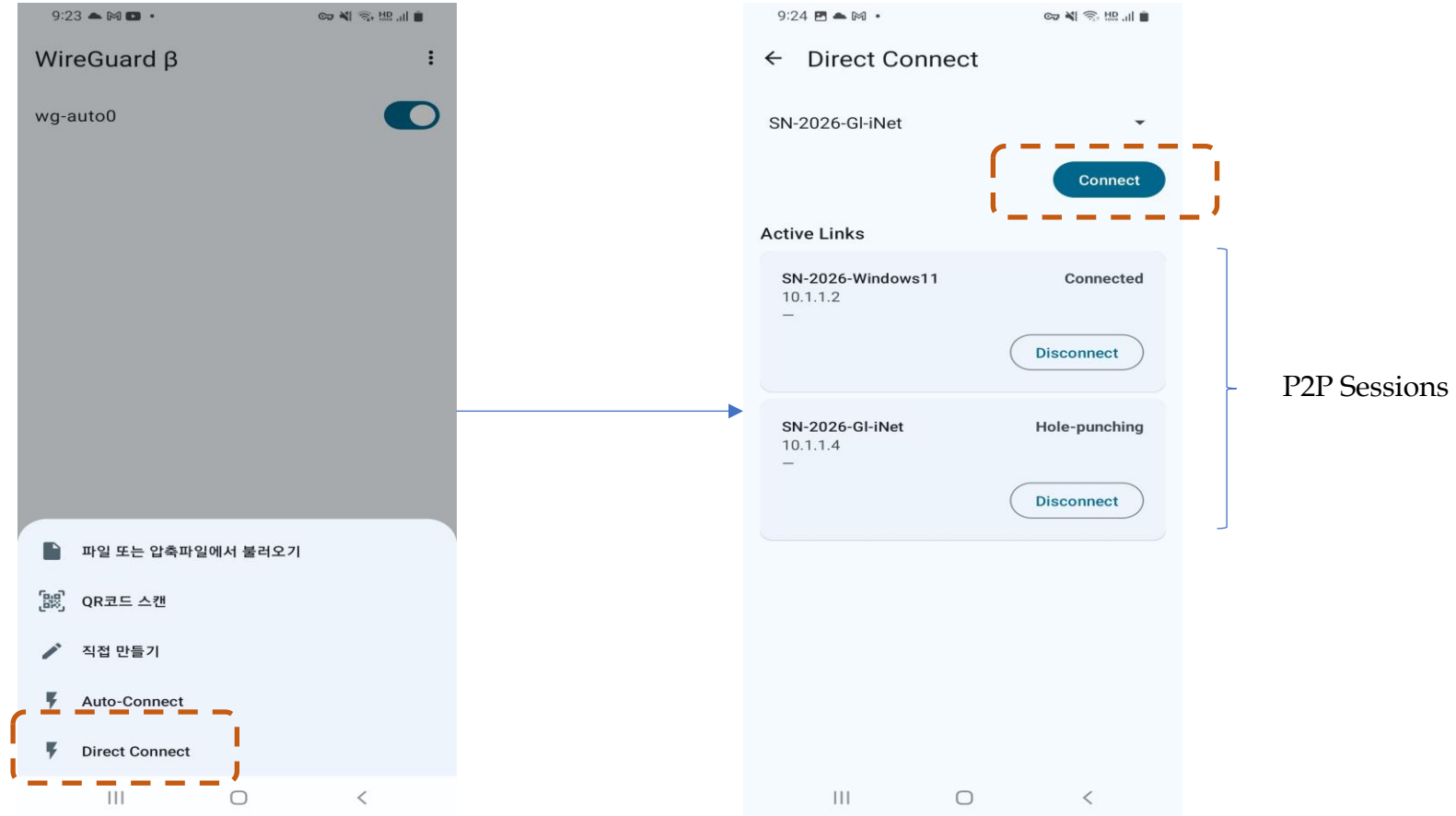
## 5.4 MightyConnect Android(1) – Auto Connect(2)



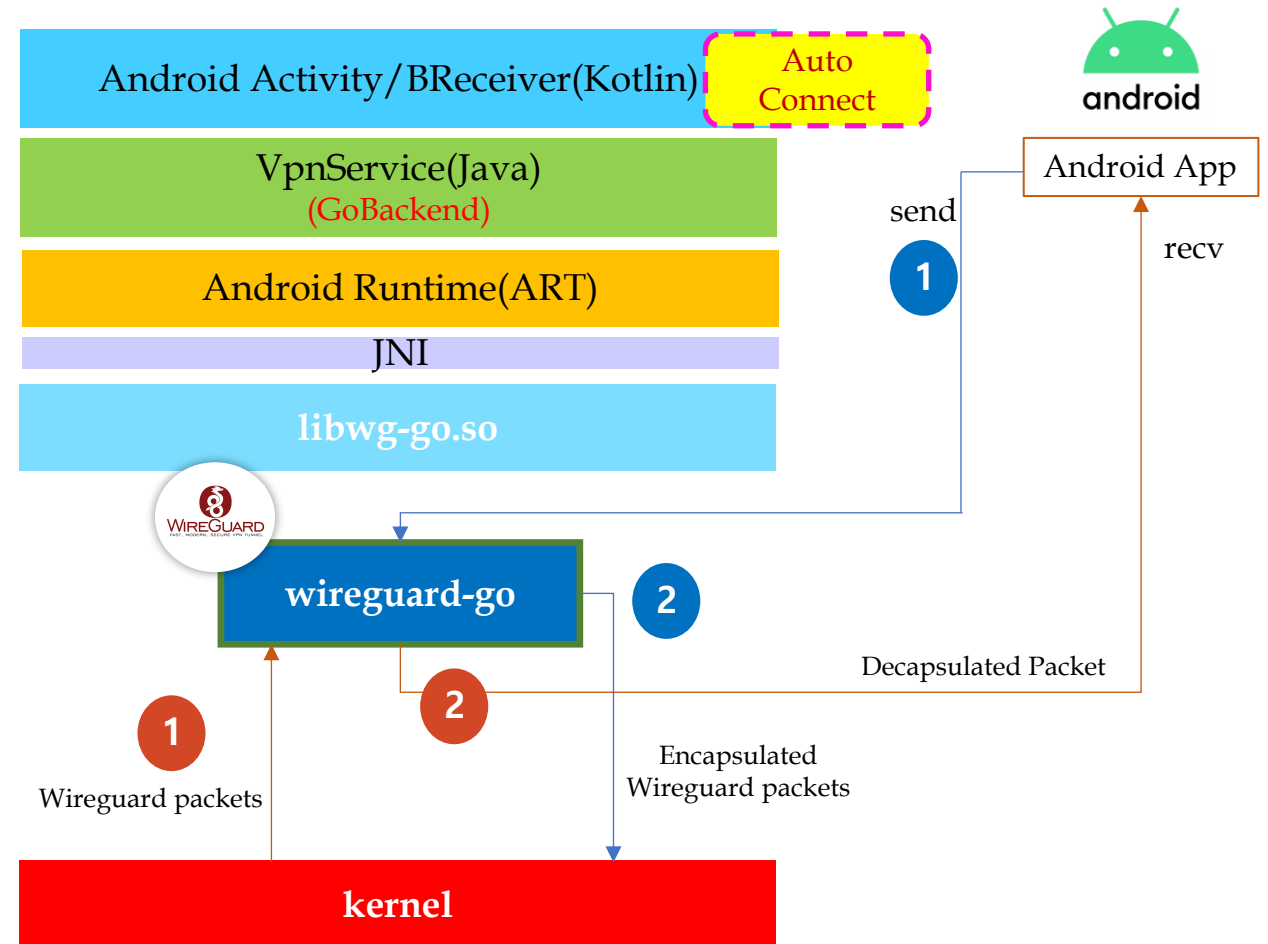
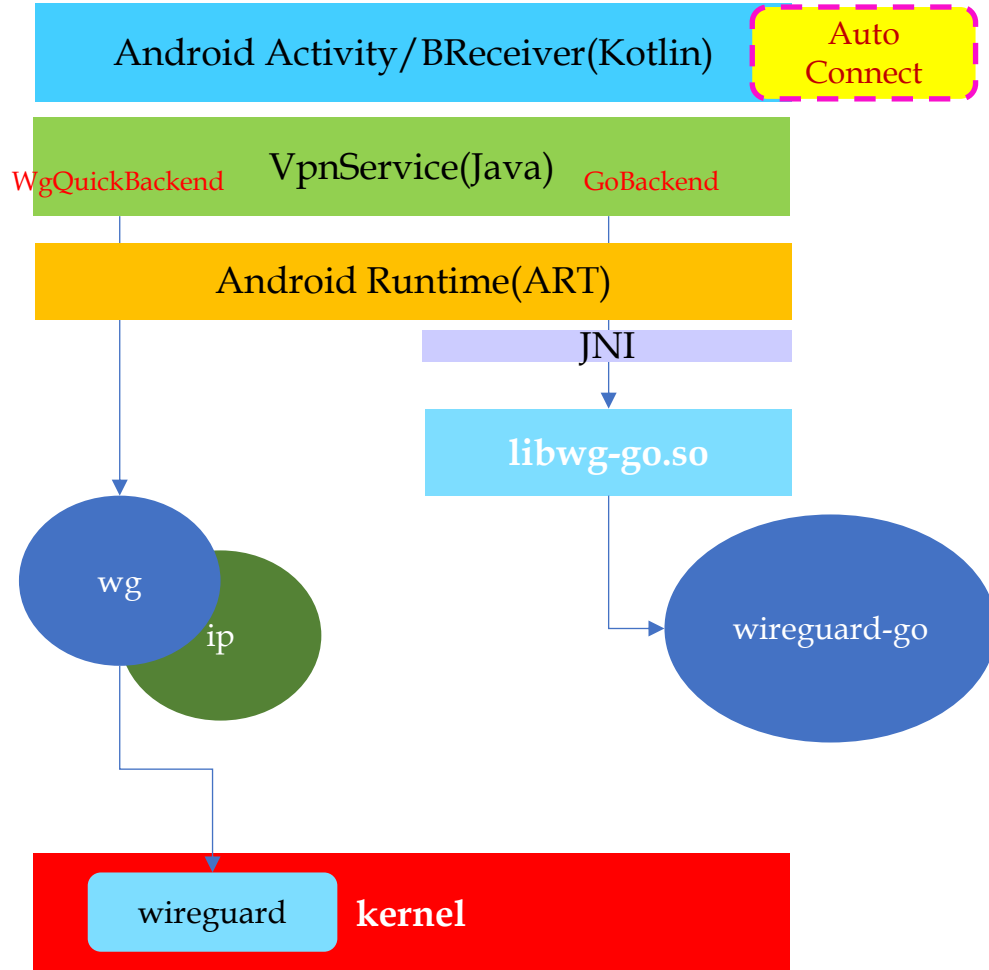
## 5.4 MightyConnect Android(1) – Auto Connect(3)



## 5.4 MightyConnect Android(2) – Direct Connect



## 5.4 MightyConnect Android(3) – Architecture



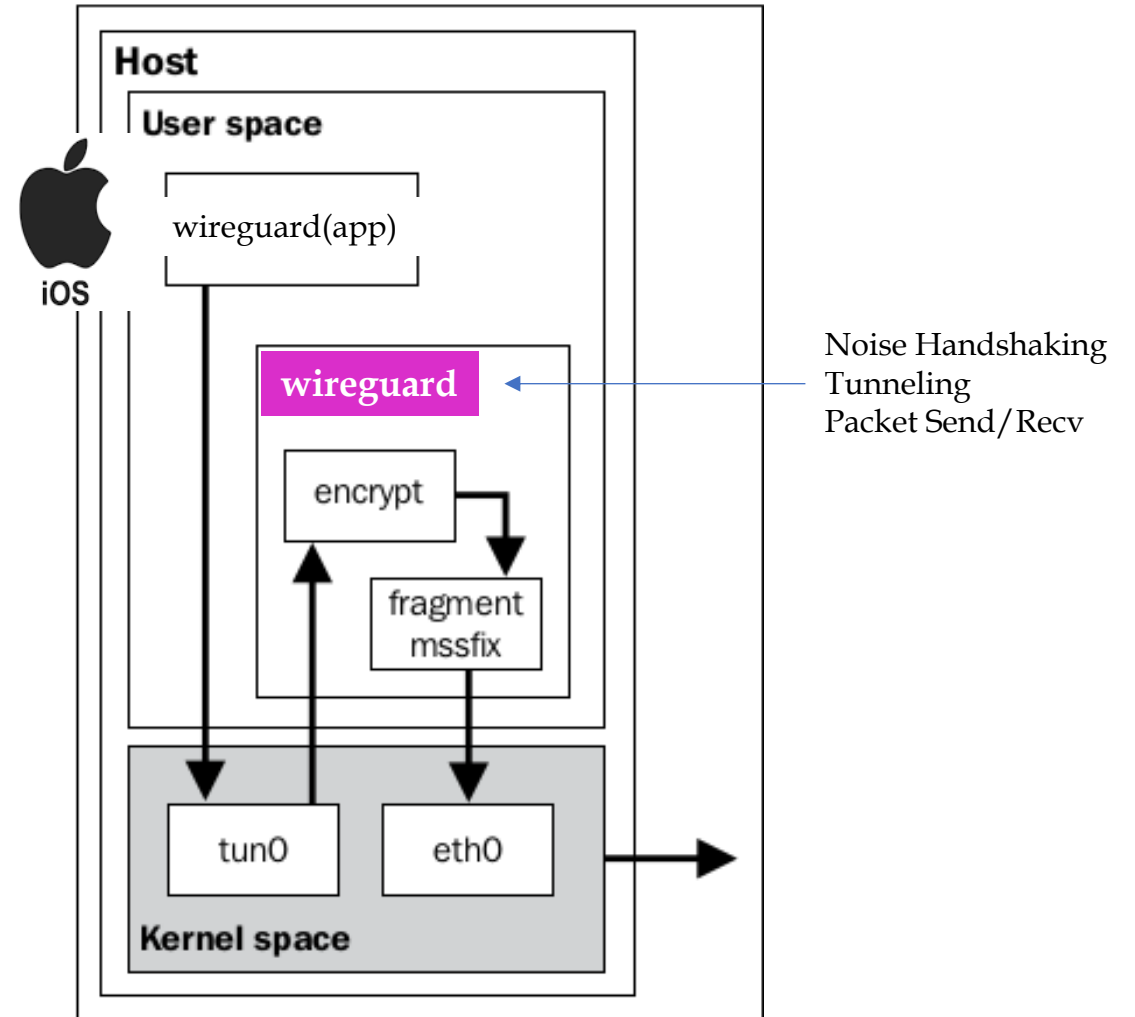
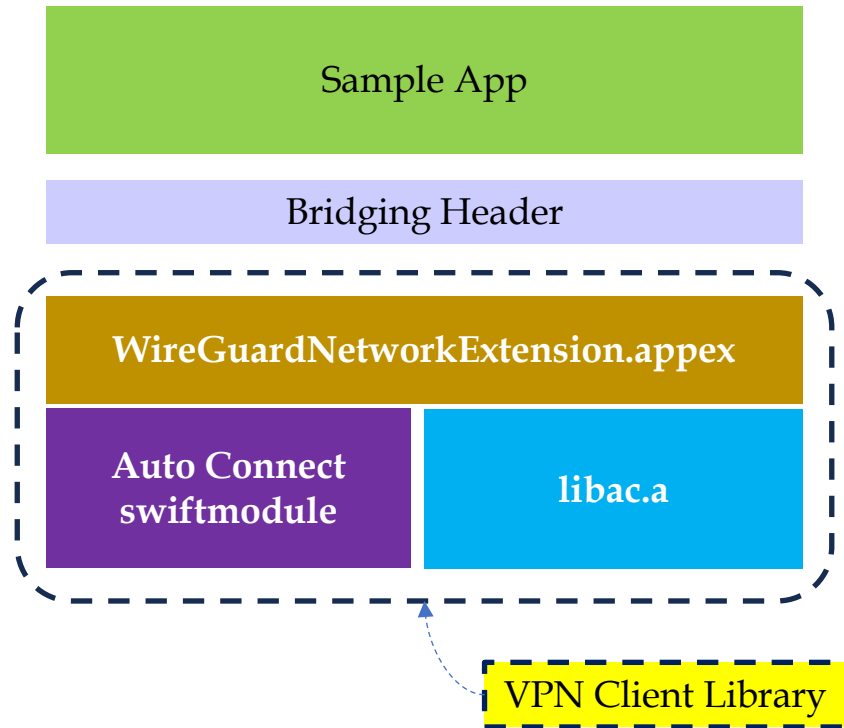


## 5.5 MightyConnect macOS/iOS

### MightyConnect macOS/iOS Agent



## 5.5 MightyConnect macOS/iOS – Architecture



## 6. References

# 6. References

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